



EM420

Production Operations Management



EM420 PRODUCTION OPERATIONS MANAGEMENT

ABET eCOURSEBOOK

AY2026

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UNITED STATES MILITARY ACADEMY
WEST POINT.

SECTION I

COURSE ASSESSMENTS



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB A

AY2026 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

December 20th, 2025

MEMORANDUM FOR COL Brandon Thompson, Engineering Management Program Director,
Department of Systems Engineering, West Point, NY 109963

SUBJECT: AT 26-1 Course Assessment for EM420: Production Operations Management

1. The attached report summarizes the EM420 assessment and curriculum change actions for Academic Term (AT) 26-1.
2. The POC for this action is the undersigned at isabella.sanders@westpoint.edu.

ISABELLA T. SANDERS
PhD, Assistant Professor
EM420 Course Director

Encls

1. Course Assessment
2. Course Instructional Memorandum, Graded Events, and Syllabus
3. OML and Final Grades

DISTRIBUTION:

1 each EM420 instructors, Program Director, and Department Head

Enclosure 1: Course Assessment

Section I – Course Information and History

1. Historic Course Offerings and Enrollments.

a. Offered.

AY14		AY15		AY16		AY17	
14-1	14-2	15-1	15-2	16-1	16-2	17-1	17-2
X	X		X	X	X	X	X
AY18		AY19		AY20		AY21	
18-1	18-2	19-1	19-2	20-1	20-2	21-1	21-2
X	X	X		X		X	
AY22		AY23		AY24		AY25	
22-1	22-2	23-1	23-2	24-1	24-2	25-1	25-2
X		X		X		X	
AY26							
26-1	25-2						
X							

b. Enrollments.

Academic Term	Enrollment
14-1	49
14-2	99
15-2	107
16-1	47
16-2	56
17-1	40
17-2	95
18-1	42
18-2	36
19-1	92
20-1	84
21-1	64
22-1	73
23-1	99
24-1	90
25-1	87
26-1	90

c. **Historical Final Grade Averages.**

Academic Term	Course Average
14-1	83.04%
14-2	79.60%
15-2	87.12%
16-1	86.81%
16-2	87.05%
17-1	91.59%
17-2	87.66%
18-1	87.22%
18-2	88.21%
19-1	88.21%
20-1	86.21%
21-1	89.0%
22-1	84.79%
23-1	87.41%
24-1	85.6%
25-1	83.0%
26-1	78.2%

d. **Course Composition.** There were 90 Cadets in the course. 84 class of 2026, 6 class of 2025. 47 Engineering Management. 25 Systems & Decision Sciences. 8 Operations Research. 3 Chemical Engineering. 3 Management. 2 Systems Engineering. 1 ASDS. 1 Econ.

e. **Course Strategy.** This course teaches the tools necessary to understand what occurs in production and service operations, i.e., what occurs between the infusion of inputs (e.g., raw materials, management) and the production of outputs (e.g., a product or service). Emphasis is on problem identification, interpretation, and solution from a systems viewpoint using applied quantitative techniques. The course looks at operations management from a strategic level and works its way down to the tactical level. 3 x Group Graded Labs, 3x Individual Quizzes, 2 x Problem Sets, 2 x WPRs, 1 x Field Trip (Replaced by Guest Lecture due to Government Shutdown), 1 x Case Study Prelim Assignment, 1 x Case Study, and the TEE help assess the student's ability to apply course material to real-world problems. The comprehensive TEE evaluates students' mastery of all course objectives.

Section II – Course Assessment

1. Achievement of Course Objectives.

a. **Course Objective Assessment Table.** Enclosure 3.

- b. **Results of Course End Feedback.** Not included.

2. Course Content.

a. **Body of Knowledge.** The course uses the 10th edition of Operations and Supply Chain Management by Roberta S. Russell and Bernard W. Taylor III.

b. **Effective Assessment.** This was primarily orchestrated through the three Group Graded Labs, Two WPR's, a TEE and Case Study. While the case study takes a long time to grade it is incredibly important to assess cadet technical writing.

3. Course Processes.

a. **Textbook.** The course uses the 10th edition of Operations and Supply Chain Management by Roberta S. Russell and Bernard W. Taylor III.

b. **Software.** None.

c. **Graded Requirements.** The 14 Graded Events for the course are seen below.

<u>Graded Events</u>	<u>Points</u>
Group Graded Lab 1	25
Individual Quiz 1	25
Problem Set 1	45
Instructor Points Part 1	30
Midterm I	150
Problem Set 2	45
Individual Quiz 2	25
Group Graded Lab 2	25
Midterm II	150
Case Study Prelim	15
Field Trip Write Up	20
Individual Quiz 3	25
Group Graded Lab 3	25
Case Study	80
TEE	300
Instructor Points Part 2	15

4. Course Pedagogy.

a. **General:** The course was delivered in person with the ability to record lessons via Teams if need be. This happened by exception only, typically when a Cadet would be absent for an excusable reason and requested it. Cadets had assigned readings and pre-lesson videos for nearly every class.

b. **Practical Exercises:** Practical Exercises take up most of the in-class time as pre-lesson videos cover much of the conceptual content.

c. **Problem Sets:** There were two Problem Sets given during the course. These problem sets aim to build student confidence in their abilities to recognize and apply the proper tools for critical course topics.

d. **Case Study:** All Cadets were given the same Case Study this year, written by Dr. Isabella Sanders. They were given a series of questions to choose from which required critical thinking and data analysis.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. Assessment of Changes to Current Academic Term (based on previous Course Assessments).

- a. **Course Objectives.** None.
- b. **Textbook.** None.
- c. **Curriculum.**
 - i. Had to remove field trips due to government shutdown. Replaced with guest lecture.
 - ii. Incorporated guest lecture into the case study.
 - iii. Broke apart GGL into an individual quiz and a GGL to have individual accountability.
- d. **Graded Requirements.**
 - i. Added individual quizzes.
 - ii. Lowered percentage of group assignments.
- e. **Other.** None.

2. Proposed Changes for Next Academic Term.

- a. **Course Objectives.** None.
- b. **Textbook.** None
- c. **Curriculum.**
 - i. As ChatGPT becomes more prevalent, incorporate more oral presentations.
- d. **Graded Requirements.**
 - i. Potentially increase the percentage value of the Case Study.
 - ii. Add a presentation for Case Study to prevent generation of results by ChatGPT.
- e. **Homework Problems.** None.
- f. **Other.**
 - i. Maintain putting C-, D, and F students in the same groups together if demographics allow. Other students can be mixed.
 - ii. DSE Majors should take this course as Firsties only.



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

17 August 2025

INSTRUCTIONAL MEMORANDUM
EM420-26-1

PRODUCTION OPERATIONS MANAGEMENT

1. **PURPOSE.** The purpose of this memorandum is to provide each cadet with the administrative details necessary to successfully complete EM420, Production Operations Management.

2. **REFERENCES.**

a. Department of Systems Engineering Administrative Memo 1-1, "General Administrative Instructions," 10 August 2023.

b. Department of Systems Engineering Administrative Memo 1-2, "Standards for Written Requirements," 10 August 2023.

c. Department of Systems Engineering Administrative Memo 1-3, "Academic Document Policy," 10 August 2023.

d. USMA publication. "Documentation of Academic Work," Office of the Dean, 25 July 2017.

e. The Little Brown Handbook, by H. Ramsey Fowler and Jane Aaron, Addison-Wesley, New York, 1998.

3. **GENERAL.**

a. This memorandum provides each cadet with the course description, course objectives, course texts, lesson assignments, grading plan, online assignment guidance, and criteria for passing the course.

b. Your instructor will inform you in advance of any deviation from the guidance provided in this memorandum.

4. SPECIFIC.

a. **Course Scope.** This course deals with the quantitative aspects of design and analysis of production operations management. Emphasis is on identification, analysis, and solution of production problems using applied quantitative techniques. Practical exercises reinforce the practical problem-solving techniques necessary for today's successful military and civilian engineering managers and systems engineers. Specific methods and techniques taught and applied are strategy and product design, process and capacity strategy, total quality management, statistical control process, location strategy, job design and work measurement, layout strategy, scheduling, maintenance and reliability, and service design and queuing. This course is required for those pursuing the Engineering Management major and an elective for the Systems Engineering and Systems and Decision Science majors. Applied technical skills are developed using Microsoft Excel®. Much of the course material directly relates to content found in the Fundamentals of Engineering Exam (FE).

b. **Course Strategy.** This course teaches the tools necessary to understand what occurs in production and service operations, i.e., what occurs between the infusion of inputs (e.g., raw materials, management) and the producing of outputs (e.g., a product or service). Emphasis is on problem identification, interpretation, and solution from a systems viewpoint using applied quantitative techniques. The course starts out looking at operations management from a strategic level and works its way down to the tactical level. The comprehensive TEE evaluates students' mastery of all course objectives. A detailed lesson syllabus is included in Enclosure 1.

c. **Course Objectives.**

1) Analyze and make operations management recommendations to a client with appropriate technology and engineering management tools.

2) Given a brief scenario description, describe the operations management mission, strategy, and competitive advantage of the business.

3) Given a brief scenario description, identify and assess statistical quality control problems and apply the appropriate tools to solve them.

4) Given a brief scenario description, identify and apply the appropriate techniques and tools to complete good product and service design.

5) Given a brief scenario description, identify and assess facility location, layout and capacity issues and apply the appropriate tools to solve them.

6) Given a brief scenario description, identify and assess job design and work measurement problems and apply the appropriate tools to solve them.

7) Given a brief scenario description, identify and assess personnel, materials, and job scheduling issues and apply the appropriate tools to solve those issues.

8) Given a brief scenario description, identify and assess queuing type problems and apply the appropriate tools to solve those issues.

9) Given a brief scenario description identify and assess quality management of production of goods and services.

10) Communicate in oral or written reports the analysis and application of course concepts to real-world scenarios.

d. Texts. Operations and Supply Chain Management, by Roberta S. Russell and Bernard W. Taylor III, Wiley & Sons Inc., 2017. 10th Edition with the online assessment tool WileyPlus.

e. Course Videos. There are a series of Pre-Lesson Videos that are to be watched prior to certain lectures. They provide fundamental material to allow for more time in class for exercises and demonstrations.

f. Additional References.

1) Parnell, G. S., Driscoll, P. J., and Henderson D. L., Editors, *Decision Making for Systems Engineering and Management*, Wiley & Sons Inc., 2011 2nd Edition.

2) FE Reference Handbook.

g. Lesson Assignments. See Enclosure 1 (Syllabus) for specific readings and assignments.

h. Performance Assessment.

1) The course consists of 1000 points.

Graded Events	Points	
Group Graded Lab 1	25	
Individual Quiz 1	25	
Problem Set 1	45	← 6 week Grades
Instructor Points Part 1	30	
Midterm I	150	← 10 week Grades
Problem Set 2	45	
Individual Quiz 2	25	
Group Graded Lab 2	25	
Midterm II	150	
Case Study Prelim	15	← 15 week Grades
Field Trip Write Up	20	
Individual Quiz 3	25	
Group Graded Lab 3	25	
Case Study	80	
TEE	300	
Instructor Points Part 2	15	

a) The three Group Graded Labs will be executed in groups of 2-3 for a total of 75 points.

b) The three individual quizzes will precede GGL's for a total of 75 points.

c) Two individual Problem Sets account for 90 points total.

d) Two Midterms (WPR's) account for 300 Points total.

e) The Field Trip Write Up will serve as a primary source for the case study 20 points.

f) Case Study prelim will allow cadets to complete 1/6 questions for the case study for early feedback to aid in completion of full case study to standard 15 Points.

g) One Individual Case Study will allow cadets to read a real-life scenario and apply concepts learned to the case. 80 points.

h) Instructor Points. 45 points total – course duration. These points come from lesson preparation in the form of watching pre-lesson videos, associated pre-lesson questions, or class preparedness. There are no make-ups or late turn-ins as preparation needs to be done prior to class for learning benefits.

i) Term End Examination. 300 points. The purpose of this exercise is to reinforce those concepts taught during the entire course in a summative assessment.

5. ADMINISTRATIVE GUIDANCE.

a. Course Structure. This is a 30-lesson course taught in 75-minute sections.

b. Minimum Requirements. Score at least a 50% on the TEE and complete the course with at least 65% of the total course points.

c. Graded Events. Unless there are exceptional circumstances, cadets will complete all graded assignments and attend all briefings and classes. Cadets who will miss an event must contact the instructor as soon as they know about the conflict. Please reference the Course Assessment Policy for prescheduled make ahead's and make ups. It is the cadet's responsibility to coordinate with the instructor to make up any material/requirements missed.

d. Re-grade Requests. If there was an arithmetic error in adding up points on your graded event, inform your instructor immediately, and the instructor will correct the grade. However, the entire assignment may be regraded and result in a lower score. Any other requests to re-grade an assignment or a portion of an assignment for any reason can also be addressed directly with instructor. Instructors will handle the request and any re-grading at their discretion.

e. Late Submissions. Failure to **attend or submit** a graded event on time for other than legitimate, compelling reasons will result in a reduced grade (**5% per day**) and may result in disciplinary action. Each assignment has a late turn-in cut off where a 0 will be awarded if work is turned in past that date. This allows for graded work to be returned to all students in a timely manner. **All graded events must be submitted to pass the course by department policy.**

f. Final Grades. Final grades will be awarded IAW with reference 2.a (Department of Systems Engineering Admin Memo 1-1).

g. Documentation. Cadets may use any available reference material in preparing course work, but use of any references **MUST BE PROPERLY DOCUMENTED**. Documentation procedures are outlined in references 2a through 2e. Pay attention to documentation allowances for each assignment.

h. Additional Instruction (AI). AI is available at mutually agreed upon times between the cadet and the instructor. Cadets are expected to have specific questions for any AI session. No AI can be given after the first version of a WPR has been given, whether that version is the primary or a make-ahead exam. No AI past the assignment due date.

i. Class. You are expected to bring your course reference manual, calculator, and laptop (as necessary) and to each class. If you are not going to be at a class, please let your instructor know *prior to the class* via email.

FOR THE HEAD OF THE DEPARTMENT:

Enclosures:

Syllabus w/ Lesson Objectives
Reference Manual

BRANDON THOMPSON
COL, FA47
Engineering Management Program Director

Enclosure 3. Final Grades

- 1. Overall Performance.** Course averages were even lower than last year. We have seen a significant uptick in cadets taking EM classes with a 2.1 GPA or lower. There were a 7 D's this semester.



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB B

AY2025 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

December 20, 2024

MEMORANDUM FOR COL Brandon Thompson, Engineering Management Program Director,
Department of Systems Engineering, West Point, NY 109963

SUBJECT: AT 25-1 Course Assessment for EM420: Production Operations Management

1. The attached report summarizes the EM420 assessment and curriculum change actions for Academic Term (AT) 25-1.
2. The POC for this action is the undersigned at isabella.sanders@westpoint.edu.

ISABELLA T. SANDERS
PhD, Assistant Professor
EM420 Course Director

Encls

1. Course Assessment
2. Course Instructional Memorandum, Graded Events, and Syllabus
3. OML and Final Grades

DISTRIBUTION:

1 each EM420 instructors, Program Director, and Department Head

Enclosure 1: Course Assessment

Section I – Course Information and History

1. Historic Course Offerings and Enrollments.

a. Offered.

AY14		AY15		AY16		AY17	
14-1	14-2	15-1	15-2	16-1	16-2	17-1	17-2
X	X		X	X	X	X	X
AY18		AY19		AY20		AY21	
18-1	18-2	19-1	19-2	20-1	20-2	21-1	21-2
X	X	X		X		X	
AY22		AY23		AY24		AY25	
22-1	22-2	23-1	23-2	24-1	24-2	25-1	25-2
X		X		X		X	

b. Enrollments.

Academic Term	Enrollment
14-1	49
14-2	99
15-2	107
16-1	47
16-2	56
17-1	40
17-2	95
18-1	42
18-2	36
19-1	92
20-1	84
21-1	64
22-1	73
23-1	99
24-1	90
25-1	87

c. **Historical Final Grade Averages.**

Academic Term	Course Average
14-1	83.04%
14-2	79.60%
15-2	87.12%
16-1	86.81%
16-2	87.05%
17-1	91.59%
17-2	87.66%
18-1	87.22%
18-2	88.21%
19-1	88.21%
20-1	86.21%
21-1	89.0%
22-1	84.79%
23-1	87.41%
24-1	85.6%
25-1	83.0%

d. **Course Composition.** There were 87 Cadets in the course. 86 class of 2025, 1 class of 2026. 47 Engineering Management. 22 Systems & Decision Sciences. 6 Chemical Engineering. 4 Management. 2 Operations Research. 1 Systems Engineering. 1 CSC. 1 Cyber. 1 Law. 1 Foreign Cadet.

e. **Course Strategy.** This course teaches the tools necessary to understand what occurs in production and service operations, i.e., what occurs between the infusion of inputs (e.g., raw materials, management) and the production of outputs (e.g., a product or service). Emphasis is on problem identification, interpretation, and solution from a systems viewpoint using applied quantitative techniques. The course looks at operations management from a strategic level and works its way down to the tactical level. 3 x Group Graded Labs, 2 x Problem Sets, 2 x WPRs, 1 x Field Trip, 1 x Case Study Prelim Assignment, 1 x Case Study, and the TEE help assess the student's ability to apply course material to real-world problems. The comprehensive TEE evaluates students' mastery of all course objectives.

Section II – Course Assessment

1. Achievement of Course Objectives.

- a. **Course Objective Assessment Table.** Enclosure 3.
- b. **Results of Course End Feedback.** Not included.

2. Course Content.

a. **Body of Knowledge.** The course uses the 10th edition of Operations and Supply Chain Management by Roberta S. Russell and Bernard W. Taylor III.

b. **Effective Assessment.** This was primarily orchestrated through the three Group Graded Labs, Two WPR's, a TEE and Case Study. While the case study takes a long time to grade it is incredibly important to assess cadet technical writing.

3. Course Processes.

a. **Textbook.** The course uses the 10th edition of Operations and Supply Chain Management by Roberta S. Russell and Bernard W. Taylor III.

b. **Software.** None.

c. **Graded Requirements.** The 12 Graded Events for the course are seen below.

<u>Graded Events</u>	<u>Points</u>
Group Graded Lab 1	40
Problem Set 1	45
WPR I	150
Problem Set 2	45
Group Graded Lab 2	40
WPR II	150
Field Trip Write Up	30
Case Study Prelim (Individual)	30
Group Graded Lab 3	40
Case Study Write Up (Individual)	90
TEE	300
Quizzes (Throughout the semester)	40

4. Course Pedagogy.

a. **General:** The course was delivered in person with the ability to record lessons via Teams if need be. This happened by exception only, typically when a Cadet would be absent for an excusable reason and requested it. Cadets had assigned readings and pre-lesson videos for nearly every class.

b. **Practical Exercises:** Practical Exercises take up most of the in-class time as pre-lesson videos cover much of the conceptual content.

c. **Problem Sets:** There were two Problem Sets given during the course. These problem sets aim to build student confidence in their abilities to recognize and apply the proper tools for critical course topics.

d. **Case Study:** All Cadets were given the same Case Study this year, written by Dr. Isabella Sanders. They were given a series of questions to choose from which required critical thinking and data analysis.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. Assessment of Changes to Current Academic Term (based on previous Course Assessments).

- a. **Course Objectives.** None.
- b. **Textbook.** None.
- c. **Curriculum.**
 - i. Added 5 mandatory field trips from production to recycling in NY and MA and a corresponding assignment.
 - ii. Incorporated field trip into the case study.
 - iii. Expanded the case study.
 - iv. Added daily quizzes to assess pre-lesson video content.
- d. **Graded Requirements.**
 - i. Added case study prelim assignment.
 - ii. Lowered percentage of group assignments.
 - iii. Added Field Trip Assignment
 - iv. Added daily quizzes.
 - v. Reverted to traditional TEE upon DEAN guidance. Notably no difference in performance.
- e. **Other.** None.

2. Proposed Changes for Next Academic Term.

- a. **Course Objectives.** None.
- b. **Textbook.** None
- c. **Curriculum.**
 - i. Include more scenario-based questions. Adding them to WPR 2 forced cadets to better understand the material, and weaker cadets performed stronger on the TEE
- d. **Graded Requirements.**
 - i. Potentially increase the percentage value of the Case Study.
- e. **Homework Problems.** None.
- f. **Other.**
 - i. Since our major now has many more people who tend to be failing classes in the middle of the semester, we should plan for a non-travel field trip (this used to be unnecessary)
 - ii. Maintain putting C-, D, and F students in the same groups together if demographics allow. Other students can be mixed.
 - iii. DSE Majors should take this course as Firsties only.

- iv. Something needs to be done about attitude and respect with cadets. There has been a notable drop-off in the last year. I don't know if this should be incorporated into IP or formally in NCORs.



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

19 August 2024

INSTRUCTIONAL MEMORANDUM
EM420-25-1

PRODUCTION OPERATIONS MANAGEMENT

1. **PURPOSE.** The purpose of this memorandum is to provide each cadet with the administrative details necessary to successfully complete EM420, Production Operations Management.

2. **REFERENCES.**

a. Department of Systems Engineering Administrative Memo 1-1, "General Administrative Instructions," 10 August 2023.

b. Department of Systems Engineering Administrative Memo 1-2, "Standards for Written Requirements," 10 August 2023.

c. Department of Systems Engineering Administrative Memo 1-3, "Academic Document Policy," 10 August 2023.

d. USMA publication. "Documentation of Academic Work," Office of the Dean, 25 July 2017.

e. The Little Brown Handbook, by H. Ramsey Fowler and Jane Aaron, Addison-Wesley, New York, 1998.

3. **GENERAL.**

a. This memorandum provides each cadet with the course description, course objectives, course texts, lesson assignments, grading plan, online assignment guidance, and criteria for passing the course.

b. Your instructor will inform you in advance of any deviation from the guidance provided in this memorandum.

4. SPECIFIC.

a. **Course Scope.** This course deals with the quantitative aspects of design and analysis of production operations management. Emphasis is on identification, analysis, and solution of production problems using applied quantitative techniques. Practical exercises reinforce the practical problem-solving techniques necessary for today's successful military and civilian engineering managers and systems engineers. Specific methods and techniques taught and applied are strategy and product design, process and capacity strategy, total quality management, statistical control process, location strategy, job design and work measurement, layout strategy, scheduling, maintenance and reliability, and service design and queuing. This course is required for those pursuing the Engineering Management major and an elective for the Systems Engineering and Systems and Decision Science majors. Applied technical skills are developed using Microsoft Excel®. Much of the course material directly relates to content found in the Fundamentals of Engineering Exam (FE).

b. **Course Strategy.** This course teaches the tools necessary to understand what occurs in production and service operations, i.e., what occurs between the infusion of inputs (e.g., raw materials, management) and the producing of outputs (e.g., a product or service). Emphasis is on problem identification, interpretation, and solution from a systems viewpoint using applied quantitative techniques. The course starts out looking at operations management from a strategic level and works its way down to the tactical level. The comprehensive TEE evaluates students' mastery of all course objectives. A detailed lesson syllabus is included in Enclosure 1.

c. **Course Objectives.**

1) Analyze and make operations management recommendations to a client with appropriate technology and engineering management tools.

2) Given a brief scenario description, describe the operations management mission, strategy, and competitive advantage of the business.

3) Given a brief scenario description, identify and assess statistical quality control problems and apply the appropriate tools to solve them.

4) Given a brief scenario description, identify and apply the appropriate techniques and tools to complete good product and service design.

5) Given a brief scenario description, identify and assess facility location, layout and capacity issues and apply the appropriate tools to solve them.

6) Given a brief scenario description, identify and assess job design and work measurement problems and apply the appropriate tools to solve them.

7) Given a brief scenario description, identify and assess personnel, materials, and job scheduling issues and apply the appropriate tools to solve those issues.

8) Given a brief scenario description, identify and assess queuing type problems and apply the appropriate tools to solve those issues.

9) Given a brief scenario description identify and assess quality management of production of goods and services.

10) Communicate in oral or written reports the analysis and application of course concepts to real-world scenarios.

d. Texts. Operations and Supply Chain Management, by Roberta S. Russell and Bernard W. Taylor III, Wiley & Sons Inc., 2017. 10th Edition with the online assessment tool WileyPlus.

e. Course Videos. There are a series of Pre-Lesson Videos that are to be watched prior to certain lectures. They provide fundamental material to allow for more time in class for exercises and demonstrations.

f. Additional References.

1) Parnell, G. S., Driscoll, P. J., and Henderson D. L., Editors, *Decision Making for Systems Engineering and Management*, Wiley & Sons Inc., 2011 2nd Edition.

2) FE Reference Handbook.

g. Lesson Assignments. See Enclosure 1 (Syllabus) for specific readings and assignments.

h. Performance Assessment.

1) The course consists of 1000 points.

Graded Events	Points	
Group Graded Lab 1	40	
Problem Set 1	45	
WPR I	150	6 week Grades
Problem Set 2	45	10 week Grades
Group Graded Lab 2	40	
WPR II	150	
Field Trip Write Up	30	
Case Study Prelim (Individual)	30	15 week Grades
Group Graded Lab 3	40	
Case Study Write Up (Individual)	90	
TEE	300	
Quizzes (Throughout the semester)	40	

a) The three Group Graded Labs will be executed in Groups for a total of (120 points).

b) Two individual Problem Sets account for (90 points).

c) Two WPR's account for (300 Points).

d) One Field Trip Write Up will serve as a primary source for the individual case study (30 points)

e) One Individual Case Study will allow cadets to read a real-life scenario and apply concepts learned to the case (120 points).

f) Quizzes. (40 points total – course duration). 22 quizzes will be given throughout the semester. The lowest 4 will be dropped from your grade, with each quiz being worth 2 points. There are no make ups. If you are not in class for an excused absence it counts as one of the drops. If you have greater than 4 excused absences for extenuating circumstances, you will have the opportunity to do a make-up extended quiz at the end of the semester. The four remaining points will be left for instructor points.

g) Term End Examination. (300 points). The purpose of this exercise is to reinforce those concepts taught during the entire course.

5. ADMINISTRATIVE GUIDANCE.

- a. Course Structure. This is a 30-lesson course taught in 75-minute sections.
- b. Minimum Requirements. Score at least a 50% on the TEE and complete the course with at least 65% of the total course points.
- c. Graded Events. Unless there are exceptional circumstances, cadets will complete all graded assignments and attend all briefings and classes. Cadets who will miss an event must contact the instructor as soon as they know about the conflict. It is the cadet's responsibility to coordinate with the instructor to make up any material/requirements missed.
- d. Re-grade Requests. If there was an arithmetic error in adding up points on your graded event, inform your instructor immediately, and the instructor will correct the grade. However, the entire assignment may be regraded and result in a lower score. Any other requests to re-grade an assignment or a portion of an assignment for any reason can also be addressed directly with instructor. Instructors will handle the request and any re-grading at their discretion.
- e. Late Submissions. Failure to ***attend or submit*** a graded event on time for other than legitimate, compelling reasons will result in a reduced grade (***5% per day***) and may result in disciplinary action. All graded events must be submitted to pass the course by department policy.
- f. Final Grades. Final grades will be awarded IAW with reference 2.a (Department of Systems Engineering Admin Memo 1-1).
- g. Documentation. Cadets may use any available reference material in preparing course work, but use of any references **MUST BE PROPERLY DOCUMENTED**. Documentation procedures are outlined in references 2a through 2e.
- h. Additional Instruction (AI). AI is available at mutually agreed upon times between the cadet and the instructor. Cadets are expected to have specific questions for any AI session. No AI can be given after the first version of a WPR has been given, whether that version is the primary or a make-ahead exam. Be prepared to show your practice problems at any AI session.
- i. Class. You are expected to bring your course reference manual, calculator, and laptop (as necessary) and to each class. If you are not going to be at a class, please let your instructor know *prior to the class* in person or via email.

FOR THE HEAD OF THE DEPARTMENT:

Enclosures:

Syllabus w/ Lesson Objectives
Reference Manual

BRANDON THOMPSON
COL, FA47
Engineering Management Program Director

Enclosure 3. Final Grades

- 1. Overall Performance.** Course averages were slightly lower than last year. We have seen a significant uptick in cadets taking EM classes with a 2.1 GPA or lower. There were no course final grades lower than a D.



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB C

AY2024 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

December 20, 2023

MEMORANDUM FOR COL James Schreiner, Engineering Management Program Director,
Department of Systems Engineering, West Point, NY 109963

SUBJECT: AT 24-1 Course Assessment for EM420: Production Operations Management

1. The attached report summarizes the EM420 assessment and curriculum change actions for Academic Term (AT) 24-1.
2. The POC for this action is the undersigned at isabella.sanders@westpoint.edu.

ISABELLA T. SANDERS
PhD, Assistant Professor
EM420 Course Director

Encls

1. Course Assessment
2. Course Instructional Memorandum, Graded Events, and Syllabus
3. OML and Final Grades

DISTRIBUTION:

1 each EM420 instructors, Program Director, and Department Head

Enclosure 1: Course Assessment

Section I – Course Information and History

1. Historic Course Offerings and Enrollments.

a. Offered.

AY14		AY15		AY16		AY17	
14-1	14-2	15-1	15-2	16-1	16-2	17-1	17-2
X	X		X	X	X	X	X
AY18		AY19		AY20		AY21	
18-1	18-2	19-1	19-2	20-1	20-2	21-1	21-2
X	X	X		X		X	
AY22		AY23					
22-1	22-2	23-1	23-2				
X		X					

AY24 – Offered 24-1

b. Enrollments.

Academic Term	Enrollment
14-1	49
14-2	99
15-2	107
16-1	47
16-2	56
17-1	40
17-2	95
18-1	42
18-2	36
19-1	92
20-1	84
21-1	64
22-1	73
23-1	99
24-1	90

c. **Historical Final Grade Averages.**

Academic Term	Course Average
14-1	83.04%
14-2	79.60%
15-2	87.12%
16-1	86.81%
16-2	87.05%
17-1	91.59%
17-2	87.66%
18-1	87.22%
18-2	88.21%
19-1	88.21%
20-1	86.21%
21-1	89.0%
22-1	84.79%
23-1	87.41%
24-1	85.6%

d. **Course Composition.** There were 89 students in the course. 78 were 2024. 9 were 2025. 2 were 2023. 50 were Engineering Management. 23 Systems & Decision Sciences. 4 Systems Engineering. 4 Economics. 3 Management. 1 Nuclear Engineering. 2 Operations Research. 1 Sociology. 1 Civil Engineering.

Course Strategy. This course teaches the tools necessary to understand what occurs in production and service operations, i.e., what occurs between the infusion of inputs (e.g., raw materials, management) and the producing of outputs (e.g., a product or service). Emphasis is on problem identification, interpretation, and solution from a systems viewpoint using applied quantitative techniques. The course starts out looking at operations management from a strategic level and works its way down to the tactical level. 3 x Group Graded Labs, 2 x Problem Sets, 1 x Mid-Term WPR, 1 x Guest Lecture Reflection, 1 x Case Study, and the TEE help assess the students' ability to apply course material to real world problems. The comprehensive TEE evaluates students' mastery of all course objectives.

Section II – Course Assessment

1. Achievement of Course Objectives.

a. **Course Objective Assessment Table.** Enclosure 3.

- b. **Results of Course End Feedback.** Not included.

2. Course Content.

a. **Body of Knowledge.** The course uses the 10th edition of Operations and Supply Chain Management by Roberta S. Russell and Bernard W. Taylor III.

b. **Effective Assessment.** This was primarily orchestrated through the three Group Graded Labs, Two WPR's, a TEE and Case Study. While the case study takes a long time to grade it is incredibly important to assess cadet technical writing.

3. Course Processes.

a. **Textbook.** The course uses the 10th edition of Operations and Supply Chain Management by Roberta S. Russell and Bernard W. Taylor III.

b. **Software.** None.

c. **Graded Requirements.** The 11 Graded Events for the course are seen below.

Graded Events	Points
Group Graded Lab 1	50
Group Graded Lab 2	50
Problem Set 1	75
WPR I	150
WPR II	150
Group Graded Lab 3	50
Problem Set 2	75
Case Study Write Up (Individual)	75
TEE	300
Instructor Points	25

4. Course Pedagogy.

a. **General:** The course was delivered in person with the ability to record lessons via Teams if need be. This happened by exception only, typically when a Cadet would be absent for an excusable reason and requested it. Cadets had assigned readings for nearly every class.

b. **Practical Exercises:** The majority of lecture based lessons ended with a PE for Cadets to complete and discuss either at the end of the class or the beginning of the next.

c. **Problem Sets:** There were two Problem Sets given during the course. The purpose of these problem sets is to build student confidence in their own abilities to recognize and apply the proper tools for critical course topics.

d. **Case Study:** All Cadets were given the same Case Study this year, written by Dr. Isabella Sanders. They were all asked to answer the same questions and present a new proposal to those questions to the class.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. Assessment of Changes to Current Academic Term (based on previous Course Assessments).

- a. **Course Objectives.** None.
- b. **Textbook.** None.
- c. **Curriculum.**
 - i. Reduced Statistical Process Control to 2 lessons (from 3) because these students have seen it in SE375
 - ii. Adjusted Learning Curves, Work Measurement, Lean, Work Sampling, Human Resources, Quality Management, Human Resources Management to include new examples
 - iii. Added a Guest Lecture
 - iv. Added an optional Field Trip
- d. **Graded Requirements.**
 - i. Added one WPR
 - ii. Made Case Study purely individual
 - iii. Made TEE split into two sections over 2 days.
- e. **Other.** None.

2. Proposed Changes for Next Academic Term.

- a. **Course Objectives.** None.
- b. **Textbook.** None
- c. **Curriculum.** None
- d. **Graded Requirements.**
 - i. The Case Study, while a large grading commitment is incredibly important for cadet writing assessment.
- e. **Homework Problems.** None.
- f. **Other.** None.



Enclosure 2: Instructional Memorandum, Graded Events



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

MADN-SE

11 August 2023

INSTRUCTIONAL MEMORANDUM
EM420-24-1

PRODUCTION OPERATIONS MANAGEMENT

1. **PURPOSE.** The purpose of this memorandum is to provide each cadet with the administrative details necessary to successfully complete EM420, Production Operations Management.
2. **REFERENCES.**
 - a. Department of Systems Engineering Administrative Memo 1-1, "General Administrative Instructions," 10 August 2023.
 - b. Department of Systems Engineering Administrative Memo 1-2, "Standards for Written Requirements," 10 August 2023.
 - c. Department of Systems Engineering Administrative Memo 1-3, "Academic Document Policy," 10 August 2023.
 - d. USMA publication. "Documentation of Academic Work," Office of the Dean, 25 July 2017.
 - e. The Little Brown Handbook, by H. Ramsey Fowler and Jane Aaron, Addison-Wesley, New York, 1998.
3. **GENERAL.**
 - a. This memorandum provides each cadet with the course description, course objectives, course texts, lesson assignments, grading plan, online assignment guidance, and criteria for passing the course.
 - b. Your instructor will inform you in advance of any deviation from the guidance provided in this memorandum.

4. SPECIFIC.

a. **Course Scope.** This course deals with the quantitative aspects of design and analysis of production operations management. Emphasis is on identification, analysis, and solution of production problems using applied quantitative techniques. Practical exercises reinforce the practical problem-solving techniques necessary for today's successful military and civilian engineering managers and systems engineers. Specific methods and techniques taught and applied are strategy and product design, process and capacity strategy, total quality management, statistical control process, location strategy, job design and work measurement, layout strategy, scheduling, maintenance and reliability, and service design and queuing. This course is required for those pursuing the Engineering Management major and an elective for the Systems Engineering and Systems and Decision Science majors. Applied technical skills are developed using Microsoft Excel®. Much of the course material directly relates to content found in the Fundamentals of Engineering Exam (FE).

b. **Course Strategy.** This course teaches the tools necessary to understand what occurs in production and service operations, i.e., what occurs between the infusion of inputs (e.g., raw materials, management) and the producing of outputs (e.g., a product or service). Emphasis is on problem identification, interpretation, and solution from a systems viewpoint using applied quantitative techniques. The course starts out looking at operations management from a strategic level and works its way down to the tactical level. The comprehensive TEE evaluates students' mastery of all course objectives. A detailed lesson syllabus is included in Enclosure 1.

c. **Course Objectives.**

1) Analyze and make operations management recommendations to a client with appropriate technology and engineering management tools.

2) Given a brief scenario description, describe the operations management mission, strategy, and competitive advantage of the business.

3) Given a brief scenario description, identify and assess statistical quality control problems and apply the appropriate tools to solve them.

4) Given a brief scenario description, identify and apply the appropriate techniques and tools to complete good product and service design.

5) Given a brief scenario description, identify and assess facility location, layout and capacity issues and apply the appropriate tools to solve them.

6) Given a brief scenario description, identify and assess job design and work measurement problems and apply the appropriate tools to solve them.

7) Given a brief scenario description, identify and assess personnel, materials, and job scheduling issues and apply the appropriate tools to solve those issues.

8) Given a brief scenario description, identify and assess queuing type problems and apply the appropriate tools to solve those issues.

9) Given a brief scenario description identify and assess quality management of production of goods and services.

10) Communicate in oral or written reports the analysis and application of course concepts to real-world scenarios.

d. Texts. Operations and Supply Chain Management, by Roberta S. Russell and Bernard W. Taylor III, Wiley & Sons Inc., 2017. 10th Edition with the online assessment tool WileyPlus.

e. Course Videos. There are a series of 12 Pre-Lesson Videos that are to be watched prior to certain lectures. They provide fundamental material to allow for more time in class for exercises and demonstrations.

f. Additional References.

1) Parnell, G. S., Driscoll, P. J., and Henderson D. L., Editors, *Decision Making for Systems Engineering and Management*, Wiley & Sons Inc., 2011 2nd Edition.

2) FE Reference Handbook.

g. Lesson Assignments. See Enclosure 1 (Syllabus) for specific readings and assignments.

h. Performance Assessment.

1) The course consists of 1000 points.

Graded Events	Points
Group Graded Lab 1	50
Group Graded Lab 2	50
Problem Set 1	75
WPR I	150
WPR II	150
Group Graded Lab 3	50
Problem Set 2	75
Case Study Write Up (Individual)	75
TEE	300
Instructor Points	25

a) The three Group Graded Labs will be executed in Groups for a total of 150 points because of group work.

b) Two individual Problem Sets account for 150 points.

c) Two WPR's account for (300 Points).

d) One Individual Case Study will allow cadets to read a real-life scenario and apply concepts learned to the case (75 points).

e) Classroom Participation and Preparation. (25 points total – course duration). Subjective measure of student's classroom preparation, participation, and overall academic professionalism.

f) Term End Examination. (300 points). The purpose of this exercise is to reinforce those concepts taught during the entire course.

5. ADMINISTRATIVE GUIDANCE.

a. Course Structure. This is a 30-lesson course taught in 75-minute sections.

b. Minimum Requirements. Score at least a 50% on the TEE and complete the course with at least 65% of the total course points.

c. Graded Events. Unless there are exceptional circumstances, cadets will complete all graded assignments and attend all briefings and classes. Cadets who will miss an event must contact the instructor as soon as they know about the conflict. It is the cadet's responsibility to coordinate with the instructor to make up any material/requirements missed.

d. Re-grade Requests. If there was an arithmetic error in adding up points on your graded event, inform your instructor immediately, and the instructor will correct the grade. However, the entire assignment may be regraded and result in a lower score. Any other requests to re-grade an assignment or a portion of an assignment for any reason can also be addressed directly with instructor. Instructors will handle the request and any re-grading at their discretion.

e. Late Submissions. Failure to ***attend or submit*** a graded event on time for other than legitimate, compelling reasons will result in a reduced grade (***5% per day***) and may result in disciplinary action. All graded events must be submitted in order to pass the course by department policy.

f. Final Grades. Final grades will be awarded IAW with reference 2.a (Department of Systems Engineering Admin Memo 1-1).

g. Documentation. Cadets may use any available reference material in preparing course work, but use of any references **MUST BE PROPERLY DOCUMENTED**. Documentation procedures are outlined in references 2a through 2e.

h. Additional Instruction (AI). AI is available at mutually agreed upon times between the cadet and the instructor. Cadets are expected to have specific questions for any AI session. No AI can be given after the first version of a WPR has been given, whether that version is the primary or a make-ahead exam. Be prepared to show your practice problems at any AI session.

i. Class. You are expected to bring your paper textbook, course reference manual, calculator, and laptop (as necessary) and to each class. If you are not going to be at a class, please let your instructor know *prior to the class* in person or via email.

FOR THE HEAD OF THE DEPARTMENT:

Enclosures:

Syllabus w/ Lesson Objectives
Reference Manual

JAMES H. SCHREINER
COL, FA47
Engineering Management Program Director

Enclosure 3. Final Grades

- 1. Overall Performance.** Course averages were slightly lower than last year. This tracks as the Spanish Cadets last year inflated the average, since they had all taken the class in Spain. The mean was 85.6%. There were no course final grades lower than a D.



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB D

AY2023 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

December 12 2023

MEMORANDUM FOR COL James Schreiner, Engineering Management Program Director,
Department of Systems Engineering, West Point, NY 10996

SUBJECT: AT 23-1 Course Assessment for EM420: Production Operations Management

1. The attached report summarizes the EM420 assessment and curriculum change actions for Academic Term (AT) 23-1.
2. The POC for this action is the undersigned at james.macgibbon@westpoint.edu or x8589.

JAMES E. MACGIBBON
CPT, EN
EM420 Course Director

Encls

1. Course Assessment
2. Course Instructional Memorandum, Graded Events, and Syllabus
3. OML and Final Grades

DISTRIBUTION:

1 each EM420 instructors, Program Director, and Department Head

Enclosure 1: Course Assessment

Section I – Course Information and History

1. Historic Course Offerings and Enrollments.

a. Offered.

AY14		AY15		AY16		AY17	
14-1	14-2	15-1	15-2	16-1	16-2	17-1	17-2
X	X		X	X	X	X	X
AY18		AY19		AY20		AY21	
18-1	18-2	19-1	19-2	20-1	20-2	21-1	21-2
X	X	X		X		X	
AY22		AY23					
22-1	22-2	23-1	23-2				
X		X					

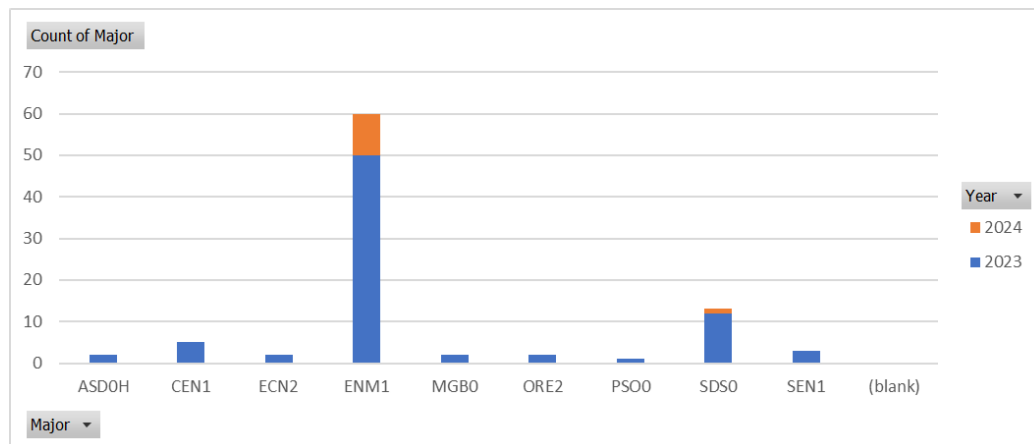
b. Enrollments.

Academic Term	Enrollment
14-1	49
14-2	99
15-2	107
16-1	47
16-2	56
17-1	40
17-2	95
18-1	42
18-2	36
19-1	92
20-1	84
21-1	64
22-1	73
23-1	99

c. **Historical Final Grade Averages.**

Academic Term	Course Average
14-1	83.04%
14-2	79.60%
15-2	87.12%
16-1	86.81%
16-2	87.05%
17-1	91.59%
17-2	87.66%
18-1	87.22%
18-2	88.21%
19-1	88.21%
20-1	86.21%
21-1	89.0%
22-1	84.79%
23-1	87.41%

d. **Course Composition.** There were 99 students in the Course. 9 were Spanish 2nd Lieutenants, 79 Firsties, and 11 Cows. 60 students were Engineering Management majors and 13 Systems Decision Sciences. Systems students, primarily EM Firsties, made up the majority of the class.



Course Strategy. This course teaches the tools necessary to understand what occurs in production and service operations, i.e., what occurs between the infusion of inputs (e.g., raw materials, management) and the producing of outputs (e.g., a product or service). Emphasis is on problem identification, interpretation, and solution from a systems

viewpoint using applied quantitative techniques. The course starts out looking at operations management from a strategic level and works its way down to the tactical level. 3 x Group Graded Labs, 2 x Problem Sets, 1 x Mid-Term WPR, 1 x Guest Lecture Reflection, 1 x Case Study, and the TEE help assess the students' ability to apply course material to real world problems. The comprehensive TEE evaluates students' mastery of all course objectives.

Section II – Course Assessment

1. Achievement of Course Objectives.

- a. **Course Objective Assessment Table.** Enclosure 3.
- b. **Results of Course End Feedback.** Not included.

2. Course Content.

- a. **Body of Knowledge.** The course uses the 10th edition of Operations and Supply Chain Management by Roberta S. Russell and Bernard W. Taylor III.
- b. **Effective Assessment.** The Course Director deliberately increased the number of smaller weighted assessments due to Cadet feedback from the previous year. This was primarily orchestrated through the three Group Graded Labs. One major WPR and a Graded Lab were removed this semester and the Case Study was made into a Group and Individual assessment.

3. Course Processes.

- a. **Textbook.** The course uses the 10th edition of Operations and Supply Chain Management by Roberta S. Russell and Bernard W. Taylor III.
- b. **Software.** None.
- c. **Graded Requirements.** The 11 Graded Events for the course are seen below.

Graded Events	Points
Group Graded Lab 1	50
Group Graded Lab 2	50
Problem Set 1	100
Field Trip Calc/Refl.	50
Mid Term	200
Group Graded Lab 3	50
Problem Set 2	100
Case Study Write Up (ind.)	50
Case Study Presentation (Group)	50
TEE	250
Instructor Points	50

4. **Course Pedagogy.**

a. **General:** The course was delivered in person with the ability to record lessons via Teams if need be. This happened by exception only, typically when a Cadet would be absent for an excusable reason and requested it. Cadets had assigned readings for nearly every class.

b. **Practical Exercises:** The majority of lecture based lessons ended with a PE for Cadets to complete and discuss either at the end of the class or the beginning of the next.

c. **Problem Sets:** There were two Problem Sets given during the course. The purpose of these problem sets is to build student confidence in their own abilities to recognize and apply the proper tools for critical course topics.

d. **Case Study:** All Cadets were given the same Case Study this year, written by Dr. Isabella Sanders. They were all asked to answer the same questions and present a new proposal to those questions to the class. In addition, each Cadet had to individually answer separate short answer questions while applying course concepts.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. Assessment of Changes to Current Academic Term (based on previous Course Assessments).

a. **Course Objectives.** None.

b. **Textbook.** None.

c. **Curriculum.**

- i. Reduced Statistical Process Control to 2 lessons (from 3) because these students have seen it in SE375
- ii. Adjusted Learning Curves, Work Measurement, Lean, Work Sampling, Human Resources, Quality Management, Human Resources Management from 6 lessons to 5 lessons and re-arranged order
- iii. Removed Layout PE (no Lego's during AY22 regardless)
- iv. Added a Field Trip that had to be cancelled and replaced by a graded Guest Lecture Reflection

d. **Graded Requirements.**

- i. Removed one WPR and one Graded Lab
- ii. Added 3 x Group Graded Labs
- iii. Made the Case Study both Group and Individual

e. **Other.** None.

2. Proposed Changes for Next Academic Term.

a. **Course Objectives.** None.

b. **Textbook.** None

c. **Curriculum.** None

d. **Graded Requirements.**

- i. The Mid-Term WPR covered too much material and was weighted too heavily. I would re-look at going back to two interim WPRs and/or reducing the weight of the Mid-Term.
- ii. The Case Study felt like an end of course project, but it wasn't. The weight of this needs to be increased and the timing in the Syllabus needs to be looked at again as it was right before TEEs.
- iii. 500 points, including Instructor Points, were allocated in the last month of the course. I would recommend smoothing that out more.

e. **Homework Problems.** None.

f. **Other.** None.



Enclosure 2: Instructional Memorandum, Graded Events



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

MADN-SE

7 June 2022

INSTRUCTIONAL MEMORANDUM
EM420-23-1

PRODUCTION OPERATIONS MANAGEMENT

1. **PURPOSE.** The purpose of this memorandum is to provide each cadet with the administrative details necessary to successfully complete EM420, Production Operations Management.
2. **REFERENCES.**
 - a. Department of Systems Engineering Administrative Memo 1-1, "General Administrative Instructions," 29 July 2022.
 - b. Department of Systems Engineering Administrative Memo 1-2, "Standards for Written Requirements," 29 July 2022.
 - c. Department of Systems Engineering Administrative Memo 1-3, "Academic Document Policy," 29 July 2022.
 - d. USMA publication. "Documentation of Academic Work," Office of the Dean, 25 July 2017.
 - e. The Little Brown Handbook, by H. Ramsey Fowler and Jane Aaron, Addison-Wesley, New York, 1998.
3. **GENERAL.**
 - a. This memorandum provides each cadet with the course description, course objectives, course texts, lesson assignments, grading plan, online assignment guidance, and criteria for passing the course.
 - b. Your instructor will inform you in advance of any deviation from the guidance provided in this memorandum.

4. SPECIFIC.

a. **Course Scope.** This course deals with the quantitative aspects of design and analysis of production operations management. Emphasis is on identification, analysis, and solution of production problems using applied quantitative techniques. Practical exercises reinforce the practical problem-solving techniques necessary for today's successful military and civilian engineering managers and systems engineers. Specific methods and techniques taught and applied are strategy and product design, process and capacity strategy, total quality management, statistical control process, location strategy, job design and work measurement, layout strategy, scheduling, maintenance and reliability, and service design and queuing. This course is required for those pursuing the Engineering Management major and an elective for the Systems Engineering and Systems and Decision Science majors. Applied technical skills are developed using Microsoft Excel®. Much of the course material directly relates to content found in the Fundamentals of Engineering Exam (FE).

b. **Course Strategy.** This course teaches the tools necessary to understand what occurs in production and service operations, i.e., what occurs between the infusion of inputs (e.g., raw materials, management) and the producing of outputs (e.g., a product or service). Emphasis is on problem identification, interpretation, and solution from a systems viewpoint using applied quantitative techniques. The course starts out looking at operations management from a strategic level and works its way down to the tactical level. The comprehensive TEE evaluates students' mastery of all course objectives. A detailed lesson syllabus is included in Enclosure 1.

c. **Course Objectives.**

1) Analyze and make operations management recommendations to a client with appropriate technology and engineering management tools.

2) Given a brief scenario description, describe the operations management mission, strategy, and competitive advantage of the business.

3) Given a brief scenario description, identify and assess statistical quality control problems and apply the appropriate tools to solve them.

4) Given a brief scenario description, identify and apply the appropriate techniques and tools to complete good product and service design.

5) Given a brief scenario description, identify and assess facility location, layout and capacity issues and apply the appropriate tools to solve them.

6) Given a brief scenario description, identify and assess job design and work measurement problems and apply the appropriate tools to solve them.

7) Given a brief scenario description, identify and assess personnel, materials, and job scheduling issues and apply the appropriate tools to solve those issues.

8) Given a brief scenario description, identify and assess queuing type problems and apply the appropriate tools to solve those issues.

9) Given a brief scenario description identify and assess quality management of production of goods and services.

10) Communicate in oral or written reports the analysis and application of course concepts to real-world scenarios.

d. Texts. Operations and Supply Chain Management, by Roberta S. Russell and Bernard W. Taylor III, Wiley & Sons Inc., 2017. 10th Edition with the online assessment tool WileyPlus.

e. Additional References.

1) Parnell, G. S., Driscoll, P. J., and Henderson D. L., Editors, *Decision Making for Systems Engineering and Management*, Wiley & Sons Inc., 2011 2nd Edition.

2) EM420 Equations Reference Handbook.

f. Lesson Assignments. See Enclosure 1 (Syllabus) for specific readings and assignments.

g. Performance Assessment.

1) The course consists of 1000 points.

Graded Events	Points
Group Graded Lab 1	50
Group Graded Lab 2	50
Problem Set 1	100
Field Trip Calc/Refl.	50
Mid Term	200
Group Graded Lab 3	50
Problem Set 2	100
Case Study Write Up (ind.)	50
Case Study Presentation (Group)	50
TEE	250
Instructor Points	50

a) The three Group Graded Labs and Case Study Presentation will be executed in Groups for a total of 200 points a result of group work.

b) Classroom Participation and Preparation. (50 points total – course duration). Subjective measure of student's classroom preparation, participation, and overall academic professionalism.

c) Term End Examination. (250 points). The purpose of this exercise is to reinforce those concepts taught during the entire course.

5. ADMINISTRATIVE GUIDANCE.

a. Course Structure. This is a 30 lesson course taught in 75 minute sections.

b. Minimum Requirements. Score at least a 50% on the TEE and complete the course with at least 65% of the total course points.

c. Graded Events. Unless there are exceptional circumstances, cadets will complete all graded assignments and attend all briefings and classes. Cadets who will miss an event must contact the instructor as soon as they know about the conflict. It is the cadet's responsibility to coordinate with the instructor to make up any material/requirements missed.

d. Re-grade Requests. If there was an arithmetic error in adding up points on your graded event, inform your instructor immediately, and the instructor will correct the grade. Any other requests to re-grade an assignment or a portion of an assignment for any reason can also be addressed directly with instructor. Instructors will handle the request and any re-grading at their discretion.

e. Late Submissions. Failure to ***attend or submit*** a graded event on time for other than legitimate, compelling reasons will result in a reduced grade (***5% per day***) and may result in disciplinary action. All graded events must be submitted in order to pass the course by department policy.

f. Final Grades. Final grades will be awarded IAW with reference 2.a (Department of Systems Engineering Admin Memo 1-1).

g. Documentation. Cadets may use any available reference material in preparing course work, but use of any references **MUST BE PROPERLY DOCUMENTED**. Documentation procedures are outlined in references 2a through 2e.

h. Additional Instruction (AI). AI is available at mutually agreed upon times between the cadet and the instructor. Cadets are expected to have specific questions for any AI session. No AI can be given after the first version of a WPR has been given, whether that version is the primary or a make-ahead exam. Be prepared to show your practice problems at any AI session.

i. Class. You are expected to bring your paper textbook, course reference manual, calculator, and laptop (as necessary) and to each class. If you are not going to be at a class, please let your instructor know *prior to the class* in person or via email.

FOR THE HEAD OF THE DEPARTMENT:

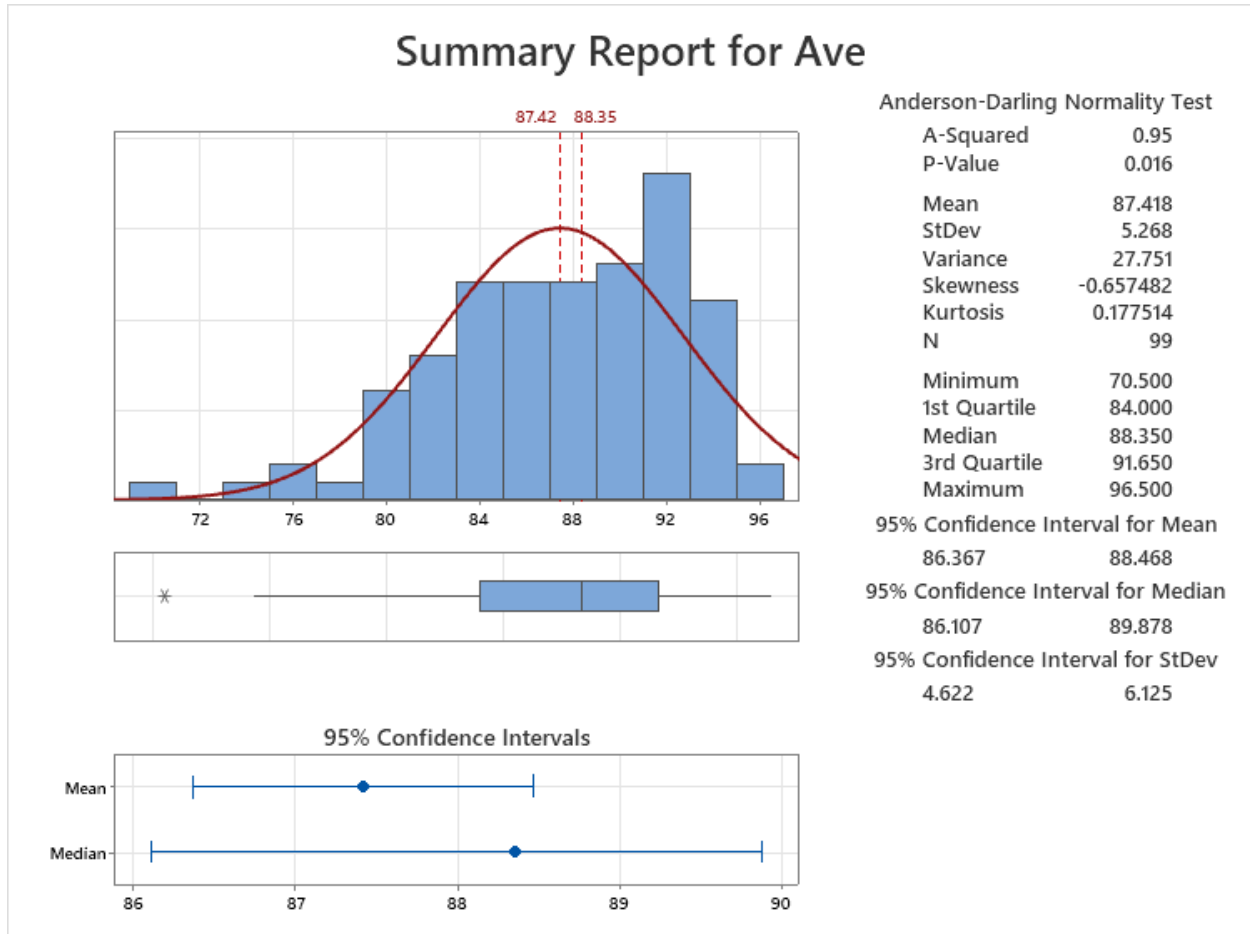
Enclosures:

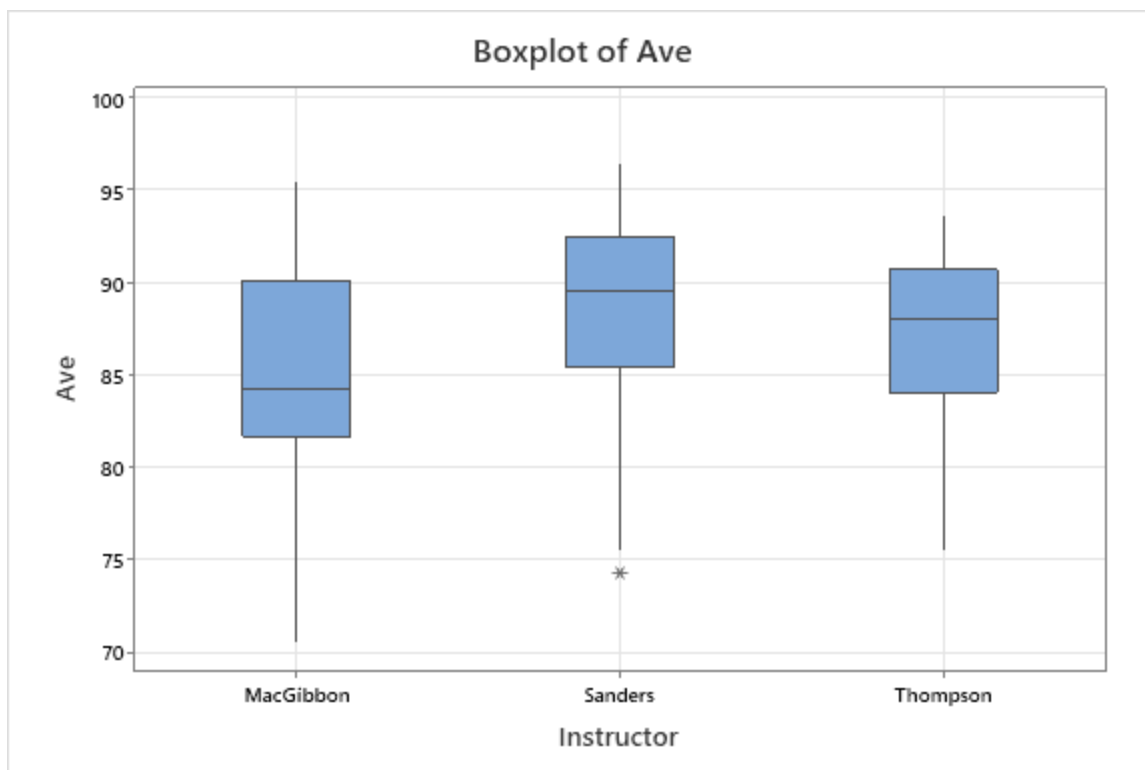
Syllabus w/ Lesson Objectives
Reference Manual

JAMES H. SCHREINER
COL, FA47
Engineering Management Program Director

Enclosure 3. Final Grades

1. **Overall Performance.** Course averages were significantly higher than last year and on par with previous years before COVID. The mean was an 87.42% and median an 88.35%. There were no course final grades lower than a C-. The boxplot below shows the distribution of final course grades by Instructor. A Mood's Median Test was completed and there is not statistical evidence to state that the median's between Instructors are different at a 95% confidence level.







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TAB E AY2022 COURSE ASSESSMENT

EM420: Production Operations Management
AY19-1

BUFF CARD LSN#	Date		In Class TEAM	Lesson	Video Assignment	Reading Assignment
1	19-Aug	Monday	NONE (all remote)	Introduction; Strategy & Operations		Chap 1, p 1-26
2	21-Aug	Wednesday	TEAM BLACK	Operational Decision Making I & II	Lesson 2 & 3 Asynch content	Edwards, Ward. "Behavioral Decision Theory". Annual Review for Psychology (1961), 12:1, pp. 473-498. / Edwards, Ward. "The Theory of Decision Making". Psychological Bulletin (1954), 51:4, pp. 380-417.
3	23-Aug	Friday	TEAM GOLD			
4	27-Aug	Tuesday	TEAM BLACK	Quality Management, Quality Tools & Cost of Quality	Lesson 4 & 5 Asynch content	Chap 2, p 53-83; 93-98
5	30-Aug	Friday (Modified)	TEAM GOLD			
6	3-Sep	Tuesday	TEAM BLACK	Statistical Process Control I & II	Lesson 6 & 7 Asynch content	Chap 3, p 107-134
7	6-Sep	Friday	TEAM GOLD			
8	10-Sep	Tuesday	TEAM BLACK	Statistical Process Control III (+ Prob Set Comp Time)	Lesson 8 & 9 Asynch content	Supplement 3, p 149-155
9	12-Sep	Thursday	TEAM GOLD			
10	14-Sep	Saturday		DROP (Graded Lab Comp)		
11	16-Sep	Monday	TEAM BLACK	Graded Lab	Review Lessons 1-9	
12	18-Sep	Wed (Modified)	TEAM GOLD			
13	20-Sep	Friday	NONE	WPR PREP		
14	23-Sep	Monday	BOTH	WPR 1 (REMOTE or in large seating area)		
15	27-Sep	Friday	TEAM BLACK	Product & Service Design	Lesson 15-16 Asynch content	Chap 4, p 157-182; 1890201
16	1-Oct	Tuesday	TEAM GOLD			
17	3-Oct	Thursday	TEAM BLACK	Queueing I , II & III	Lesson 17-18 Asynch content	Chap 5, p 201-216, FE Handbook, p 215-216, Industrial Engineering FE Reference Manual Excerpt PDF
18	7-Oct	Monday	TEAM GOLD			
19	9-Oct	Wednesday		DROP		
20	11-Oct	Friday	TEAM BLACK	Processes and Technology / Capacity Planning Intro	Lesson 20-21 Asynch Content	Chap 6, p 224-247; Chap 7, p 254-264
21	16-Oct	Wednesday	TEAM GOLD			
22	18-Oct	Friday	TEAM BLACK	Capacity and Facility Design: Designing Process & Product Layouts	Lesson 22-23 Asynch Content	Chap 7, p 264-281
23	22-Oct	Tuesday	TEAM GOLD			
24	24-Oct	Thursday	TEAM BLACK	Layout II / Facility Location Models	Lesson 24-25 Asynch Content	Review Chap 7 / Supplement 7, p 292-305
25	28-Oct	Monday	TEAM GOLD			
26	30-Oct	Wednesday	NONE	WPR PREP		
27	1-Nov	Friday	BOTH	WPR 1 (REMOTE or in large seating area)		
28	4-Nov	Monday		DROP		
29	8-Nov	Friday	TEAM BLACK	Human Resources and Quality Management / Job Design & Ergo	Lesson 29-30 Asynch Content	Chap 8, p 312-326, Chap 8, p 326-333, Brightwood FE Industrial Engineering Review Manual Exerpt
30	14-Nov	Thursday	TEAM GOLD			
31	18-Nov	Monday	TEAM BLACK	Learning Curves & Work Measurement (+ PE)	Lesson 31-32 Asynch Content	Chap 8, p 334-347 Supplement 8, p 343-351
32	20-Nov	Wednesday	TEAM GOLD			
33	22-Nov	Friday	TEAM BLACK	Work Sampling & Lean Systems	Lesson 33-34 Asynch Content	Supplement 8, p 31-354 / Chap 16, p 707-735
34	25-Nov	Monday	TEAM GOLD			Chap 16, p 707-735
35	27-Nov	Wednesday (Modified)		DROP		
36	3-Dec	Tuesday	TEAM BLACK	Scheduling I & II	Lesson 36-37 Async content	Chap 17, p 741-754
37	5-Dec	Thursday	TEAM GOLD			
38	9-Dec	Monday		DROP (Case Study Sue 11 Dec)		
39	11-Dec	Wednesday				
40	43812	Friday (Modified)	NONE (all remote)	Course Review		

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[illegible]



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TAB F AY2021 COURSE ASSESSMENT

EM420: Production Operations Management

AY19-1

LSN#	Date		Lesson	Lesson Objectives	Reading Assignment	Major Event
1	17-Aug	Monday	LSN 1: Introduction to Operations and Operation Strategy	1. Define Operations, Operations Management, and Value Chain. 2. Identify the key factors that contributed to the evolution of operations and supply chain management. 3. Explain how and why businesses operate globally and the importance of globalization in supply chain management 4. Calculate and assess productivity measures used for measuring competitiveness. 5. Identify a firm's primary task, core competencies, order winners, and qualifiers, and how the firm places to position itself in the market.	Chap 1, p 1-26	
2	19-Aug	Wednesday	LSN 2: Operational Decision Making I	1. Understand objective and subjective probabilities; compare and contrast objective and subjective probabilities. 2. Understand decision utility and the relation of decision utility to value focused thinking. 3. Understand why the psychology of decision-making matters. 4. Understand the subjectively expected utility maximization model (SEU model) and the differences of the SEU model compared to the SDP. 5. Understand the fundamental concept of risk and how it relates to decision making.	Edwards, Ward. "Behavioral Decision Theory". Annual Review for Psychology (1961), 12:1, pp. 473-498. Edwards, Ward. "The Theory of Decision Making". Psychological Bulletin (1954), 51:4, pp. 380-417.	Problem Set #1 Posted
3	21-Aug	Friday	LSN 3: Operational Decision Making II	1. Apply decision analysis for decision-making situations in which uncertainty without probabilities exists using the following techniques: Maximax, Maximin, Minimax Regret, Hurwicz, and Equal Likelihood. 2. Apply decision analysis for decision-making situations in which uncertainty with probabilities exists using expected value in Sequential Decision Trees. 3. Calculate the Expected Value of Perfect Information (EVPI).	Supplement 1, p 32-41	
4	25-Aug	Tuesday	LSN 4: Quality Management and Quality Tools	1. Identify the dimensions of quality for products and services and how quality changes with perspective. 2. Identify the key factors that contributed to the evolution of quality management in the 20th century. 3. Use Pareto Analysis in solving problems of quality management. 4. Define Six Sigma and explain its origin.	Chap 2, p 52-82	
5	27-Aug	Thursday	LSN 5: The Cost of Quality	1. Identify the two costs of achieving good quality and provide examples of each. 2. Identify the two costs of achieving bad quality and provide examples of each. 3. Calculate quality indexes, product yield for both a single stage and multistage process, product costs per unit, and Quality Productivity Ratio (QPR). 4. Explain International Organization for Standardization (ISO) certifications and the importance of industry awards for quality.	Chap 2, p 83-98	
6	31-Aug	Monday	LSN 6: Statistical Process Control I	1. Explain how attribute and variable measures are applied to SPC. 2. What are the four criteria used to determine whether the process measured in a control chart is in control? 3. Construct and evaluate a P-chart for a sample process and evaluate whether the process is in control.	Chap 3, p 106-115	
7	2-Sep	Wednesday	LSN 7: Statistical Process Control II	1. Construct and evaluate a C-chart for a sample process and evaluate whether the process is in control. 2. Construct and evaluate a Mean and Range Chart for a sample process and evaluate whether the process is in control. 3. Apply criteria for identifying control chart patterns.	Chap 3, p 115-126	
8	4-Sep	Friday	LSN 8: Statistical Process Control III	1. Explain how Process Capability is used to determine process variability. 2. Calculate Process Capability Ratio (Cp) and Process Capability Index (Cpk). 3. Develop a single-sample attribute plan and calculate the acceptable number of defective items in the sample.	Chap 3, p 126-134 Supplement 3, p 149-156	
9	8-Sep	Tuesday	DROP			
10	10-Sep	Thursday	LSN 9: Statistical Process Control Graded Lab	1. Use computer software (MS Excel) to measure SPC for given problems, use the appropriate control charts to determine whether the processes are in control and determine whether process variation is acceptable.		Graded Lab
11	14-Sep	Monday	LSN 10: WPR 1	1. Demonstrate understanding of lesson objectives from lessons 1-8.		WPR 1
12	17-Sep	Thursday	LSN 11: Product Design	1. Explain the product design process to include the various idea generation methodologies. 2. Explain the process of concurrent design, identify its three types, and calculate product reliability and maintainability. 3. Explain the techniques for reviewing design quality: Failure Mode and Effects Analysis (FMEA), Fault Tree Analysis, and Value Analysis.	Chap 4, p 158-170	
13	21-Sep	Monday	LSN 12: Service Design	1. Evaluate the impact of services on jobs and the economy. 2. Articulate the differences between products and services characteristics. 3. Describe the service-process matrix and provide examples.	Chap 5, p 190-201	
14	25-Sep	Friday	LSN 13: Queueing I	1. Describe the elements of a queueing system to include the waiting line, calling population arrival rate, service time, and queue discipline. 2. Understand the components of Kendall notation and how they relate to a queueing system. 3. Describe the operating characteristics of a queueing system. 4. Evaluate a basic single-server queueing system model.	Chap 5, p 201-210 FE Reference Handbook, p 215-216 Industrial Engineering FE Reference Manual Excerpt PDF	Problem Set #2 Posted
15	29-Sep	Tuesday	LSN 14: Queueing II	1. Evaluate a basic single-server queueing system model with a finite calling population, a finite queue, and with constant service times. 2. Evaluate a basic multiple-server queueing system model.	Chap 5, p 210-215 FE Reference Handbook, p 215-216 Industrial Engineering FE Reference Manual Excerpt PDF	

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16	1-Oct	Thursday	LSN 15: Processes and Technology	1. Define a process and a process strategy. 2. Define outsourcing and describe what factors go into evaluating whether to outsource. 3. Describe the product-process matrix and provide examples. 4. Perform a breakeven analysis and identify the point(s) of indifference.	Chap 6, p 224-232	
17	5-Oct	Monday	LSN 16: Capacity and Facility Design: Capacity Planning	1. Define capacity and capacity planning. Evaluate different strategies for capacity expansion. 2. Define economies of scale, best operating level, and cycle time. 3. Describe the objectives of facility layout, the three basic forms of layout, and discuss their relative advantages and disadvantages.	Chap 7, p 254-264	
18	7-Oct	Wednesday	DROP			
19	9-Oct	Friday	LSN 17: Capacity and Facility Design: Designing Process Layouts	1. Use block diagramming and relationship diagramming to design a facility's process layout.	Chap 7, p 264-270	

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LSN#	Date		Lesson	Lesson Objectives	Reading Assignment	Major Event
20	14-Oct	Wednesday	LSN 18: Capacity and Facility Design: Designing Product Layouts	1. Define line balancing, desired cycle time, and actual cycle time. 2. Explain precedence requirements as they relate to product layouts. 3. Use line balancing to design a facility's product layout.	Chap 7, p 270-274	
21	16-Oct	Friday	LSN 19: Layout Practical Exercise	1. Use line balancing to evaluate and redesign a facility's product layout.	Review Chap 7	
22	20-Oct	Tuesday	LSN 20: Facility Location Models	1. Use the center-of-gravity and load-distance techniques to determine where best to locate a facility.	Supplement 7, p 293-306	
23	22-Oct	Thursday	DROP			
24	26-Oct	Monday	LSN 21: WPR 2	1. Demonstrate understanding of lesson objectives from lessons 11-20.		WPR 2
25	28-Oct	Wednesday	LSN 22: Human Factors & Ergonomics I	1. Describe the process for identifying hazards and how to use the risk assessment matrix. 2. Calculate permissible noise exposure and identify the elements of an OSHA approved hearing conservation program. 3. Define ergonomics and describe its relationship to cumulative trauma disorder (CTD). 4. Quantitatively solve problems of biomechanics.	Chap 8, p 328-335 & Brightwood FE Industrial Engineering Review Manual Exerpt	
26	30-Oct	Friday	LSN 23: Human Factors & Ergonomics II	1. Quantitatively solve problems of anthropometry. 2. Calculate the National Institute of Occupational Safety and Health (NIOSH) Recommended Weight Limit (RWL) for a lifting job. 3. Define cognitive engineering and describe how information processing, situational awareness, and mental models fit under the larger subset of human factors.	Chap 8, p 328-335 & Brightwood FE Industrial Engineering Review Manual Exerpt	Case Study Posted
27	2-Nov	Monday	LSN 24: Learning Curves & Work Measurement	1. Explain the learning curve effect for work improvement. Calculate the time to make the nth product and the learning rate. 2. Use the learning curve effect to improve a product layout.	Chap 8, p 335-340 Supplement 8, p 343-351	
28	6-Nov	Friday	LSN 25: Time Study and Work Sampling	1. Calculate the standard time to perform a job and the number of cycles needed for a time study for a given level of confidence. 2. Calculate the number of samples needed in a work sampling study.	Supplement 8, p 345-356	
29	9-Nov	Monday	LSN 26: Lean Systems	1. Explain how the basic elements of lean systems eliminate waste, create flow, and continuously improve operations. 2. Assess the benefits of lean, as well as the difficulties in implementing lean systems. 3. Calculate the number of kanbans necessary to implement a lean system.	Chap 16, p 717-745	
30	12-Nov	Thursday	LSN 27: Scheduling I	1. Define loading, sequencing, and monitoring. 2. Solve scheduling problems using the loading assignment method. 3. Define flow time, makespan, and tardiness. 4. Solve scheduling problems by sequencing jobs through one process.	Chap 17, p 750-760	
31	16-Nov	Monday	LSN 28: Scheduling II	1. Explain Johnson's Rule. 2. Solve scheduling problems by sequencing jobs through two serial processes.	Chap 17, p 760-768	
32	18-Nov	Wednesday	LSN 29: Human Resources Management		Chap 8, p 313-328	
33	20-Nov	Friday	DROP -- Case Study Work Time			
34	23-Nov	Monday	DROP -- Case Study Work Time			
35	25-Nov	Wednesday	DROP -- Case Study Work Time			
36	1-Dec	Tuesday	DROP -- Case Study Work Time			
37	3-Dec	Thursday	DROP -- Case Study Work Time			
38	7-Dec	Monday	DROP -- Case Study Work Time			
39	9-Dec	Wednesday	DROP -- Case Study Work Time			
40	11-Dec	Friday	LSN 30: Course Review	1. Review lesson objectives from the entire course.		Case Study DUE



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SECTION II

EXAMINATIONS



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TAB G

Written Partial Review I

EM420 Production and Operations Management**Written Partial Review****Version A**

- 1) This is a 75-minute graded event worth 150 points.
- 2) This is a closed book and closed notes exam. You may use one single sided 3x5" index card and the provided reference book (DO NOT MARK IN IT!) and an authorized scientific calculator. No collaboration is allowed.
- 3) You may disassemble this document if you'd like. You may also use extra paper provided at the front of the classroom. Please ensure you staple the exam back together in the correct order with extra sheets labeled and at the back.
- 4) **SHOW ALL WORK!** Unsupported or missing work earns little or no partial credit.
- 5) **Rounding.** Show dollars and cents for any monetary answer (\$11.75). Show two significant digits beyond the decimal for percentage answers (11.75%) and at least two significant digits beyond the decimal for all other answers (0.13).
- 6) Upon completion, sign the acknowledgment statement below:

"I did not use any sources, nor did I receive any assistance while completing this WPR. This WPR will be released from academic security on 30OCT2025. Prior to that time, I may not discuss any aspects of this exam with anyone except an EM420 instructor."

Printed Name: _____ Signature: _____ Date Signed: _____

8) GRADING SUMMARY:

Section	Topic	Total Points Available	Total Points Earned
1	Situational Assessment	20	
2	Multiple Choice	25	
3	Ammo Plants	20	
4	Winter Jacket	20	
5	Field Kitchens	15	
6	Drive Thru	30	
7	Backpacks	20	
BONUS	Western Electric	5	
	Total	150	

Section I: Situational Assessment (No Partial Credit) [20 points]

Please choose the BEST problem-solving strategy, chart, or calculation that would aid in making a decision for the prompt. There is one best answer for each situation. No partial credit. Explanation is not necessary. You must write your answers on the lines below for grading

1A _____

1B _____

1C _____

1D _____

1E _____

1F _____

Order Winner	Order Qualifier
EVPI	Hurwicz
Equal likelihood	Maximax
Maximin	Priority
Minimax Regret	Pareto Analysis
Quality Productivity Ratio	P Chart
X bar Chart	R Chart
C Chart	FMEA
Fault Tree Analysis	House Of Quality
Service Blueprint	Calling Population
FIFO	LIFO
General	Degenerate
Markovian	

1G _____

1A [3 Points] A program manager wants to compute the expected monetary value of hiring a market research firm before making a major sourcing decision, to see whether additional information is worth the cost. Which calculation should they perform?

1B [3 Points] An aircraft mishap investigation team wants to model how combinations of electrical, hydraulic, and pilot-control failures could lead to loss of flight control. They want a logical diagram that starts with the top event (loss of control) & maps pathways downward to root causes.

1C [3 Points] A quality engineer at an Army depot notices that production of Humvee transmissions has been missing daily output targets for the last two weeks. Reviewing the data, they see eight different types of process failures, including torque wrench calibration errors, mislabeled parts etc. Management asks them to figure out which issues are causing the majority of lost throughput so they can direct maintenance to fix those first. What should they use?

1D [3 Points] An operations officer notices repeat breakdowns on forklifts. They want to systematically rank which failure means (hydraulic leak, steering failure, sensor malfunction) are most critical so they can focus limited maintenance funds on the highest risk items first.

1E [3 points] An ammo supply point (ASP) officer is modeling wait times for convoys arriving to pick up munitions before a training exercise. Historically, truck arrivals are random and independent, with interarrival times well-fit by an exponential distribution. The officer also confirms that service times (loading each truck) are memoryless and follow an exponential pattern. Which type of distribution matches service times?

1F [3 points] An Army Program Executive Office is evaluating two vendors for next-gen body armor. Both vendors meet the minimum ballistic protection and weight standards required to compete. However, Vendor A can deliver within 30 days while Vendor B's lead time is 90 days. Rapid fielding is the deciding factor for award. How can you classify rapid fielding?

1G [2 points] Please list your Group Graded Lab teammates' first names.

Section II: Multiple Choice (No Partial Credit) [25 points]

Choose the best answer. You must write your letter answers on the lines below for grading

2A _____

2B _____

2C _____

2D _____

2E _____

2A (5 Points) Consider an espresso stand with a single barista. Customers arrive at the stand at the rate of 28 per hour according to a Poisson distribution. Service times are exponentially distributed with a service rate of 35 customers per hour. The probability that there are more than 2 customers in the system is:

- a. 0.128 b. 0.488 c. 0.512 d. 0.640

2B (4 Points) A process that is in control as indicated by its Statistical Process Control chart indicates that some _____ in the process may be _____.

- a. variation, natural b. variation, assignable c. attributes, random d. defects, measurable

2C (4 Points) Decision criteria are interrelated, often depend on the eye of the beholder and settings. _____ criteria is actually the same as the _____ criteria when $\alpha = 1$.

- a. Hurwicz, EVPI b. Minimin, Maximin c. Minimax Regret, Hurwicz d. Maximax, Hurwicz

2D (5 Points) You are a quality engineer at a small-arms ammunition plant. Your team has measured the reliability of three critical machines in series that form a cartridge assembly line:

Case Feeder: $R_1=0.96$ Primer Inserter: $R_2=0.94$ Bullet Seater: $R_3=0.9$

What is the reliability of the overall system (assume machines are arranged in series)?

- a. 0.831 b. 0.850 c. 0.900 d. 0.922

2E (5 Points) During a recent 4,000-hour production run, the primer inserter broke down 12 times. Each repair took on shift or 4 hours to complete on average. What was its availability during this period?

- a. 0.88 b. 0.92 c. 0.95 d. 0.99

Section III: Ammo Plants [20 Points]

Problem 3 You are a commander managing two Army ammo plants which operate **320 days/year** at **10 hours/day**. Each unit is sold at the plant's listed price.

Item	Plant Alpha	Plant Bravo
Output (units/year)	1,120,000	1,050,000
Selling price (\$/unit)	\$38.00	\$40.00
Labor hours/year	480,000	420,000
Labor cost (\$/hour)	\$28.00	\$31.00
Material cost/year	\$18,900,000	\$19,800,000
Energy cost/year	\$2,450,000	\$2,200,000

3A (7.5 Points). Calculate the Total Productivity for each plant. Interpret your results in words

Item	Plant Alpha	Plant Bravo
Output (units/year)	1,120,000	1,050,000
Selling price (\$/unit)	\$38.00	\$40.00
Labor hours/year	480,000	420,000
Labor cost (\$/hour)	\$28.00	\$31.00
Material cost/year	\$18,900,000	\$19,800,000
Energy cost/year	\$2,450,000	\$2,200,000

3B (12.5 Points). The Army can invest **\$15,000** total:

Option 1 (\$7,500): **+5% labor productivity** (same hours, but effective output increases 5%).

Option 2 (\$7,500): **-10% material cost.**

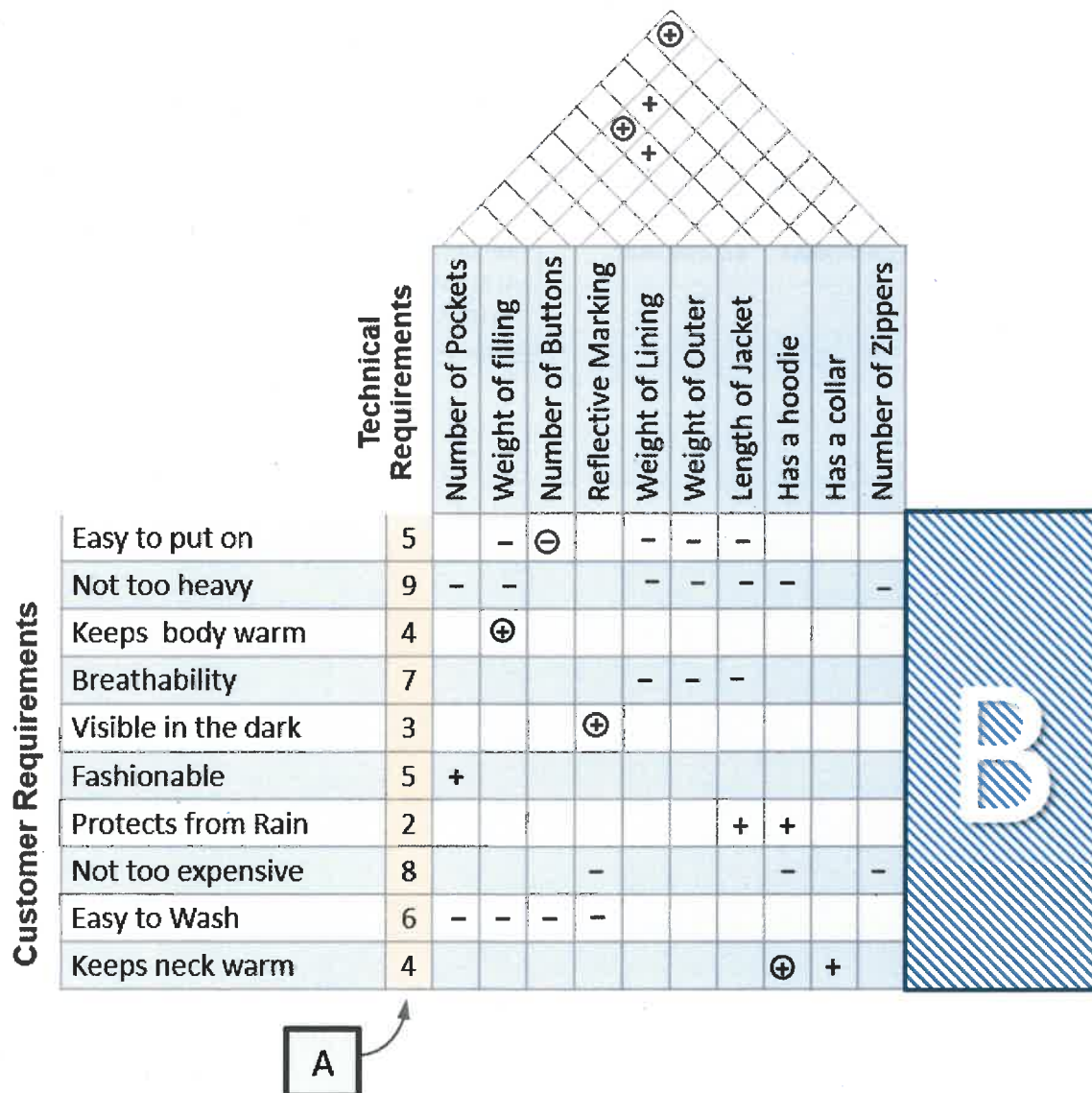
You have the following options for investment:

- A. Apply option 1 to both plants
- B. Apply option 2 for both plants
- C. Apply Option 1 for Plant A & Option 2 for Plant B.
- D. Apply Option 1 for Plant B & Option 2 for Plant A.

Recommend which investment strategy the Army should choose and support with quantitative analysis.

Section IV: Winter Jacket [20 points]

You are part of a product design team tasked with developing a new winter jacket for Soldiers operating in cold-weather environments. Your product design team has created the following matrix to help you understand the relationship between customer and technical requirements. Use this diagram to answer the questions that follow about trade-offs and technical focus areas for the jacket design.



4A (5 Points) Please explain the relationship in words between the customer requirement “Easy to Wash” and the technical requirement “number of buttons.”

4B (5 Points) Identify one customer requirement that has weak or few relationships to the technical requirements. How might you redesign the technical approach to better address that requirement?

4C (5 Points) What does A (the label for the orange numbers) represent? Interpret one of the numbers in reference to the customer requirement.

4D (5 Points) What else is typically included in a house of quality that is not shown? Hint: You could identify the area hidden by B on the diagram or any other missing pieces of the diagram. Think about the purpose of the house of quality.

Section V: Field Kitchens [15 Points]

You are the operations manager for a field kitchen deciding whether to invest in a new blast chiller.

- If you buy the blast chiller, there is a 60% chance demand is high, generating a profit of \$50,000, and a 40% chance demand is low, leading to a profit of \$10,000 (because the chiller is under-utilized).
- If you do not buy, you save the purchase cost but must rely on the old cooling method. In this case, there is a 60% chance of high demand, earning \$30,000, and a 40% chance of low demand, earning \$15,000.

5A (15 Points) Calculate the Expected Monetary Value (EMV) for each decision and explain the option you would choose, and the circumstance. Be sure to write all assumptions.

Section VI: Drive Thru [30 Points]

The arrival rate of cars at a drive-thru restaurant is 20 cars per hour. The service rate is 18 cars per hour. The available space for cars in the drive-thru is restricted to 6 cars only. Otherwise, the queue goes into the street and cars cannot queue. The arrival rate and service rate follow the Poisson distribution. Identify the type of queue and then determine the performance measures of this queue. *Be sure to interpret each answer in words!*

6A [10 points] What is the probability that the system is full?

6B [10 points] How many customers are lost per hour on average?

6C [10 points] What is the effective arrival rate? What does that mean in words?

Section VII: Backpacks [20 Points]

You have recently retired from the military and are working for a tactical backpack company and much count the blemishes on each backpack (stitch undone, loose thread, bleach spot etc.). A sample includes one backpack from a batch of 50 backpacks. The following samples are collected at random in a one month period. Assume 95% confidence.

Sample	Defects
1	12
2	6
3	7
4	5
5	2

7A [5 points] What type of control chart is best used in this situation.

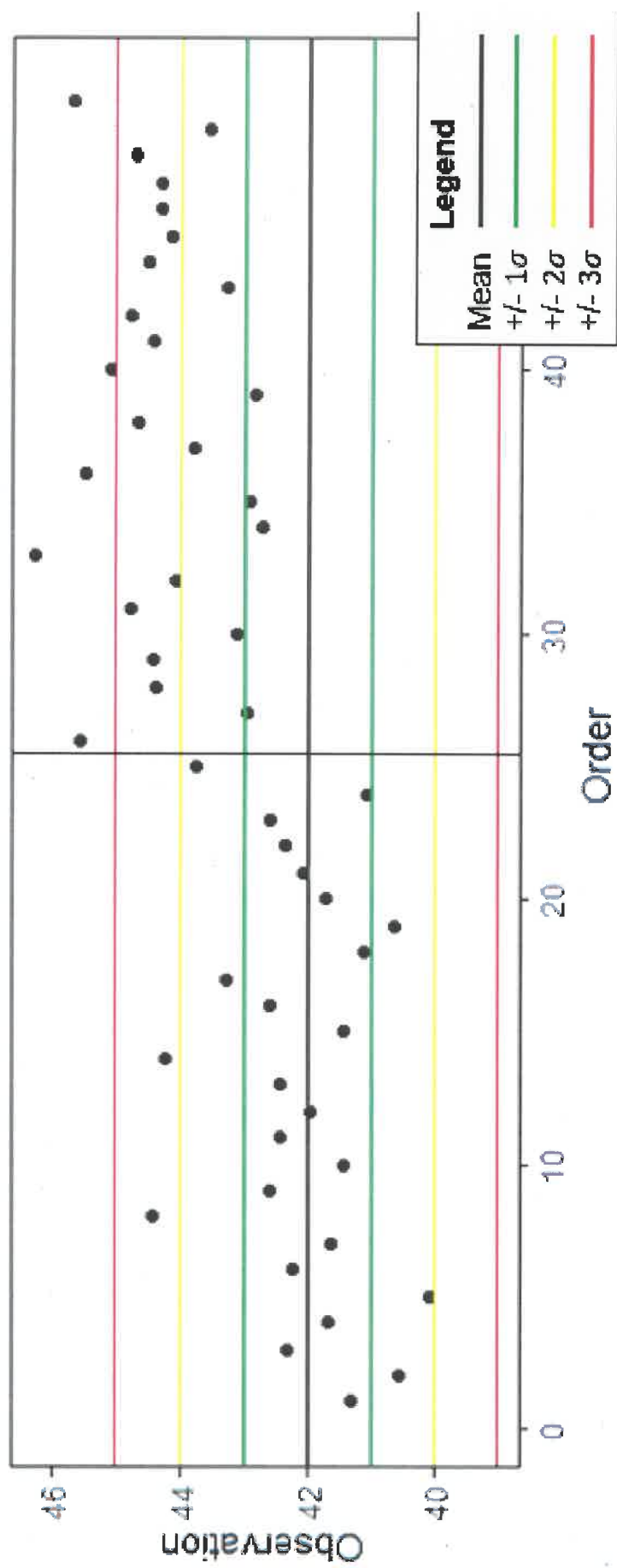
7B [5 points] Calculate the UCL, UCL and CL

7C [5 points] Draw the Control Chart, be sure to include all labels.

7D [5 points] Please interpret your control chart in words. Use context from the prompt.

Bonus (5 points): Many types of rules can be considered to determine whether a process is out of control. Use the Western Electric Rules to determine if the process is out of control (if any rule holds true the process is out of control). Indicate which points break which rule by circling and labeling them with the rule number broken.

1. The most recent point plots outside one of the 3-sigma control limits.
2. Two of the three most recent points plot outside and on the same side as one of the 2-sigma control limits.
3. Four of the five most recent points plot outside and on the same side as one of the 1-sigma control limits.
4. Eight out of the last eight points plot on the same side of the center line, or target value.
5. Six points in a row increasing or decreasing.
6. Fifteen points in a row within one sigma.
7. Fourteen points in a row alternating direction.
8. Eight points in a row outside one sigma.



Outstanding Student Work

WPR1

● Graded

Student

Ethan Ellis

Total Points

121.5 / 120 pts

Question 1

Question 1: Situational Assessment

20 / 20 pts

1.1 — 1A

3 / 3 pts

✓ - 0 pts Correct

- 1.5 pts Incorrect, however, response is relevant to decision-making under risk/uncertainty

- 2.5 pts Incorrect

- 3 pts Blank

1.2 — 1B

3 / 3 pts

✓ - 0 pts Correct

- 1 pt Close, but incorrect

- 2.5 pts Incorrect

- 3 pts Blank

1.3 — 1C

3 / 3 pts

✓ - 0 pts Correct

- 1 pt Incorrect, but it is a quality/reliability tool that also prioritizes failure modes

- 1.5 pts Incorrect, but they are quality control charts, at least within the SPC domain

- 2.5 pts Incorrect

- 3 pts Blank

1.4 — 1D

3 / 3 pts

✓ - 0 pts Correct

- 0.5 pts Incorrect, but very plausible misinterpretation since both prioritize causes

- 1 pt Incorrect, but still in reliability/failure analysis family

- 2.5 pts Incorrect

- 3 pts Blank

1.5 — 1E

3 / 3 pts

✓ - 0 pts Correct

- 1.5 pts Conceptual Mislabel; chose different distribution indicator

- 2.5 pts Incorrect

- 3 pts Blank

1.6

1F

3 / 3 pts

✓ - 0 pts Correct

- 0.5 pts Incorrect, but in the same competitive strategy family, so understandable misinterpretation

- 1.5 pts Incorrect, but indicates some degree of prioritization/importance (was used in class in the context of queuing discipline)

- 2.5 pts Incorrect

- 3 pts Blank

1.7

1G

2 / 2 pts

✓ - 0 pts Correct

- 2 pts Incorrect

Question 2

Question 2: Multiple Choice

25 / 25 pts

2.1 — **2A**

5 / 5 pts

✓ **- 0 pts** Correct

- 4 pts Incorrect

- 0.5 pts Failure to Follow Instructions

- 1.5 pts Calculated P_2 rather than $P_n > 2$

- 1 pt Calculated $P_n < 2$

2.2 — **2B**

5 / 5 pts

✓ **- 0 pts** Correct

- 1.5 pts Partially correct.

- 4 pts Incorrect

2.3 — **2C**

5 / 5 pts

✓ **- 0 pts** Correct

- 2 pts Incorrect, but identified connection of alpha value to Hurwicz

- 4 pts Incorrect

2.4 — **2D**

5 / 5 pts

✓ **- 0 pts** Correct

- 5 pts [Click here to replace this description.](#)

- 2.5 pts Failure to Follow Instructions

2.5 — **2E**

5 / 5 pts

✓ **- 0 pts** Correct

- 4 pts Incorrect

3.1 — 3A

7 / 7.5 pts

– 0 pts Correct

Plant A -Either:

0.032 Units/\$ or 1.22 (dmnl)

Plant B - Either:

0.030 units/\$ or 1.20 (dmnl)

Interpretation:

A>B and why

✓ – 0.5 pts Missing/Incorrect Units

– 0.5 pts Incorrect Plant A Calculation

Plant A Productivity:

Method 1 (units per dollar):

Output/Input = 1,120,000 units ÷ [(480,000 labor hrs/yr × \$28/hr) + \$18,900,000/yr + \$2,450,000/yr]

= 1,120,000 units ÷ \$34,790,000

= 0.032 units per dollar

Method 2 (value of output per dollar input):

(1,120,000 units × \$38/unit) ÷ \$34,790,000

= \$42,560,000 ÷ \$34,790,000

= 1.22 (dimensionless ratio)

– 0.5 pts Incorrect Plant B Calculation

Plant B Productivity:

Method 1 (units per dollar):

Output/Input = 1,050,000 units ÷ [(420,000 labor hrs/yr × \$31/hr) + \$19,800,000/yr + \$2,200,000/yr]

= 1,050,000 units ÷ \$34,820,000

= 0.030 units per dollar

Method 2 (value of output per dollar input):

(1,050,000 units × \$40/unit) ÷ \$34,820,000

= \$42,000,000 ÷ \$34,820,000

= 1.21 (dimensionless ratio)

– 2.5 pts No TFP Plant A

– 2.5 pts No TFP Plant B

– 1 pt TFP Incorrectly Calculated

– 0.5 pts Incorrect/Incomplete Interpretation

Follows

– 1.5 pts Incorrect/Incomplete Interpretation

Does not follow or does not address question properly

– 2.5 pts No Interpretation

✓ - 0 pts Correct Recommendation; Logic Follows
or Sufficiently Rigorous Quantitative Support

- 2.5 pts Correct Recommendation, Logic Does not Follow or Insufficient Quantitative Support
- 5 pts Incorrect Recommendation, Logic Follows or Sufficiently Rigorous Quantitative Support
- 7.5 pts Incorrect/No Recommendation, Logic Does not Follow or Work is insufficient to arrive at conclusion
- 12.5 pts No Recommendation or Incorrect Guess with no/little support
- 1 pt You arrived at the correct results, but then misinterpreted the question or your outcomes. It was asking both Plant A and Plant B, not Both option 1 and option 2. Numerical answer supports that you understood the question.

Question 4

Question 4: Winter Jacket

19 / 20 pts

4.1 — 4A

5 / 5 pts

✓ - 0 pts Correct

- 0.5 pts Minor error in explanation

- 1 pt Did not explain that as "Number of Buttons" Increases, "Ease of Wash" Decreases (Negative or Inverse Relationship)

- 3 pts Insufficient/Incorrect Response

- 5 pts Did not answer

4.2 — 4B

4 / 5 pts

- 0 pts Correct

- 0.5 pts Minor Error in explanation

- 1 pt Incorrectly identified customer requirement vs technical requirement

✓ - 1 pt Identified customer requirement with strong negative or strong positive relationship to technical requirement

- 1 pt Identified a customer requirement with large number of relationships

- 2 pts Weak/Missing Explanation

- 5 pts Did not answer the question

4.3 — 4C

5 / 5 pts

✓ - 0 pts Correct

- 0.75 pts Insufficient/Unclear interpretation of the Customer Importance or Incorrect Interpretation of the values (Higher is better)

- 1.5 pts Did not interpret or incorrectly interpreted in reference to customer requirement

- 2.5 pts Incorrect/Missing interpretation of A

- 5 pts Missing response

4.4 — 4D

5 / 5 pts

✓ - 0 pts Correct

- 1 pt Incomplete/Incorrect Response or Does not consider other components of HOQ

- 5 pts Missing Response

Question 5

Question 5: Field Kitchens

15 / 15 pts

5.1 5A

15 / 15 pts

✓ - 0 pts Correct

- 0.5 pts Units

- 1 pt Minor Math Error

- 1 pt Incorrect/Incomplete EMV for Blast Chiller

- 1 pt Incorrect/Incomplete EMV for Old Cooling Method

- 1 pt Incorrect Interpretation
Follows

- 5 pts Missing EMV for Blast Chiller

- 5 pts Missing EMV for Old Cooling Method

- 5 pts Incorrect/Missing Interpretation
Does not Follow

- 2.5 pts Provided explanation, but did not provide recommended course of action

Question 6

Question 6: Drive Thru

0 / 0 pts

6.1 — 6A

0 / 0 pts

– 0 pts Correct

– 1 pt Minor Math/Formulation Error

✓ – 2.5 pts Did not identify solution as P6
Probability that there are 6 vehicles in the system = probability that the system is full

– 2.5 pts Used incorrect Formulation of P6
Should be M/M/1/6 Kendall Notation, resulting in:
 $P_6 = ((1-\rho)/(1-\rho^{6+1})) \cdot \rho^6$

– 2.5 pts Did not arrive at solution as a probability (probabilities cannot be negative or exceed 100%) or did not sufficiently demonstrate understanding of tested concept

– 10 pts Did not answer question or showed insufficient work to demonstrate understanding of the subject

– 1 pt Should use 6 for M and n, resulting in incorrect response. Should be 19.17%

– 0.5 pts Transcription Error

6.2 — 6B

0 / 0 pts

– 0 pts Correct

– 0 pts Incorrect, but follows

– 1 pt Incorrect; follows with some minor error in math or logic

✓ – 5 pts Incorrect; answer follows some logic

– 7.5 pts Incorrect; answer does not follow logic or numbers do not appear to have justification

– 2.5 pts Correct (or nearly correct) response, but work does not adequately demonstrate how you arrived at answer or arrived at correct (or nearly correct) answer through incorrect means

– 10 pts Missing Response

6.3 — 6C

0 / 0 pts

– 0 pts Correct

– 0 pts Incorrect effective arrival rate, but follows from previous

✓ – 2.5 pts Incorrect/Incomplete Numerical Solution;

– 5 pts Formula only

– 7.5 pts No numerical Solution or entirely incorrect approach or work shown/not shown does not demonstrate understanding of core concept

– 2.5 pts No/insufficient/incorrect interpretation

– 1 pt Minor math error or incorrect input
examples include incorrect λ (should be 20 cars/hr), incorrect P6 (should be 0.19, but may follow from previous question)

– 0.5 pts Units

– 1 pt Minor issues with interpretation

Question 7

Question 7: Backpacks

20 / 20 pts

7.1 — 7A

5 / 5 pts

✓ - 0 pts Correct

- 3 pts Correctly identified as attribute, but misidentified as p-chart (should be c chart)

- 5 pts Incorrect

7.2 — 7B

5 / 5 pts

✓ - 0 pts Correct

- 0 pts Incorrect Chart, but Follows

- 1 pt Incorrect Z Values
95% -> $z=1.96$ or ~ 2

- 1.5 pts Incorrect CL
If C Chart, should be $(12+6+7+5+2)/5 = 6.4$
If (incorrectly) using P Chart, should still be $(12/50+6/50+7/50+5/50+2/50)/5 = 0.128$

- 1 pt LCL Cannot be Negative

- 1 pt Incorrect CL
Should be $\bar{C}$

- 1.5 pts Used Incorrect Sigma Values
Should be $\sqrt{\bar{C}}$

- 3 pts Incomplete Calculations

- 5 pts No/Insufficient response

- 1.5 pts Correct CL (6.4) for a C chart, but then proceeded to convert to incorrect P chart CL and then use P chart UCL and LCL

- 4 pts No CL, incorrect/Incomplete UCL and LCL formulations

- 0.5 pts Minor Math Error

- 3 pts Incorrect UCL and LCL
If C Chart, should be $\bar{C} \pm z\sqrt{\bar{C}}$
If (incorrectly) P Chart, should be $\bar{P} \pm z\sqrt{\bar{P}(1-\bar{P})/n}$

7.3 — 7C

5 / 5 pts

✓ - 0 pts Correct or Follows

- 1 pt Missing/Incorrect Labels

- 1 pt Missing/Incorrect Axes or Axis Labels

- 1 pt Missing/Incorrect Points

- 1 pt Missing/Incorrect CL

- 1 pt Missing/Incorrect LCL/UCL

✓ - 0 pts Correct

- 1 pt Incomplete interpretation or minor issues with interpretation

- 2.5 pts Incorrect Interpretation based on control chart

- 5 pts Missing Interpretation

Question 8

Bonus

3 / 0 pts

- 0 pts Did not Attempt or None Correct or did not label with rule

+ 1 pt Some Correct

✓ + 3 pts Mostly Correct

+ 5 pts All correct
1, 2, 3, 4, 8

EM420 Written Partial Review

Name: Ethan Ellis

Academic Term 26-1

Hour: G-1

EM420 Production and Operations Management

Written Partial Review

Version A

- 1) This is a 75-minute graded event worth 150 points.
- 2) This is a closed book and closed notes exam. You may use one single sided 3x5" index card and the provided reference book (DO NOT MARK IN IT!) and an authorized scientific calculator. No collaboration is allowed.
- 3) You may disassemble this document if you'd like. You may also use extra paper provided at the front of the classroom. Please ensure you staple the exam back together in the correct order with extra sheets labeled and at the back.
- 4) **SHOW ALL WORK!** Unsupported or missing work earns little or no partial credit.
- 5) **Rounding.** Show dollars and cents for any monetary answer (\$11.75). Show two significant digits beyond the decimal for percentage answers (11.75%) and at least two significant digits beyond the decimal for all other answers (0.13).
- 6) Upon completion, sign the acknowledgment statement below:

"I did not use any sources, nor did I receive any assistance while completing this WPR. This WPR will be released from academic security on 30OCT2025. Prior to that time, I may not discuss any aspects of this exam with anyone except an EM420 instructor."

Printed Name: Ethan Ellis Signature: E. Ellis Date Signed: 9/29/25

8) GRADING SUMMARY:

Section	Topic	Total Points Available	Total Points Earned
1	Situational Assessment	20	
2	Multiple Choice	25	
3	Ammo Plants	20	
4	Winter Jacket	20	
5	Field Kitchens	15	
6	Drive Thru	30	
7	Backpacks	20	
BONUS	Western Electric	5	
	Total	150	

Section I: Situational Assessment (No Partial Credit) [20 points]

Please choose the BEST problem-solving strategy, chart, or calculation that would aid in making a decision for the prompt. There is one best answer for each situation. No partial credit. Explanation is not necessary. You must write your answers on the lines below for grading

- 1A EVPI
- 1B Fault Tree Analysis
- 1C Pareto Analysis
- 1D FMEA
- 1E Markovian
- 1F Order Winner

Order Winner	Order Qualifier
EVPI	Hurwicz
Equal likelihood	Maximax
Maximin	Priority
Minimax Regret	Pareto Analysis
Quality Productivity Ratio	P Chart
X bar Chart	R Chart
C Chart	FMEA
Fault Tree Analysis	House Of Quality
Service Blueprint	Calling Population
FIFO	LIFO
General	Degenerate
Markovian	

- 1G Sarah, All:

1A [3 Points] A program manager wants to compute the expected monetary value of hiring a market research firm before making a major sourcing decision, to see whether additional information is worth the cost. Which calculation should they perform?

1B [3 Points] An aircraft mishap investigation team wants to model how combinations of electrical, hydraulic, and pilot-control failures could lead to loss of flight control. They want a logical diagram that starts with the top event (loss of control) & maps pathways downward to root causes.

1C [3 Points] A quality engineer at an Army depot notices that production of Humvee transmissions has been missing daily output targets for the last two weeks. Reviewing the data, they see eight different types of process failures, including torque wrench calibration errors, mislabeled parts etc. Management asks them to figure out which issues are causing the majority of lost throughput so they can direct maintenance to fix those first. What should they use?

1D [3 Points] An operations officer notices repeat breakdowns on forklifts. They want to systematically rank which failure means (hydraulic leak, steering failure, sensor malfunction) are most critical so they can focus limited maintenance funds on the highest risk items first.

1E [3 points] An ammo supply point (ASP) officer is modeling wait times for convoys arriving to pick up munitions before a training exercise. Historically, truck arrivals are random and independent, with interarrival times well-fit by an exponential distribution. The officer also confirms that service times (loading each truck) are memoryless and follow an exponential pattern. Which type of distribution matches service times?

1F [3 points] An Army Program Executive Office is evaluating two vendors for next-gen body armor. Both vendors meet the minimum ballistic protection and weight standards required to compete. However, Vendor A can deliver within 30 days while Vendor B's lead time is 90 days. Rapid fielding is the deciding factor for award. How can you classify rapid fielding?

1G [2 points] Please list your Group Graded Lab teammates' first names.

Section II: Multiple Choice (No Partial Credit) [25 points]

Choose the best answer. You must write your letter answers on the lines below for grading

- 2A _____ C
- 2B _____ A
- 2C _____ D
- 2D _____ A
- 2E _____ D

2A (5 Points) Consider an espresso stand with a single barista. Customers arrive at the stand at the rate of 28 per hour according to a Poisson distribution. Service times are exponentially distributed with a service rate of 35 customers per hour. The probability that there are more than 2 customers in the system is:

a. 0.128

b. 0.488

☒ c. 0.512

d. 0.640

$P > 2 \rightarrow$

$$1 - P(X \leq 2)$$

$$P = \frac{\lambda}{\mu}$$

$$P_0 = (1 - \rho) \cdot \rho^0 = 0.2$$

$$P_1 = (1 - \rho) \cdot \rho^1 = 0.16$$

$$P_2 = (1 - \rho) \cdot \rho^2 = 0.128$$

$$1 - 0.488 = 0.512$$

$$\lambda = \frac{28 \text{ cust}}{1 \text{ hr}}$$

$$\mu = \frac{35 \text{ cust}}{1 \text{ hr}}$$

2B (4 Points) A process that is in control as indicated by its Statistical Process Control chart indicates that some _____ in the process may be _____.

☒ a. variation, natural

b. variation, assignable

c. attributes, random

d. defects, measurable

2C (4 Points) Decision criteria are interrelated, often depend on the eye of the beholder and settings. _____ criteria is actually the same as the _____ criteria when $\alpha = 1$.

a. Hurwicz, EVPI

b. Minimin, Maximin

c. Minimax Regret, Hurwicz

☒ d. Maximax, Hurwicz

2D (5 Points) You are a quality engineer at a small-arms ammunition plant. Your team has measured the reliability of three critical machines in series that form a cartridge assembly line:

Case Feeder: $R_1 = 0.96$

Primer Insertor: $R_2 = 0.94$

Bullet Seater: $R_3 = 0.92$

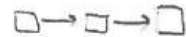
What is the reliability of the overall system (assume machines are arranged in series)?

☒ a. 0.831

b. 0.850

c. 0.900

d. 0.922



2E (5 Points) During a recent 4,000-hour production run, the primer inserter broke down 12 times. Each repair took on shift or 4 hours to complete on average. What was its availability during this period?

a. 0.88

b. 0.92

c. 0.95

☒ d. 0.99

$$SA = \frac{4000}{\frac{4000}{12} + 4} = 0.99$$

Section III: Ammo Plants [20 Points]

Problem 3 You are a commander managing two Army ammo plants which operate **320 days/year** at **10 hours/day**. Each unit is sold at the plant's listed price.

Item	Plant Alpha	Plant Bravo
Output (units/year)	1,120,000	1,050,000
Selling price (\$/unit)	\$38.00	\$40.00
Labor hours/year	480,000	420,000
Labor cost (\$/hour)	\$28.00	\$31.00
Material cost/year	\$18,900,000	\$19,800,000
Energy cost/year	\$2,450,000	\$2,200,000

3A (7.5 Points). Calculate the Total Productivity for each plant. Interpret your results in words

$$\text{Total Productivity} = \frac{\text{Output}}{\text{Total inputs}}$$

$$\text{Total Productivity}_A = \frac{1,120,000}{(480,000)(28) + 18,900,000 + 2,450,000} = .03219 \text{ or } 3.22\%$$

$$\text{Total Productivity}_B = \frac{1,050,000}{(420,000)(31) + 19,800,000 + 2,200,000} = .02998 \text{ or } 3.00\%$$

Plant Alpha produces more outputs per the cost of total inputs (3.22%) than does Plant Bravo (3.00%). This means that Plant A can more efficiently use their total production costs to create output than Plant B.

Material

$$18,900,000 \times .1 = 1,890,000$$

$$19,800,000 \times .1 = 1,980,000$$

$$1,120,000 \times .05 = 56,000$$

$$1,050,000 \times .05 = 52,500$$

Item	Plant Alpha	Plant Bravo
Output (units/year)	1,120,000	1,050,000
Selling price (\$/unit)	\$38.00	\$40.00
Labor hours/year	480,000	420,000
Labor cost (\$/hour)	\$28.00	\$31.00
Material cost/year	\$18,900,000	\$19,800,000
Energy cost/year	\$2,450,000	\$2,200,000

3B (12.5 Points). The Army can invest \$15,000 total:

Option 1 (\$7,500): +5% labor productivity (same hours, but effective output increases 5%).

Option 2 (\$7,500): -10% material cost.

You have the following options for investment:

- A. Apply option 1 to both plants
- B. Apply option 2 for both plants
- C. Apply Option 1 for Plant A & Option 2 for Plant B.
- D. Apply Option 1 for Plant B & Option 2 for Plant A.

Recommend which investment strategy the Army should choose and support with quantitative analysis.

Oris

$$TP_A = 3.22\%$$

$$TP_B = 3.00\%$$

	Plant A	Plant B
+5% labor productivity output	1,176,000	1,102,500
-10% Mat cost	17,010,000	17,820,000

$$A. TP_A = 1,176,000$$

$$\frac{(480,000)(28) + 18,900,000 + 2,450,000}{1,176,000} = .0338 = 3.38\%$$

$$TP_B = 1,102,500$$

$$\frac{(420,000)(31) + 19,800,000 + 2,200,000}{1,102,500} = .0315 = 3.15\%$$

$$B. TP_A = 1,120,000$$

$$\frac{(480,000)(28) + 17,010,000 + 2,450,000}{1,120,000} = .0340 = 3.40\%$$

$$TP_B = 1,050,000$$

$$\frac{(420,000)(31) + 17,820,000 + 2,200,000}{1,050,000} = .03177 = 3.18\%$$

I recommend choosing option B for both plants. Option 2 raises the total productivity ratio for Plant A higher by .02% versus option 1 (3.40% vs 3.38%) and Option 2 raises total productivity ratio for plant B higher by .03% versus option 1 (3.18% vs 3.15%).

Section IV: Winter Jacket [20 points]

You are part of a product design team tasked with developing a new winter jacket for Soldiers operating in cold-weather environments. Your product design team has created the following matrix to help you understand the relationship between customer and technical requirements. Use this diagram to answer the questions that follow about trade-offs and technical focus areas for the jacket design.

Customer Requirements

		Technical Requirements									
		Number of Pockets	Weight of filling	Number of Buttons	Reflective Marking	Weight of Lining	Weight of Outer	Length of Jacket	Has a hoodie	Has a collar	Number of Zippers
Easy to put on	5	-	-	⊖		-	-	-			
Not too heavy	9	-	-			-	-	-	-		-
Keeps body warm	4		⊕								
Breathability	7					-	-	-			
Visible in the dark	3				⊕						
Fashionable	5	+									
Protects from Rain	2							+	+		
Not too expensive	8				-				-		-
Easy to Wash	6	-	-	-	-						
Keeps neck warm	4								⊕	+	

A

B

4A (5 Points) Please explain the relationship in words between the customer requirement "Easy to Wash" and the technical requirement "number of buttons."

There is a negative relationship between the customer requirement "Easy to Wash" and the technical requirement "number of buttons". This means that as the number of buttons on the jacket increases, the jacket becomes more difficult to wash.

4B (5 Points) Identify one customer requirement that has weak or few relationships to the technical requirements. How might you redesign the technical approach to better address that requirement?

The visible in the dark customer requirement has few relationships to the technical requirements. I could redesign the technical approach to include number of reflective markers, size of reflective markers, or distance reflective markers can be seen.

4C (5 Points) What does A (the label for the orange numbers) represent? Interpret one of the numbers in reference to the customer requirement.

Label A means the value out of 10 the customer requirement holds to the customer. For example, the customer values a "Not too heavy" (9) jacket much more than a "protects from rain" (2) jacket, and would be more willing to sacrifice less protection from rain for a lighter jacket.

4D (5 Points) What else is typically included in a house of quality that is not shown? Hint: You could identify the area hidden by B on the diagram or any other missing pieces of the diagram. Think about the purpose of the house of quality.

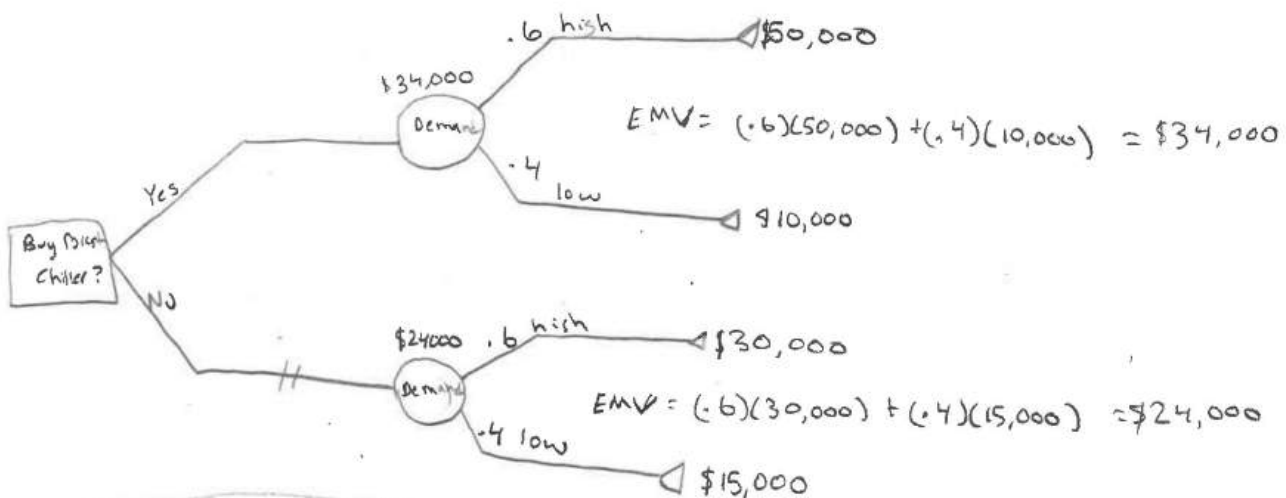
Usually, a house of quality contains how the current product compares to competitors products. This allows the producer to see what needs to be improved to distinguish their product from competitors, and evaluate what needs to be improved in reference to consumer needs.

Section V: Field Kitchens [15 Points]

You are the operations manager for a field kitchen deciding whether to invest in a new blast chiller.

- If you buy the blast chiller, there is a 60% chance demand is high, generating a profit of \$50,000, and a 40% chance demand is low, leading to a profit of \$10,000 (because the chiller is under-utilized).
- If you do not buy, you save the purchase cost but must rely on the old cooling method. In this case, there is a 60% chance of high demand, earning \$30,000, and a 40% chance of low demand, earning \$15,000.

5A (15 Points) Calculate the Expected Monetary Value (EMV) for each decision and explain the option you would choose, and the circumstance. Be sure to write all assumptions.



$$EMV_1 = \$34,000$$

$$EMV_2 = \$24,000$$

I would choose decision 1 to buy the blast chiller
Since it has a higher EMV at \$34,000.

Effective

$$S=1$$

Section VI: Drive Thru [30 Points]

The arrival rate of cars at a drive-thru restaurant is 20 cars per hour. The service rate is 18 cars per hour. The available space for cars in the drive-thru is restricted to 6 cars only. Otherwise, the queue goes into the street and cars cannot queue. The arrival rate and service rate follow the Poisson distribution. Identify the type of queue and then determine the performance measures of this queue. Be sure to interpret each answer in words!

$$\lambda = \frac{20 \text{ cars}}{1 \text{ hr}} \quad \mu = \frac{18 \text{ cars}}{1 \text{ hr}}$$

6A [10 points] What is the probability that the system is full?

$$M/M/1 : 6 / \infty / \text{FIFO}$$

$$(P > 6) \Rightarrow (1 - P < 6)$$

$$\rho = \frac{20}{18} = 1.11$$

There is an 100% probability that the system is full since utilization is greater than 1. Arrivals outpace the service rate.

$$P_0 = \left(\frac{1-\rho}{1-\rho^7} \right) \rho^0 = \left(\frac{1-1.11}{1-(1.11)^7} \right) (1) = .102$$

$$P_1 = (.1022)(1.11)^1 = .113$$

$$P_2 = (.1022)(1.11)^2 = .126$$

$$P_3 = (.1022)(1.11)^3 = .139$$

$$P_4 = (.1022)(1.11)^4 = .154$$

$$P_5 = (.1022)(1.11)^5 = .170$$

Over 100%

6B [10 points] How many customers are lost per hour on average?

There are 2 customers lost per hour on average. Since the arrival rate is 20 cars per hour but the server can only serve 18 cars per hour, 2 customers are consistently not served per hour, and are therefore lost.

6C [10 points] What is the effective arrival rate? What does that mean in words?

$$\bar{\lambda} = 20(1 - P_m)$$

$$\bar{\lambda} = 18$$

The effective arrival rate is the rate in which the queue can be effective. In this case, the arrival rate must be 18 cars per hour or less so utilization is less than or equal to 1.

Section VII: Backpacks [20 Points]

You have recently retired from the military and are working for a tactical backpack company and much count the blemishes on each backpack (stitch undone, loose thread, bleach spot etc.). A sample includes one backpack from a batch of 50 backpacks. The following samples are collected at random in a one month period. Assume 95% confidence.

Sample	Defects
1	12
2	6
3	7
4	5
5	2

7A [5 points] What type of control chart is best used in this situation.

$\bar{c}$ - chart.

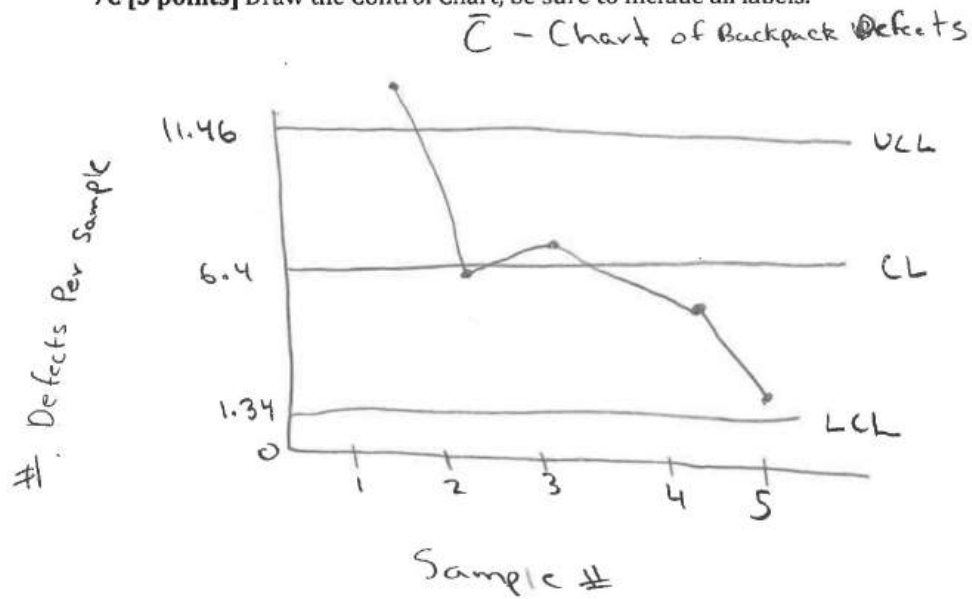
7B [5 points] Calculate the UCL, UCL and CL

$$\underline{CL} = \frac{\text{total \# defects}}{\text{total \# samples}} = \frac{32}{5} = \underline{6.4}$$

$$\underline{UCL} = 6.4 + 2\sqrt{6.4} = \underline{11.46}$$

$$\underline{LCL} = 6.4 - 2\sqrt{6.4} = \underline{1.34}$$

7C [5 points] Draw the Control Chart, be sure to include all labels.

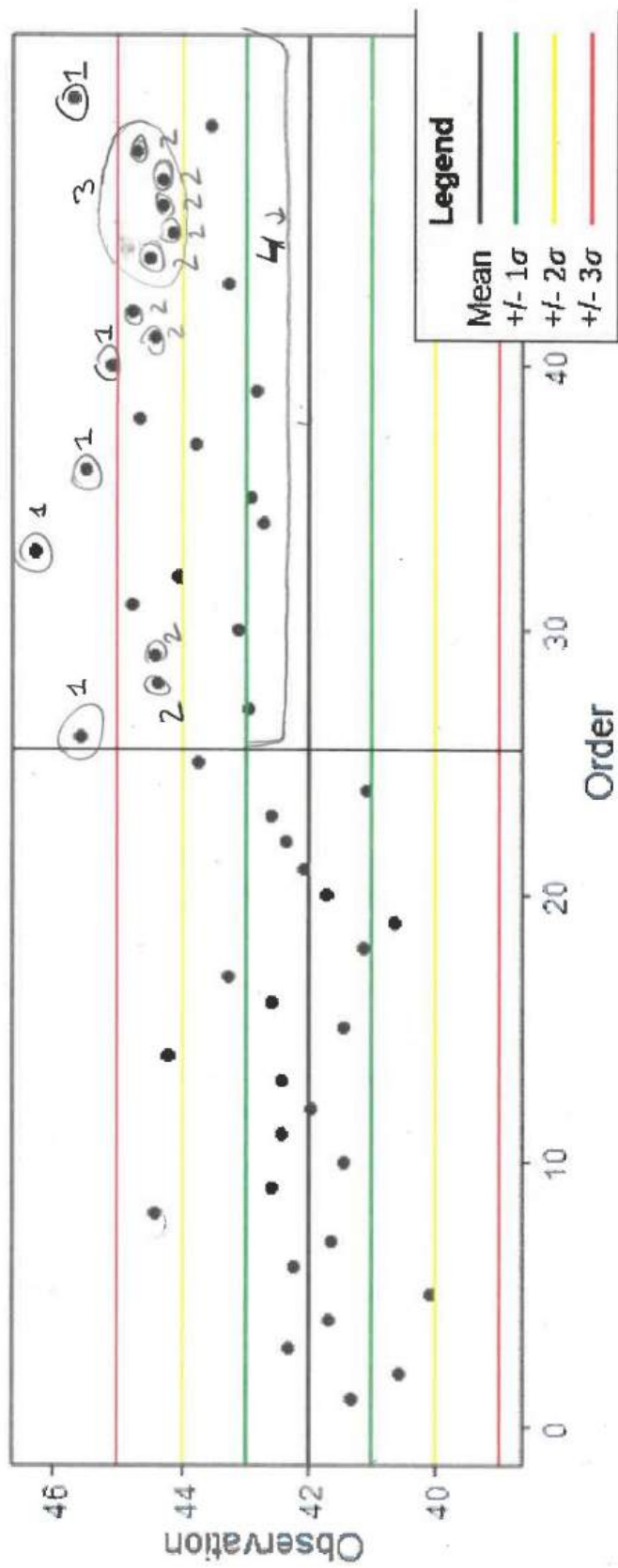


7D [5 points] Please interpret your control chart in words. Use context from the prompt.

This process is out of control. Since there is a sample with a backpack with a number of blemishes outside our UCL, we must classify our backpack blemishes as too high to be in control.

Bonus (5 points): Many types of rules can be considered to determine whether a process is out of control. Use the Western Electric Rules to determine if the process is out of control (if any rule holds true the process is out of control). Indicate which points break which rule by circling and labeling them with the rule number broken.

1. The most recent point plots outside one of the 3-sigma control limits.
2. Two of the three most recent points plot outside and on the same side as one of the 2-sigma control limits.
3. Four of the five most recent points plot outside and on the same side as one of the 1-sigma control limits.
4. Eight out of the last eight points plot on the same side of the center line, or target value.
5. Six points in a row increasing or decreasing.
6. Fifteen points in a row within one sigma.
7. Fourteen points in a row alternating direction.
8. Eight points in a row outside one sigma.



Fair-to-Good Student Work

WPR1

● Graded

Student

Terrence Carter

Total Points

104.5 / 120 pts

Question 1

Question 1: Situational Assessment

17 / 20 pts

1.1 — 1A

3 / 3 pts

✓ - 0 pts Correct

- 1.5 pts Incorrect, however, response is relevant to decision-making under risk/uncertainty

- 2.5 pts Incorrect

- 3 pts Blank

1.2 — 1B

3 / 3 pts

✓ - 0 pts Correct

- 1 pt Close, but incorrect

- 2.5 pts Incorrect

- 3 pts Blank

1.3 — 1C

0.5 / 3 pts

- 0 pts Correct

- 1 pt Incorrect, but it is a quality/reliability tool that also prioritizes failure modes

- 1.5 pts Incorrect, but they are quality control charts, at least within the SPC domain

✓ - 2.5 pts Incorrect

- 3 pts Blank

1.4 — 1D

2.5 / 3 pts

- 0 pts Correct

✓ - 0.5 pts Incorrect, but very plausible misinterpretation since both prioritize causes

- 1 pt Incorrect, but still in reliability/failure analysis family

- 2.5 pts Incorrect

- 3 pts Blank

1.5 — 1E

3 / 3 pts

✓ - 0 pts Correct

- 1.5 pts Conceptual Mislabel; chose different distribution indicator

- 2.5 pts Incorrect

- 3 pts Blank

1.6

1F

3 / 3 pts

✓ - 0 pts Correct

- 0.5 pts Incorrect, but in the same competitive strategy family, so understandable misinterpretation

- 1.5 pts Incorrect, but indicates some degree of prioritization/importance (was used in class in the context of queuing discipline)

- 2.5 pts Incorrect

- 3 pts Blank

1.7

1G

2 / 2 pts

✓ - 0 pts Correct

- 2 pts Incorrect

Question 2

Question 2: Multiple Choice

19 / 25 pts

2.1 — 2A

5 / 5 pts

✓ - 0 pts Correct

- 4 pts Incorrect

- 0.5 pts Failure to Follow Instructions

- 1.5 pts Calculated P2 rather than $P_n > 2$

- 1 pt Calculated $P_n < 2$

2.2 — 2B

1 / 5 pts

- 0 pts Correct

- 1.5 pts Partially correct.

✓ - 4 pts Incorrect

2.3 — 2C

3 / 5 pts

- 0 pts Correct

✓ - 2 pts Incorrect, but identified connection of alpha value to Hurwicz

- 4 pts Incorrect

2.4 — 2D

5 / 5 pts

✓ - 0 pts Correct

- 5 pts [Click here to replace this description.](#)

- 2.5 pts Failure to Follow Instructions

2.5 — 2E

5 / 5 pts

✓ - 0 pts Correct

- 4 pts Incorrect

3.1 — 3A

6 / 7.5 pts

- 0 pts Correct

Plant A -Either:

0.032 Units/\$ or 1.22 (dmnl)

Plant B - Either:

0.030 units/\$ or 1.20 (dmnl)

Interpretation:

A>B and why

✓ - 0.5 pts Missing/Incorrect Units

✓ - 0.5 pts Incorrect Plant A Calculation

Plant A Productivity:

Method 1 (units per dollar):

Output/Input = 1,120,000 units ÷ [(480,000 labor hrs/yr × \$28/hr) + \$18,900,000/yr + \$2,450,000/yr]

= 1,120,000 units ÷ \$34,790,000

= 0.032 units per dollar

Method 2 (value of output per dollar input):

(1,120,000 units × \$38/unit) ÷ \$34,790,000

= \$42,560,000 ÷ \$34,790,000

= 1.22 (dimensionless ratio)

✓ - 0.5 pts Incorrect Plant B Calculation

Plant B Productivity:

Method 1 (units per dollar):

Output/Input = 1,050,000 units ÷ [(420,000 labor hrs/yr × \$31/hr) + \$19,800,000/yr + \$2,200,000/yr]

= 1,050,000 units ÷ \$34,820,000

= 0.030 units per dollar

Method 2 (value of output per dollar input):

(1,050,000 units × \$40/unit) ÷ \$34,820,000

= \$42,000,000 ÷ \$34,820,000

= 1.21 (dimensionless ratio)

- 2.5 pts No TFP Plant A

- 2.5 pts No TFP Plant B

- 1 pt TFP Incorrectly Calculated

- 0.5 pts Incorrect/Incomplete Interpretation

Follows

- 1.5 pts Incorrect/Incomplete Interpretation

Does not follow or does not address question properly

- 2.5 pts No Interpretation

- **0 pts** Correct Recommendation; Logic Follows or Sufficiently Rigorous Quantitative Support
- **2.5 pts** Correct Recommendation, Logic Does not Follow or Insufficient Quantitative Support
- **5 pts** Incorrect Recommendation, Logic Follows or Sufficiently Rigorous Quantitative Support
- **7.5 pts** Incorrect/No Recommendation, Logic Does not Follow or Work is insufficient to arrive at conclusion
- **12.5 pts** No Recommendation or Incorrect Guess with no/little support

✓ - **1 pt** You arrived at the correct results, but then misinterpreted the question or your outcomes. It was asking both Plant A and Plant B, not Both option 1 and option 2. Numerical answer supports that you understood the question.

Question 4

Question 4: Winter Jacket

20 / 20 pts

4.1 — 4A

5 / 5 pts

✓ - 0 pts Correct

- 0.5 pts Minor error in explanation

- 1 pt Did not explain that as "Number of Buttons" Increases, "Ease of Wash" Decreases (Negative or Inverse Relationship)

- 3 pts Insufficient/Incorrect Response

- 5 pts Did not answer

4.2 — 4B

5 / 5 pts

✓ - 0 pts Correct

- 0.5 pts Minor Error in explanation

- 1 pt Incorrectly identified customer requirement vs technical requirement

- 1 pt Identified customer requirement with strong negative or strong positive relationship to technical requirement

- 1 pt Identified a customer requirement with large number of relationships

- 2 pts Weak/Missing Explanation

- 5 pts Did not answer the question

4.3 — 4C

5 / 5 pts

✓ - 0 pts Correct

- 0.75 pts Insufficient/Unclear interpretation of the Customer Importance or Incorrect Interpretation of the values (Higher is better)

- 1.5 pts Did not interpret or incorrectly interpreted in reference to customer requirement

- 2.5 pts Incorrect/Missing interpretation of A

- 5 pts Missing response

4.4 — 4D

5 / 5 pts

✓ - 0 pts Correct

- 1 pt Incomplete/Incorrect Response or Does not consider other components of HOQ

- 5 pts Missing Response

Question 5

Question 5: Field Kitchens

15 / 15 pts

5.1 5A

15 / 15 pts

✓ - 0 pts Correct

- 0.5 pts Units

- 1 pt Minor Math Error

- 1 pt Incorrect/Incomplete EMV for Blast Chiller

- 1 pt Incorrect/Incomplete EMV for Old Cooling Method

- 1 pt Incorrect Interpretation
Follows

- 5 pts Missing EMV for Blast Chiller

- 5 pts Missing EMV for Old Cooling Method

- 5 pts Incorrect/Missing Interpretation
Does not Follow

- 2.5 pts Provided explanation, but did not provide recommended course of action

Question 6

Question 6: Drive Thru

0 / 0 pts

6.1 — 6A

0 / 0 pts

– 0 pts Correct

– 1 pt Minor Math/Formulation Error

✓ – 2.5 pts Did not identify solution as P6
Probability that there are 6 vehicles in the system = probability that the system is full

– 2.5 pts Used incorrect Formulation of P6
Should be M/M/1/6 Kendall Notation, resulting in:
 $P_6 = ((1-\rho)/(1-\rho^{(6+1)})) \cdot \rho^6$

✓ – 2.5 pts Did not arrive at solution as a probability (probabilities cannot be negative or exceed 100%) or did not sufficiently demonstrate understanding of tested concept

– 10 pts Did not answer question or showed insufficient work to demonstrate understanding of the subject

– 1 pt Should use 6 for M and n, resulting in incorrect response. Should be 19.17%

– 0.5 pts Transcription Error

6.2 — 6B

0 / 0 pts

– 0 pts Correct

– 0 pts Incorrect, but follows

– 1 pt Incorrect; follows with some minor error in math or logic

✓ – 5 pts Incorrect; answer follows some logic

– 7.5 pts Incorrect; answer does not follow logic or numbers do not appear to have justification

– 2.5 pts Correct (or nearly correct) response, but work does not adequately demonstrate how you arrived at answer or arrived at correct (or nearly correct) answer through incorrect means

– 10 pts Missing Response

– 0 pts Correct

✓ – 0 pts Incorrect effective arrival rate, but follows from previous

– 2.5 pts Incorrect/Incomplete Numerical Solution;

– 5 pts Formula only

– 7.5 pts No numerical Solution or entirely incorrect approach or work shown/not shown does not demonstrate understanding of core concept

– 2.5 pts No/insufficient/incorrect interpretation

– 1 pt Minor math error or incorrect input
examples include incorrect λ (should be 20 cars/hr), incorrect P_6 (should be 0.19, but may follow from previous question)

– 0.5 pts Units

✓ – 1 pt Minor issues with interpretation

Question 7

Question 7: Backpacks

15 / 20 pts

7.1 — 7A

2 / 5 pts

- 0 pts Correct

✓ - 3 pts Correctly identified as attribute, but misidentified as p-chart (should be c chart)

- 5 pts Incorrect

7.2 — 7B

4 / 5 pts

- 0 pts Correct

✓ - 0 pts Incorrect Chart, but Follows

✓ - 1 pt Incorrect Z Values
95% -> $z=1.96$ or ~ 2

- 1.5 pts Incorrect CL

If C Chart, should be $(12+6+7+5+2)/5 = 6.4$ If (incorrectly) using P Chart, should still be $(12/50+6/50+7/50+5/50+2/50)/5 = 0.128$

- 1 pt LCL Cannot be Negative

- 1 pt Incorrect CL

Should be $\bar{C}$

- 1.5 pts Used Incorrect Sigma Values

Should be $\sqrt{\bar{C}}$

- 3 pts Incomplete Calculations

- 5 pts No/Insufficient response

- 1.5 pts Correct CL (6.4) for a C chart, but then proceeded to convert to incorrect P chart CL and then use P chart UCL and LCL

- 4 pts No CL, incorrect/Incomplete UCL and LCL formulations

- 0.5 pts Minor Math Error

- 3 pts Incorrect UCL and LCL

If C Chart, should be $\bar{C} \pm z\sqrt{\bar{C}}$ If (incorrectly) P Chart, should be $\bar{P} \pm z\sqrt{\bar{P}(1-\bar{P})/n}$

7.3 — 7C

5 / 5 pts

✓ - 0 pts Correct or Follows

- 1 pt Missing/Incorrect Labels

- 1 pt Missing/Incorrect Axes or Axis Labels

- 1 pt Missing/Incorrect Points

- 1 pt Missing/Incorrect CL

- 1 pt Missing/Incorrect LCL/UCL

– 0 pts Correct

✓ – 1 pt Incomplete interpretation or minor issues with interpretation

– 2.5 pts Incorrect Interpretation based on control chart

– 5 pts Missing Interpretation

Question 8

Bonus

1 / 0 pts

– 0 pts Did not Attempt or None Correct or did not label with rule

✓ + 1 pt Some Correct

+ 3 pts Mostly Correct

+ 5 pts All correct
1, 2, 3, 4, 8

EM420 Written Partial Review

Name: THREWS CANTAR

Academic Term 26-1

Hour: G1**EM420 Production and Operations Management****Written Partial Review****Version A**

- 1) This is a 75-minute graded event worth 150 points.
- 2) This is a closed book and closed notes exam. You may use one single sided 3x5" index card and the provided reference book (DO NOT MARK IN IT!) and an authorized scientific calculator. No collaboration is allowed.
- 3) You may disassemble this document if you'd like. You may also use extra paper provided at the front of the classroom. Please ensure you staple the exam back together in the correct order with extra sheets labeled and at the back.
- 4) **SHOW ALL WORK!** Unsupported or missing work earns little or no partial credit.
- 5) **Rounding.** Show dollars and cents for any monetary answer (\$11.75). Show two significant digits beyond the decimal for percentage answers (11.75%) and at least two significant digits beyond the decimal for all other answers (0.13).
- 6) Upon completion, sign the acknowledgment statement below:

"I did not use any sources, nor did I receive any assistance while completing this WPR. This WPR will be released from academic security on 30CT2025. Prior to that time, I may not discuss any aspects of this exam with anyone except an EM420 instructor."

Printed Name: THREWS CANTAR Signature: [Signature] Date Signed: 3 Oct 25

8) GRADING SUMMARY:

Section	Topic	Total Points Available	Total Points Earned
1	Situational Assessment	20	
2	Multiple Choice	25	
3	Ammo Plants	20	
4	Winter Jacket	20	
5	Field Kitchens	15	
6	Drive Thru	30	
7	Backpacks	20	
BONUS	Western Electric	5	
	Total	150	

Section I: Situational Assessment (No Partial Credit) [20 points]

Please choose the BEST problem-solving strategy, chart, or calculation that would aid in making a decision for the prompt. There is one best answer for each situation. No partial credit. Explanation is not necessary. You must write your answers on the lines below for grading

- 1A EVPI
- 1B FAULT TREE ANALYSIS
- 1C SERVICE BLUEPRINT
- 1D PARETO ANALYSIS
- 1E MARKOVIAN
- 1F ORDER WINNER
- 1G JASAM or LOGAN or SEN or JAMISON

Order Winner	Order Qualifier
EVPI	Hurwicz
Equal likelihood	Maximax
Maximin	Priority
Minimax Regret	Pareto Analysis
Quality Productivity Ratio	P Chart
X bar Chart	R Chart
C Chart	FMEA
Fault Tree Analysis	House Of Quality
Service Blueprint	Calling Population
FIFO	LIFO
General	Degenerate
Markovian	

1A [3 Points] A program manager wants to compute the expected monetary value of hiring a market research firm before making a major sourcing decision, to see whether additional information is worth the cost. Which calculation should they perform? EVPI

1B [3 Points] An aircraft mishap investigation team wants to model how combinations of electrical, hydraulic, and pilot-control failures could lead to loss of flight control. They want a logical diagram that starts with the top event (loss of control) & maps pathways downward to root causes.

1C [3 Points] A quality engineer at an Army depot notices that production of Humvee transmissions has been missing daily output targets for the last two weeks. Reviewing the data, they see eight different types of process failures, including torque wrench calibration errors, mislabeled parts etc. Management asks them to figure out which issues are causing the majority of lost throughput so they can direct maintenance to fix those first. What should they use?

1D [3 Points] An operations officer notices repeat breakdowns on forklifts. They want to systematically rank which failure means (hydraulic leak, steering failure, sensor malfunction) are most critical so they can focus limited maintenance funds on the highest risk items first.

1E [3 points] An ammo supply point (ASP) officer is modeling wait times for convoys arriving to pick up munitions before a training exercise. Historically, truck arrivals are random and independent, with interarrival times well-fit by an exponential distribution. The officer also confirms that service times (loading each truck) are memoryless and follow an exponential pattern. Which type of distribution matches service times?

1F [3 points] An Army Program Executive Office is evaluating two vendors for next-gen body armor. Both vendors meet the minimum ballistic protection and weight standards required to compete. However, Vendor A can deliver within 30 days while Vendor B's lead time is 90 days. Rapid fielding is the deciding factor for award. How can you classify rapid fielding?

1G [2 points] Please list your Group Graded Lab teammates' first names.

Section II: Multiple Choice (No Partial Credit) [25 points]

Choose the best answer. You must write your letter answers on the lines below for grading

2A _____ c _____

2B _____ d _____

2C _____ c _____

2D _____ a _____

2E _____ d _____

2A (5 Points) Consider an espresso stand with a single barista. Customers arrive at the stand at the rate of 28 per hour according to a Poisson distribution. Service times are exponentially distributed with a service rate of 35 customers per hour. The probability that there are more than 2 customers in the system is:

- a. 0.128 b. 0.488 c. 0.512 d. 0.640

2B (4 Points) A process that is in control as indicated by its Statistical Process Control chart indicates that some _____ in the process may be _____.

- a. variation, natural b. variation, assignable c. attributes, random d. defects, measurable

2C (4 Points) Decision criteria are interrelated, often depend on the eye of the beholder and settings. _____ criteria is actually the same as the _____ criteria when $\alpha = 1$.

- a. Hurwicz, EVPI b. Minimin, Maximin c. Minimax Regret, Hurwicz d. Maximax, Hurwicz

2D (5 Points) You are a quality engineer at a small-arms ammunition plant. Your team has measured the reliability of three critical machines in series that form a cartridge assembly line:

Case Feeder: $R_1=0.96$ Primer Inserter: $R_2=0.94$ Bullet Seater: $R_3=0.92$

What is the reliability of the overall system (assume machines are arranged in series)?

- a. 0.831 b. 0.850 c. 0.900 d. 0.922

2E (5 Points) During a recent 4,000-hour production run, the primer inserter broke down 12 times. Each repair took on shift or 4 hours to complete on average. What was its availability during this period?

- a. 0.88 b. 0.92 c. 0.95 d. 0.99

Section III: Ammo Plants [20 Points]

Problem 3 You are a commander managing two Army ammo plants which operate **320 days/year** at **10 hours/day**. Each unit is sold at the plant's listed price.

Item	Plant Alpha	Plant Bravo
Output (units/year)	1,120,000	1,050,000
Selling price (\$/unit)	\$38.00	\$40.00
Labor hours/year	480,000	420,000
Labor cost (\$/hour)	\$28.00	\$31.00
Material cost/year	\$18,900,000	\$19,800,000
Energy cost/year	\$2,450,000	\$2,200,000

3A (7.5 Points). Calculate the Total Productivity for each plant. Interpret your results in words

$$\text{PLANT ALPHA} \rightarrow \frac{1,120,000}{\$38 + \$480,000 + \$28 + \$18,900,000 + \$2,450,000} \rightarrow 0.051365308$$

$$\text{PLANT BRAVO} \rightarrow \frac{1,050,000}{\$40 + \$420,000 + \$31 + \$19,800,000 + \$2,200,000} \rightarrow 0.046833036$$

PLANT ALPHA HAS A HIGHER PRODUCTIVITY RATE THAN PLANT BRAVO WHILE MAINTAINING THE SAME WORK SCHEDULE. A POSSIBLE REASON FOR THIS ARE A BETTER MATERIAL VENDOR AND LOWER TAXES ALLOW FOR CHEAPER SELLING PRICES / LABOR COST AND MATERIAL COST.

OC

Item	Plant Alpha 0.513	Plant Bravo 0.468
Output (units/year)	1,120,000	1,050,000
Selling price (\$/unit)	\$38.00	\$40.00
Labor hours/year	480,000	420,000
Labor cost (\$/hour)	\$28.00	\$31.00
Material cost/year	\$18,900,000	\$19,800,000
Energy cost/year	\$2,450,000	\$2,200,000

3B (12.5 Points). The Army can invest \$15,000 total:

Option 1 (\$7,500): +5% labor productivity (same hours, but effective output increases 5%).

Option 2 (\$7,500): -10% material cost.

You have the following options for investment:

- A. Apply option 1 to both plants
- B. Apply option 2 for both plants
- C. Apply Option 1 for Plant A & Option 2 for Plant B.
- D. Apply Option 1 for Plant B & Option 2 for Plant A.

Recommend which investment strategy the Army should choose and support with quantitative analysis.

most
likely
for
both

PLANT BRAVO

$$\begin{aligned}
 &10\% \text{ MATERIAL COST} \rightarrow \frac{1050000}{40 + 420000 + 31 + 17820000 + 2200000} \rightarrow 0.514 \\
 &15\% \text{ LABOR} \rightarrow \frac{1102500}{40 + 420000 + 31 + 19800000 + 2200000} \rightarrow 0.49174689 \\
 &\text{BOTH} \rightarrow \frac{1102500}{40 + 420000 + 31 + 17820000 + 2200000} \rightarrow 0.53938169
 \end{aligned}$$

PLANT ALPHA

$$\begin{aligned}
 &10\% \text{ MAT} \rightarrow \frac{1120000}{38 + 480000 + 28 + 17010000 + 2450000} \rightarrow 0.5616832 \\
 &15\% \text{ LABOR} \rightarrow \frac{1176000}{38 + 480000 + 28 + 18900000 + 2450000} \rightarrow 0.53870657 \\
 &\text{BOTH} \rightarrow \frac{1176000}{38 + 480000 + 28 + 17010000 + 2450000} \rightarrow 0.58476256
 \end{aligned}$$

Section IV: Winter Jacket [20 points]

You are part of a product design team tasked with developing a new winter jacket for Soldiers operating in cold-weather environments. Your product design team has created the following matrix to help you understand the relationship between customer and technical requirements. Use this diagram to answer the questions that follow about trade-offs and technical focus areas for the jacket design.

		Technical Requirements								
		Number of Pockets	Weight of filling	Number of Buttons	Reflective Marking	Weight of Lining	Weight of Outer	Length of Jacket	Has a hoodie	Has a collar
Customer Requirements	Easy to put on	5	-	⊖		-	-	-		
	Not too heavy	9	-	-		-	-	-	-	-
	Keeps body warm	4	⊕							
	Breathability	7				-	-	-		
	Visible in the dark	3			⊕					
	Fashionable	5	+							
	Protects from Rain	2						+	+	
	Not too expensive	8			-				-	-
	Easy to Wash	6	-	-	-					
	Keeps neck warm	4							⊕	+
		B								

A

4A (5 Points) Please explain the relationship in words between the customer requirement "Easy to Wash" and the technical requirement "number of buttons."

THE NEGATIVE MEANS THE NUMBER OF BUTTONS NEGATIVELY AFFECTS THE EASE OF WASHING THE WINTER JACKET,

4B (5 Points) Identify one customer requirement that has weak or few relationships to the technical requirements. How might you redesign the technical approach to better address that requirement?

FASHIONABLE ; I WOULD LIKE MORE TOWARDS SOMETHING SUCH AS SYMMETRICAL APPEARANCE OR SOMETHING TO DO WITH THE STRUCTURE OF THE JACKET.

4C (5 Points) What does A (the label for the orange numbers) represent? Interpret one of the numbers in reference to the customer requirement.

THE VALUE OF THE CUS REQ , THE MOST IMPORTANT ASPECT IS THE JACKET IS NOT TOO HEAVY SINCE IT IS RATED HIGHEST W/ A 9.

4D (5 Points) What else is typically included in a house of quality that is not shown? Hint: You could identify the area hidden by B on the diagram or any other missing pieces of the diagram. Think about the purpose of the house of quality.

B HOUSES HOW THE ITEM COMPARES TO COMPETITORS IN THE VARIABLE OF A, AB, B AND A X W/ A LINE GOING DOWN REPRESENTING THE PRODUCT YOU ARE PRODUCING. BELOW THE CENTER SQUARE SHOWS THE DIMENSIONS OF THE JACKET.

Section V: Field Kitchens [15 Points]

You are the operations manager for a field kitchen deciding whether to invest in a new blast chiller.

- If you buy the blast chiller, there is a 60% chance demand is high, generating a profit of \$50,000, and a 40% chance demand is low, leading to a profit of \$10,000 (because the chiller is under-utilized).
- If you do not buy, you save the purchase cost but must rely on the old cooling method. In this case, there is a 60% chance of high demand, earning \$30,000, and a 40% chance of low demand, earning \$15,000.

5A (15 Points) Calculate the Expected Monetary Value (EMV) for each decision and explain the option you would choose, and the circumstance. Be sure to write all assumptions.

$$EV : \sum \text{PROB} (\text{STATE}) \cdot \text{PAVoff} : \text{CHOOSE MAX}$$

$$\text{BUY} \quad .60 (50,000) + .40 (10,000) \rightarrow \$34,000$$

$$\text{DON'T BUY} \quad .60 (30,000) + .40 (15,000) \rightarrow \$24,000$$

I WOULD CHOOSE TO BUY THE BLAST CHILLER BECAUSE
THE EXPECTED VALUE OF BUYING IS \$34,000 PROFIT
WHILE THE LATTER OPTION TO NOT BUY IS \$24,000 PROFIT
A \$10,000 MARGIN OF DIFFERENCE IN FAVOR OF BUYING

Section VI: Drive Thru [30 Points]

The arrival rate of cars at a drive-thru restaurant is 20 cars per hour. The service rate is 18 cars per hour. The available space for cars in the drive-thru is restricted to 6 cars only. Otherwise, the queue goes into the street and cars cannot queue. The arrival rate and service rate follow the Poisson distribution. Identify the type of queue and then determine the performance measures of this queue. *Be sure to interpret each answer in words!*

FINISH
QUE

6A [10 points] What is the probability that the system is full?

M/M/1 / U / ∞ / FIFO

ARRIVE 20 CARS PER HR

SERVICE 18 CARS PER HR

2 CARS IN QUEUE 1 HR

4 CARS IN QUEUE 2 HR

6 CARS IN QUEUE 3 HR

20/18

1.1 → 111 %

6B [10 points] How many customers are lost per hour on average?

2 CARS PER HOUR THAT EXPONENTIALLY INCREASES PER HOUR

6C [10 points] What is the effective arrival rate? What does that mean in words?

$$\tilde{\lambda} = \lambda (1 - p_m)$$

$$20 (1 - 0.11)$$

22.2 CARS PER HOUR

→ NO CLUE WHAT PM MEANS

THE EFFECTIVE ARRIVAL RATE IS THE ACTUAL AMOUNT OF CARS ARRIVING AT THE WINDOW AND MAKING IT THROUGH

Section VII: Backpacks [20 Points]

You have recently retired from the military and are working for a tactical backpack company and must count the blemishes on each backpack (stitch undone, loose thread, bleach spot etc.). A sample includes one backpack from a batch of 50 backpacks. The following samples are collected at random in a one month period. Assume 95% confidence.

Sample	Defects
1	12
2	6
3	7
4	5
5	2

7A [5 points] What type of control chart is best used in this situation.

P - CHART

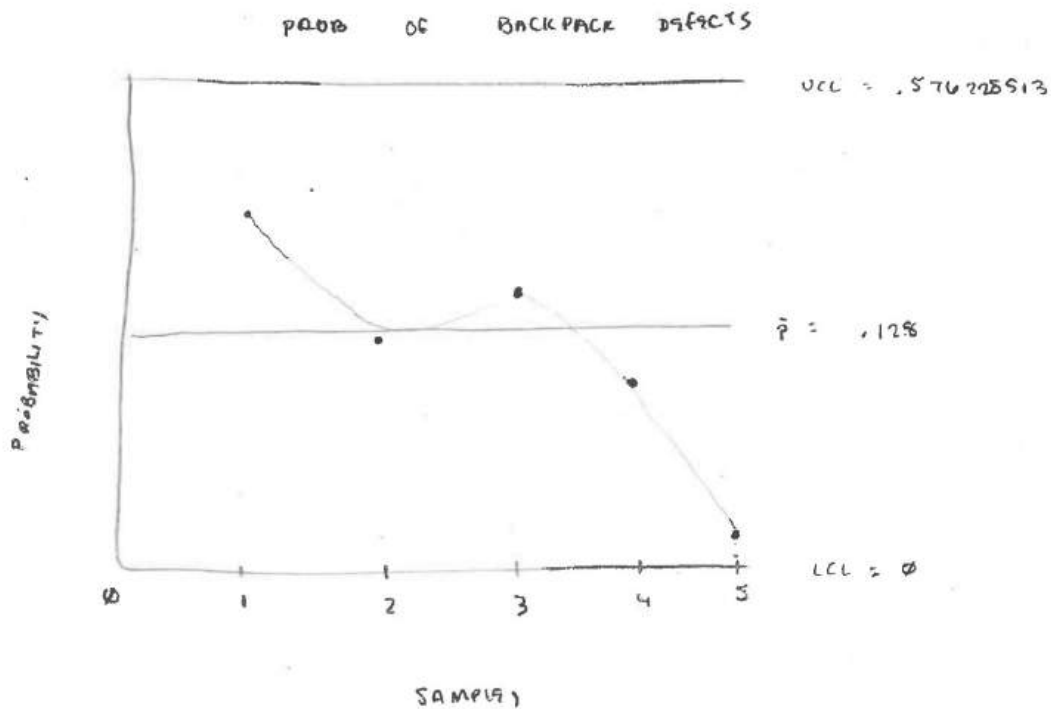
7B [5 points] Calculate the UCL, ^{LCL}UCL and CL

$$UCL \rightarrow \bar{p} + 3\sqrt{\bar{p}(1-\bar{p})/n} \rightarrow 0.428 + 3\sqrt{0.428(1-0.428)/5} \rightarrow 0.576228913$$

$$LCL \rightarrow \bar{p} - 3\sqrt{\bar{p}(1-\bar{p})/n} \rightarrow 0.428 - 3\sqrt{0.428(1-0.428)/5} \rightarrow -0.2$$

$$CL \rightarrow \bar{p} = \frac{\sum \text{DEFECTIVES}}{\sum n \text{ ALL OVER SAMPLES}} = \frac{32}{250} = 0.128$$

7C [5 points] Draw the Control Chart, be sure to include all labels.

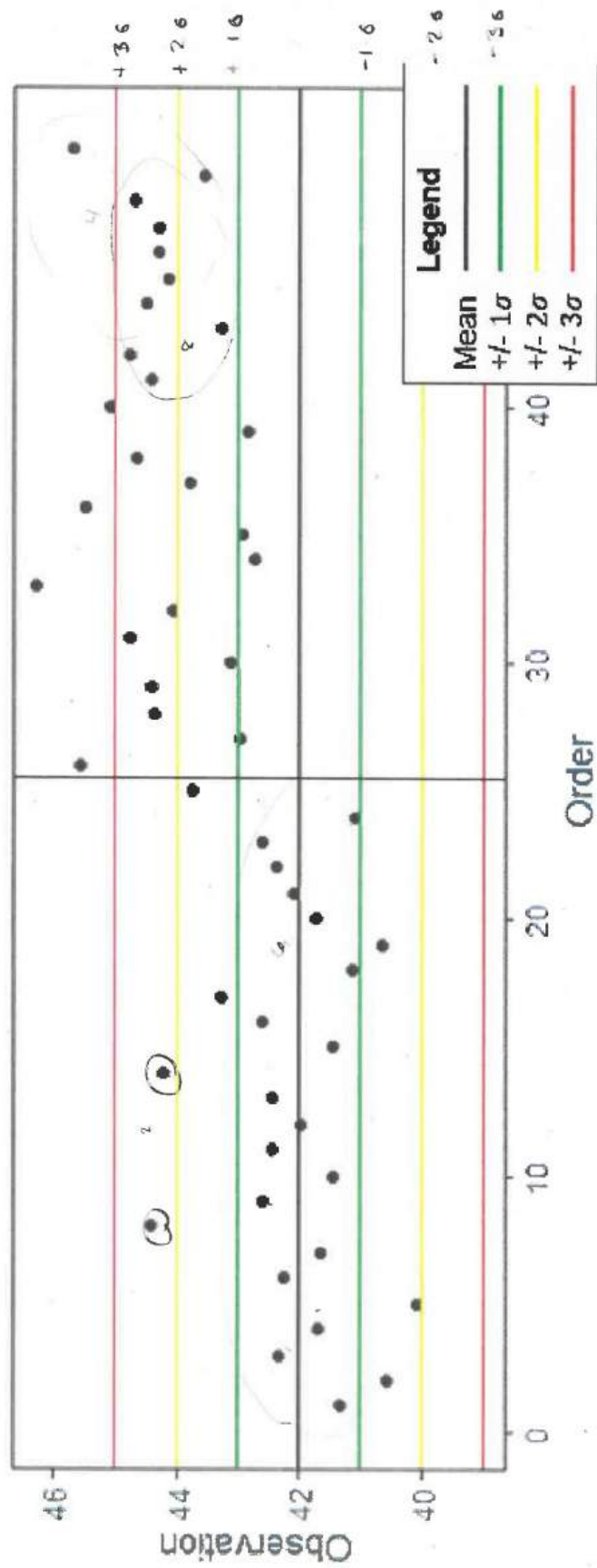


7D [5 points] Please interpret your control chart in words. Use context from the prompt.

THE PROBABILITY OF DEFECTS DECREASES WITH EVERY NEW SAMPLE MADE. THE ONLY OUTLIER WOULD BE SAMPLE 1, SO I WOULD RERUN THIS TEST TO SEE IF THE SAME RESULT OCCURS.

Bonus (5 points): Many types of rules can be considered to determine whether a process is out of control. Use the Western Electric Rules to determine if the process is out of control (if any rule holds true the process is out of control). Indicate which points break which rule by circling and labeling them with the rule number broken.

1. The most recent point plots outside one of the 3-sigma control limits.
2. Two of the three most recent points plot outside and on the same side as one of the 2-sigma control limits.
3. Four of the five most recent points plot outside and on the same side as one of the 1-sigma control limits.
4. Eight out of the last eight points plot on the same side of the center line, or target value.
5. Six points in a row increasing or decreasing.
6. Fifteen points in a row within one sigma.
7. Fourteen points in a row alternating direction.
8. Eight points in a row outside one sigma.



Poor Student Work

WPR1

● Graded

Student

Eric Ford

Total Points

62 / 120 pts

Question 1

Question 1: Situational Assessment

8 / 20 pts

1.1 — 1A

0.5 / 3 pts

– 0 pts Correct

– 1.5 pts Incorrect, however, response is relevant to decision-making under risk/uncertainty

✓ – 2.5 pts Incorrect

– 3 pts Blank

1.2 — 1B

2 / 3 pts

– 0 pts Correct

✓ – 1 pt Close, but incorrect

– 2.5 pts Incorrect

– 3 pts Blank

1.3 — 1C

0.5 / 3 pts

– 0 pts Correct

– 1 pt Incorrect, but it is a quality/reliability tool that also prioritizes failure modes

– 1.5 pts Incorrect, but they are quality control charts, at least within the SPC domain

✓ – 2.5 pts Incorrect

– 3 pts Blank

1.4 — 1D

2 / 3 pts

– 0 pts Correct

– 0.5 pts Incorrect, but very plausible misinterpretation since both prioritize causes

✓ – 1 pt Incorrect, but still in reliability/failure analysis family

– 2.5 pts Incorrect

– 3 pts Blank

1.5 — 1E

0.5 / 3 pts

– 0 pts Correct

– 1.5 pts Conceptual Mislabel; chose different distribution indicator

✓ – 2.5 pts Incorrect

– 3 pts Blank

1.6

1F

0.5 / 3 pts

– 0 pts Correct

– 0.5 pts Incorrect, but in the same competitive strategy family, so understandable misinterpretation

– 1.5 pts Incorrect, but indicates some degree of prioritization/importance (was used in class in the context of queuing discipline)

✓ – 2.5 pts Incorrect

– 3 pts Blank

1.7

1G

2 / 2 pts

✓ – 0 pts Correct

– 2 pts Incorrect

Question 2

Question 2: Multiple Choice

19.5 / 25 pts

2.1 — 2A

5 / 5 pts

✓ - 0 pts Correct

- 4 pts Incorrect

- 0.5 pts Failure to Follow Instructions

- 1.5 pts Calculated P2 rather than $P_n > 2$

- 1 pt Calculated $P_n < 2$

2.2 — 2B

3.5 / 5 pts

- 0 pts Correct

✓ - 1.5 pts Partially correct.

- 4 pts Incorrect

2.3 — 2C

5 / 5 pts

✓ - 0 pts Correct

- 2 pts Incorrect, but identified connection of alpha value to Hurwicz

- 4 pts Incorrect

2.4 — 2D

5 / 5 pts

✓ - 0 pts Correct

- 5 pts [Click here to replace this description.](#)

- 2.5 pts Failure to Follow Instructions

2.5 — 2E

1 / 5 pts

- 0 pts Correct

✓ - 4 pts Incorrect

3.1 — 3A

6 / 7.5 pts

- 0 pts Correct

Plant A -Either:

0.032 Units/\$ or 1.22 (dmnl)

Plant B - Either:

0.030 units/\$ or 1.20 (dmnl)

Interpretation:

A>B and why

✓ - 0.5 pts Missing/Incorrect Units

✓ - 0.5 pts Incorrect Plant A Calculation

Plant A Productivity:

Method 1 (units per dollar):

Output/Input = $1,120,000 \text{ units} \div [(480,000 \text{ labor hrs/yr} \times \$28/\text{hr}) + \$18,900,000/\text{yr} + \$2,450,000/\text{yr}]$ = $1,120,000 \text{ units} \div \$34,790,000$

= 0.032 units per dollar

Method 2 (value of output per dollar input):

 $(1,120,000 \text{ units} \times \$38/\text{unit}) \div \$34,790,000$ = $\$42,560,000 \div \$34,790,000$

= 1.22 (dimensionless ratio)

✓ - 0.5 pts Incorrect Plant B Calculation

Plant B Productivity:

Method 1 (units per dollar):

Output/Input = $1,050,000 \text{ units} \div [(420,000 \text{ labor hrs/yr} \times \$31/\text{hr}) + \$19,800,000/\text{yr} + \$2,200,000/\text{yr}]$ = $1,050,000 \text{ units} \div \$34,820,000$

= 0.030 units per dollar

Method 2 (value of output per dollar input):

 $(1,050,000 \text{ units} \times \$40/\text{unit}) \div \$34,820,000$ = $\$42,000,000 \div \$34,820,000$

= 1.21 (dimensionless ratio)

- 2.5 pts No TFP Plant A

- 2.5 pts No TFP Plant B

- 1 pt TFP Incorrectly Calculated

- 0.5 pts Incorrect/Incomplete Interpretation

Follows

- 1.5 pts Incorrect/Incomplete Interpretation

Does not follow or does not address question properly

- 2.5 pts No Interpretation

– 0 pts Correct Recommendation; Logic Follows
or Sufficiently Rigorous Quantitative Support

– 2.5 pts Correct Recommendation, Logic Does not Follow or Insufficient Quantitative Support

✓ – 5 pts Incorrect Recommendation, Logic Follows or Sufficiently Rigorous Quantitative Support

– 7.5 pts Incorrect/No Recommendation, Logic Does not Follow or Work is insufficient to arrive at conclusion

– 12.5 pts No Recommendation or Incorrect Guess with no/little support

– 1 pt You arrived at the correct results, but then misinterpreted the question or your outcomes. It was asking both Plant A and Plant B, not Both option 1 and option 2. Numerical answer supports that you understood the question.

Question 4

Question 4: Winter Jacket

15 / 20 pts

4.1 — 4A

2 / 5 pts

– 0 pts Correct

– 0.5 pts Minor error in explanation

– 1 pt Did not explain that as "Number of Buttons" Increases, "Ease of Wash" Decreases (Negative or Inverse Relationship)

✓ – 3 pts Insufficient/Incorrect Response

– 5 pts Did not answer

4.2 — 4B

4 / 5 pts

– 0 pts Correct

– 0.5 pts Minor Error in explanation

– 1 pt Incorrectly identified customer requirement vs technical requirement

✓ – 1 pt Identified customer requirement with strong negative or strong positive relationship to technical requirement

– 1 pt Identified a customer requirement with large number of relationships

– 2 pts Weak/Missing Explanation

– 5 pts Did not answer the question

4.3 — 4C

5 / 5 pts

✓ – 0 pts Correct

– 0.75 pts Insufficient/Unclear interpretation of the Customer Importance or Incorrect Interpretation of the values (Higher is better)

– 1.5 pts Did not interpret or incorrectly interpreted in reference to customer requirement

– 2.5 pts Incorrect/Missing interpretation of A

– 5 pts Missing response

4.4 — 4D

4 / 5 pts

– 0 pts Correct

✓ – 1 pt Incomplete/Incorrect Response or Does not consider other components of HOQ

– 5 pts Missing Response

Question 5

Question 5: Field Kitchens

5 / 15 pts

5.1 5A

5 / 15 pts

– 0 pts Correct

– 0.5 pts Units

– 1 pt Minor Math Error

– 1 pt Incorrect/Incomplete EMV for Blast Chiller

– 1 pt Incorrect/Incomplete EMV for Old Cooling Method

– 1 pt Incorrect Interpretation
Follows

✓ – 5 pts Missing EMV for Blast Chiller

✓ – 5 pts Missing EMV for Old Cooling Method

– 5 pts Incorrect/Missing Interpretation
Does not Follow

– 2.5 pts Provided explanation, but did not provide recommended course of action

Question 6

Question 6: Drive Thru

0 / 0 pts

6.1 — 6A

0 / 0 pts

– 0 pts Correct

– 1 pt Minor Math/Formulation Error

– 2.5 pts Did not identify solution as P6
Probability that there are 6 vehicles in the system = probability that the system is full

✓ – 2.5 pts Used incorrect Formulation of P6
Should be M/M/1/6 Kendall Notation, resulting in:
 $P_6 = ((1-\rho)/(1-\rho^{(6+1)})) \cdot \rho^6$

✓ – 2.5 pts Did not arrive at solution as a probability (probabilities cannot be negative or exceed 100%) or did not sufficiently demonstrate understanding of tested concept

– 10 pts Did not answer question or showed insufficient work to demonstrate understanding of the subject

– 1 pt Should use 6 for M and n, resulting in incorrect response. Should be 19.17%

– 0.5 pts Transcription Error

6.2 — 6B

0 / 0 pts

– 0 pts Correct

– 0 pts Incorrect, but follows

– 1 pt Incorrect; follows with some minor error in math or logic

✓ – 5 pts Incorrect; answer follows some logic

– 7.5 pts Incorrect; answer does not follow logic or numbers do not appear to have justification

– 2.5 pts Correct (or nearly correct) response, but work does not adequately demonstrate how you arrived at answer or arrived at correct (or nearly correct) answer through incorrect means

– 10 pts Missing Response

– 0 pts Correct

– 0 pts Incorrect effective arrival rate, but follows from previous

✓ – 2.5 pts Incorrect/Incomplete Numerical Solution;

– 5 pts Formula only

– 7.5 pts No numerical Solution or entirely incorrect approach or work shown/not shown does not demonstrate understanding of core concept

✓ – 2.5 pts No/insufficient/incorrect interpretation

– 1 pt Minor math error or incorrect input
examples include incorrect λ (should be 20 cars/hr), incorrect P_6 (should be 0.19, but may follow from previous question)

✓ – 0.5 pts Units

– 1 pt Minor issues with interpretation

Question 7

Question 7: Backpacks

1 / 20 pts

7.1 — 7A

0 / 5 pts

- 0 pts Correct

- 3 pts Correctly identified as attribute, but misidentified as p-chart (should be c chart)

✓ - 5 pts Incorrect

7.2 — 7B

1 / 5 pts

- 0 pts Correct

- 0 pts Incorrect Chart, but Follows

- 1 pt Incorrect Z Values
95% -> $z=1.96$ or ~ 2 - 1.5 pts Incorrect CL
If C Chart, should be $(12+6+7+5+2)/5 = 6.4$
If (incorrectly) using P Chart, should still be $(12/50+6/50+7/50+5/50+2/50)/5 = 0.128$

- 1 pt LCL Cannot be Negative

- 1 pt Incorrect CL
Should be $\bar{C}$ - 1.5 pts Used Incorrect Sigma Values
Should be $\sqrt{\bar{C}}$

- 3 pts Incomplete Calculations

- 5 pts No/Insufficient response

- 1.5 pts Correct CL (6.4) for a C chart, but then proceeded to convert to incorrect P chart CL and then use P chart UCL and LCL

✓ - 4 pts No CL, incorrect/Incomplete UCL and LCL formulations

- 0.5 pts Minor Math Error

- 3 pts Incorrect UCL and LCL
If C Chart, should be $\bar{C} \pm z\sqrt{\bar{C}}$
If (incorrectly) P Chart, should be $\bar{P} \pm z\sqrt{\bar{P}(1-\bar{P})/n}$

7.3 — 7C

0 / 5 pts

- 0 pts Correct or Follows

✓ - 1 pt Missing/Incorrect Labels

✓ - 1 pt Missing/Incorrect Axes or Axis Labels

✓ - 1 pt Missing/Incorrect Points

✓ - 1 pt Missing/Incorrect CL

✓ - 1 pt Missing/Incorrect LCL/UCL

– 0 pts Correct

– 1 pt Incomplete interpretation or minor issues with interpretation

– 2.5 pts Incorrect Interpretation based on control chart

✓ – 5 pts Missing Interpretation

Question 8

Bonus

0 / 0 pts

✓ – 0 pts Did not Attempt or None Correct or did not label with rule

+ 1 pt Some Correct

+ 3 pts Mostly Correct

+ 5 pts All correct
1, 2, 3, 4, 8

EM420 Production and Operations Management**Written Partial Review****Version A**

- 1) This is a 75-minute graded event worth 150 points.
- 2) This is a closed book and closed notes exam. You may use one single sided 3x5" index card and the provided reference book (DO NOT MARK IN IT!) and an authorized scientific calculator. No collaboration is allowed.
- 3) You may disassemble this document if you'd like. You may also use extra paper provided at the front of the classroom. Please ensure you staple the exam back together in the correct order with extra sheets labeled and at the back.
- 4) **SHOW ALL WORK!** Unsupported or missing work earns little or no partial credit.
- 5) **Rounding.** Show dollars and cents for any monetary answer (\$11.75). Show two significant digits beyond the decimal for percentage answers (11.75%) and at least two significant digits beyond the decimal for all other answers (0.13).
- 6) Upon completion, sign the acknowledgment statement below:

"I did not use any sources, nor did I receive any assistance while completing this WPR. This WPR will be released from academic security on 30OCT2025. Prior to that time, I may not discuss any aspects of this exam with anyone except an EM420 instructor."

Printed Name: Eric Ford Signature: Eric Ford Date Signed: 9/15/15

8) GRADING SUMMARY:

Section	Topic	Total Points Available	Total Points Earned
1	Situational Assessment	20	
2	Multiple Choice	25	
3	Ammo Plants	20	
4	Winter Jacket	20	
5	Field Kitchens	15	
6	Drive Thru	30	
7	Backpacks	20	
BONUS	Western Electric	5	
	Total	150	

Section I: Situational Assessment (No Partial Credit) [20 points]

Please choose the BEST problem-solving strategy, chart, or calculation that would aid in making a decision for the prompt. There is one best answer for each situation. No partial credit. Explanation is not necessary. You must write your answers on the lines below for grading

- 1A order qualifier
- 1B FMEA
- 1C quality productivity ratio
- 1D fault tree analysis
- 1E X bar chart
- 1F FIFO

Order Winner	Order Qualifier
EVPI	Hurwicz
Equal likelihood	Maximax
Maximin	Priority
Minimax Regret	Pareto Analysis
Quality Productivity Ratio	P Chart
X bar Chart	R Chart
C Chart	FMEA
Fault Tree Analysis	House Of Quality
Service Blueprint	Calling Population
FIFO	LIFO
General	Degenerate
Markovian	

1G Charlie Finner

1A [3 Points] A program manager wants to compute the expected monetary value of hiring a market research firm before making a major sourcing decision, to see whether additional information is worth the cost. Which calculation should they perform?

1B [3 Points] An aircraft mishap investigation team wants to model how combinations of electrical, hydraulic, and pilot-control failures could lead to loss of flight control. They want a logical diagram that starts with the top event (loss of control) & maps pathways downward to root causes.

1C [3 Points] A quality engineer at an Army depot notices that production of Humvee transmissions has been missing daily output targets for the last two weeks. Reviewing the data, they see eight different types of process failures, including torque wrench calibration errors, mislabeled parts etc. Management asks them to figure out which issues are causing the majority of lost throughput so they can direct maintenance to fix those first. What should they use?

1D [3 Points] An operations officer notices repeat breakdowns on forklifts. They want to systematically rank which failure means (hydraulic leak, steering failure, sensor malfunction) are most critical so they can focus limited maintenance funds on the highest risk items first.

1E [3 points] An ammo supply point (ASP) officer is modeling wait times for convoys arriving to pick up munitions before a training exercise. Historically, truck arrivals are random and independent, with interarrival times well-fit by an exponential distribution. The officer also confirms that service times (loading each truck) are memoryless and follow an exponential pattern. Which type of distribution matches service times?

1F [3 points] An Army Program Executive Office is evaluating two vendors for next-gen body armor. Both vendors meet the minimum ballistic protection and weight standards required to compete. However, Vendor A can deliver within 30 days while Vendor B's lead time is 90 days. Rapid fielding is the deciding factor for award. How can you classify rapid fielding?

1G [2 points] Please list your Group Graded Lab teammates' first names.

Section II: Multiple Choice (No Partial Credit) [25 points]

Choose the best answer. You must write your letter answers on the lines below for grading

2A c

2B b

2C d

2D a

2E a

2A (5 Points) Consider an espresso stand with a single barista. Customers arrive at the stand at the rate of 28 per hour according to a Poisson distribution. Service times are exponentially distributed with a service rate of 35 customers per hour. The probability that there are more than 2 customers in the system is:

- a. 0.128 b. 0.488 c. 0.512 d. 0.640

2B (4 Points) A process that is in control as indicated by its Statistical Process Control chart indicates that some _____ in the process may be _____.

- a. variation, natural b. variation, assignable c. attributes, random d. defects, measurable

2C (4 Points) Decision criteria are interrelated, often depend on the eye of the beholder and settings. _____ criteria is actually the same as the _____ criteria when $\alpha = 1$.

- a. Hurwicz, EVPI b. Minimin, Maximin c. Minimax Regret, Hurwicz d. Maximax, Hurwicz

2D (5 Points) You are a quality engineer at a small-arms ammunition plant. Your team has measured the reliability of three critical machines in series that form a cartridge assembly line:
Case Feeder: $R_1=0.96$ Primer Insertor: $R_2=0.94$ Bullet Seater: $R_3=0.90$
What is the reliability of the overall system (assume machines are arranged in series)?

- a. 0.831 b. 0.850 c. 0.900 d. 0.922

2E (5 Points) During a recent 4,000-hour production run, the primer inserter broke down 12 times. Each repair took on shift or 4 hours to complete on average. What was its availability during this period?

- a. 0.88 b. 0.92 c. 0.95 d. 0.99

Section III: Ammo Plants [20 Points]

Problem 3 You are a commander managing two Army ammo plants which operate **320 days/year** at **10 hours/day**. Each unit is sold at the plant's listed price.

Item	Plant Alpha	Plant Bravo
Output (units/year)	1,120,000	1,050,000
Selling price (\$/unit)	\$38.00	\$40.00
Labor hours/year	480,000	420,000
Labor cost (\$/hour)	\$28.00	\$31.00
Material cost/year	\$18,900,000	\$19,800,000
Energy cost/year	\$2,450,000	\$2,200,000

3A (7.5 Points). Calculate the Total Productivity for each plant. Interpret your results in words

Plant Alpha

$$TFP = \frac{1,120,000}{42.56m + 13.44m + 18.9m + 2.45m}$$

$$= 0.014$$

TFP =

Plant Bravo

$$TFP = \frac{1,050,000}{42m + 13.02m + 19.8m + 2.2m}$$

$$= 0.0136$$

Plant Alpha has a higher total factor productivity than Plant Bravo

Item	Plant Alpha	Plant Bravo
Output (units/year)	1,120,000	1,050,000
Selling price (\$/unit)	\$38.00	\$40.00
Labor hours/year	480,000	420,000
Labor cost (\$/hour)	\$28.00	\$31.00
Material cost/year	\$18,900,000	\$19,800,000
Energy cost/year	\$2,450,000	\$2,200,000

3B (12.5 Points). The Army can invest \$15,000 total:

Option 1 (\$7,500): **+5% labor productivity** (same hours, but effective output increases 5%).

Option 2 (\$7,500): **-10% material cost.**

You have the following options for investment:

- A. Apply option 1 to both plants
- B. Apply option 2 for both plants
- C. Apply Option 1 for Plant A & Option 2 for Plant B.
- D. Apply Option 1 for Plant B & Option 2 for Plant A.

Recommend which investment strategy the Army should choose and support with quantitative analysis.

A +5%

$$1,120,000 \times 1.05 = 1,176,000$$

$$1,176,000 \times 38 = 44,688,000$$

B

$$1,120,000 \times 40 = 44,800,000$$

B -10% material cost

$$10\% \text{ of } 19,800,000 = 1,980,000$$

$$19,800,000 - 1,980,000 = 17,820,000$$

A

$$10\% \text{ of } 18,900,000 = 1,890,000$$

$$18,900,000 - 1,890,000 = 17,010,000$$

I would choose option **A** because the profit of the +5% Labor productivity for both outweighs how much money they save from -10% material cost

Section IV: Winter Jacket [20 points]

You are part of a product design team tasked with developing a new winter jacket for Soldiers operating in cold-weather environments. Your product design team has created the following matrix to help you understand the relationship between customer and technical requirements. Use this diagram to answer the questions that follow about trade-offs and technical focus areas for the jacket design.

		Technical Requirements										
			Number of Pockets	Weight of filling	Number of Buttons	Reflective Marking	Weight of Lining	Weight of Outer	Length of Jacket	Has a hoodie	Has a collar	Number of Zippers
Customer Requirements	Easy to put on	5	-	-	⊖		-	-	-			
	Not too heavy	9	-	-			-	-	-	-		-
	Keeps body warm	4		⊕								
	Breathability	7					-	-	-			
	Visible in the dark	3				⊕						
	Fashionable	5	+									
	Protects from Rain	2							+	+		
	Not too expensive	8				-				-		-
	Easy to Wash	6	-	-	-	-						
	Keeps neck warm	4								⊕	+	

B

A

4A (5 Points) Please explain the relationship in words between the customer requirement "Easy to Wash" and the technical requirement "number of buttons."

those two requirements do not have a direct correlation meaning # of buttons does not effect the customers wash standards. they do not effect each other. it is weak

4B (5 Points) Identify one customer requirement that has weak or few relationships to the technical requirements. How might you redesign the technical approach to better address that requirement?

keeps body warm and the requirement of has a hoodie.
I might redesign this by taking less importance on the weight of the jacket and distribute it to the (lower) pants and hoodie to help keep the head warm

4C (5 Points) What does A (the label for the orange numbers) represent? Interpret one of the numbers in reference to the customer requirement.

it represents the ranking for customer requirements.
for ex. not too heavy being #9 means that it is most important out of all customer requirements but the jacket is not too heavy

4D (5 Points) What else is typically included in a house of quality that is not shown? Hint: You could identify the area hidden by B on the diagram or any other missing pieces of the diagram. Think about the purpose of the house of quality.

the actual quality of materials instead of just labeling them

Section V: Field Kitchens [15 Points]

You are the operations manager for a field kitchen deciding whether to invest in a new blast chiller.

- If you buy the blast chiller, there is a 60% chance demand is high, generating a profit of \$50,000, and a 40% chance demand is low, leading to a profit of \$10,000 (because the chiller is under-utilized).
- If you do not buy, you save the purchase cost but must rely on the old cooling method. In this case, there is a 60% chance of high demand, earning \$30,000, and a 40% chance of low demand, earning \$15,000.

5A (15 Points) Calculate the Expected Monetary Value (EMV) for each decision and explain the option you would choose, and the circumstance. Be sure to write all assumptions.

- assuming that there will never be a time of no demand
- assuming I will always receive a profit
- assuming the chiller will not break
- there is no purchase cost
- expected high demand

in this case I would invest in the new chiller
because I am expecting a high demand which is
leading to me wanting to invest

Section VI: Drive Thru [30 Points]

$$s = 1$$

The arrival rate of cars at a drive-thru restaurant is 20 cars per hour. The service rate is 18 cars per hour. The available space for cars in the drive-thru is restricted to 6 cars only. Otherwise, the queue goes into the street and cars cannot queue. The arrival rate and service rate follow the Poisson distribution. Identify the type of queue and then determine the performance measures of this queue. *Be sure to interpret each answer in words!*

6A [10 points] What is the probability that the system is full?

$$P_0 = 1 - \frac{\frac{20}{60 \text{ min}}}{\frac{18}{60 \text{ min}}}$$

$$= 0.11$$

$$P_n = (1 - P)P^n$$

$$= (1 - 0.33) \cdot 0.33^6$$

$$= 0.0005133$$

this means the system
is most likely full

6B [10 points] How many customers are lost per hour on average?

$$\frac{20 \text{ car}}{60 \text{ min}} - \frac{18 \text{ car}}{60 \text{ min}} = \frac{2 \text{ car}}{60 \text{ min}}$$

2 cars are lost per hour

6C [10 points] What is the effective arrival rate? What does that mean in words?

$$\bar{\lambda} = \lambda(1 - P_n)$$

$$= \frac{20}{60} (1 -)$$

$$= 0.333$$

Section VII: Backpacks [20 Points]

You have recently retired from the military and are working for a tactical backpack company and much count the blemishes on each backpack (stitch undone, loose thread, bleach spot etc.). A sample includes one backpack from a batch of 50 backpacks. The following samples are collected at random in a one month period. Assume 95% confidence.

Sample	Defects
1	12
2	6
3	7
4	5
5	2

7A [5 points] What type of control chart is best used in this situation.

process control

7B [5 points] Calculate the UCL, UCL and CL

$$UCL = \bar{x} + A_2 \cdot \bar{r}$$

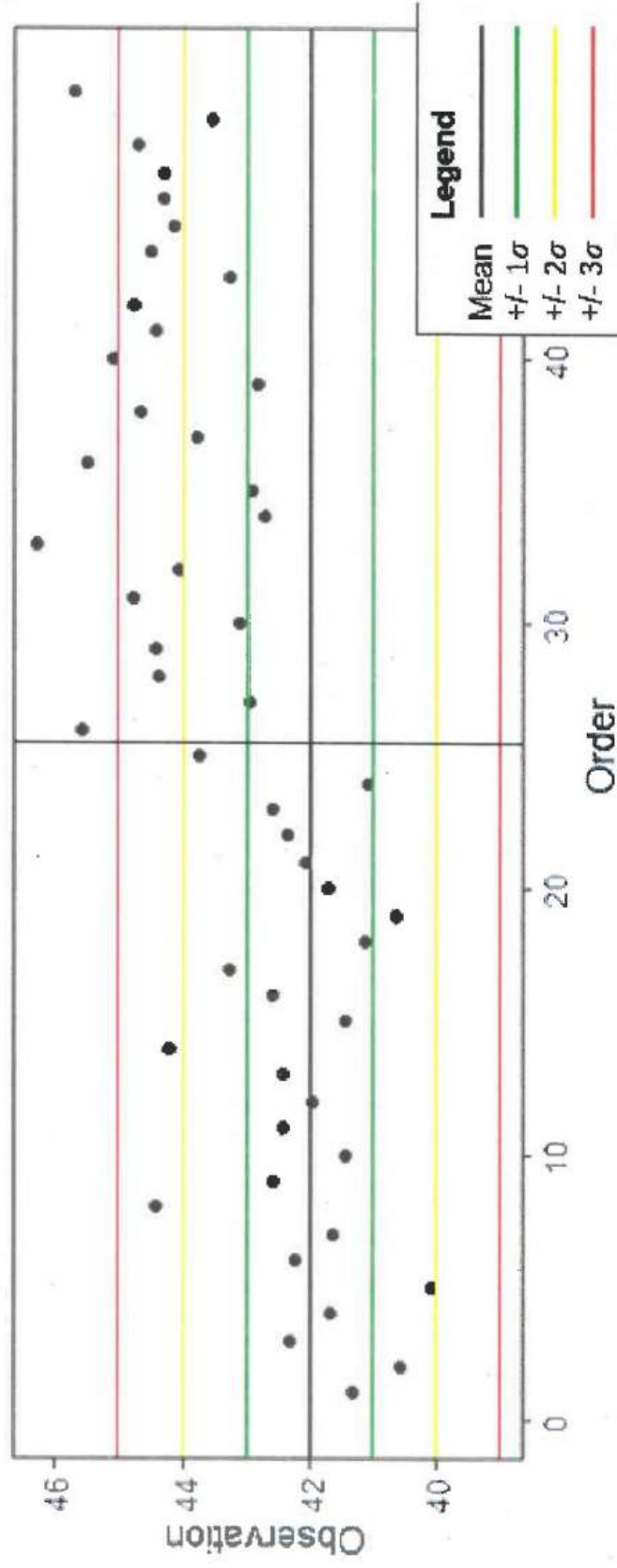
$$LCL = \max(\bar{x} - A_2 \cdot \bar{r}, 0)$$

7C [5 points] Draw the Control Chart, be sure to include all labels.

7D [5 points] Please interpret your control chart in words. Use context from the prompt.

Bonus (5 points): Many types of rules can be considered to determine whether a process is out of control. Use the Western Electric Rules to determine if the process is out of control (if any rule holds true the process is out of control). Indicate which points break which rule by circling and labeling them with the rule number broken.

1. The most recent point plots outside one of the 3-sigma control limits.
2. Two of the three most recent points plot outside and on the same side as one of the 2-sigma control limits.
3. Four of the five most recent points plot outside and on the same side as one of the 1-sigma control limits.
4. Eight out of the last eight points plot on the same side of the center line, or target value.
5. Six points in a row increasing or decreasing.
6. Fifteen points in a row within one sigma.
7. Fourteen points in a row alternating direction.
8. Eight points in a row outside one sigma.





UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB H

Written Partial Review II

EM420 Written Partial Review II

Name: _____

Academic Term 26-1

Hour: _____

EM420 Production and Operations Management
Written Partial Review
Version C

- 1) This is a 75 minute graded event worth 150 points.
- 2) This is a closed book and closed notes exam. You may use one single sided 3x5" index cards and the provided FE book (DO NOT MARK IN IT!) and an authorized scientific calculator. No collaboration is allowed.
- 3) You may disassemble this document if you'd like. You may also use extra paper provided at the front of the classroom. Please ensure you staple the exam back together in the correct order with extra sheets labeled and at the back.
- 4) **SHOW ALL WORK!** Unsupported or missing work earns little or no partial credit.
- 5) **Rounding.** Show dollars and cents for any monetary answer (\$12.75). Show two significant digits beyond the decimal for percentage answers (12.75%) and at least two significant digits beyond the decimal for all other answers (0.13) unless otherwise noted
- 7) Upon completion, sign the acknowledgment statement below:

"I did not use any sources, nor did I receive any assistance while completing this WPR. This WPR will be released from academic security on 21NOV2025. Prior to that time, I may not discuss any aspects of this exam with anyone except an EM420 instructor."

Printed Name: _____

Signature: _____

Date Signed: _____

8) GRADING SUMMARY:

Section	Topic	Total Points Available	Total Points Earned
1	Situation Assessment	30	
2	Multiple Choice & Short Answer	45	
3	Operational Readiness	30	
4	Joint Logistics Task Force	25	
5	Drones	20	
	Total	150	



Answer Sheet for Section 1 & Section 2

1A _____

1B _____

1C _____

1D _____

1E _____

1F _____

1G _____

1G Situation:

2A _____

2B _____

2C _____

2D _____

2E _____

2F _____

2G _____

**Section 1: Situation Assessment [30 points]**

In the following situations, please choose the BEST problem-solving strategy, chart, or calculation that would aid in making a decision for the problem at hand. A glossary is provided below for your reference. There is one best answer for each situation. No partial credit will be provided. Explanation is not necessary.

Alternative Workplace	FMEA	Load Distance
Block Diagramming	Gantt Chart	Location Factors
Break Even Analysis	Heijunka	Movement
Center of Gravity	Horizontal Enlargement	Muther's Diagram
Commissions	House of Quality	Outsourcing
Cross Training	Hygiene Factors	Point(s) of Indifference
Cycle Time	Individual Incentive	Product Process Matrix
Defects	Inventory	Profit Sharing
Economies of Scale	Job Rotation	Service Blueprint
Ellsberg's Paradox	Job Training	Service-Process Matrix
Empowerment	Johnson's Rule	Simple Sequencing Rules
EVPI	Kaizen	Theory X Employee
Fixed-Position Layout	Kanban System	Theory Y Employee
Flexible Work Schedules	Line Balancing	Vertical Enlargement
Flow Time	Load Assignment Method	Value Stream Mapping

Record all answers on Page 2, on the answer sheet.

1A [4 Points] – A defense electronics plant is assembling communication modules on a high-volume line. Some stations are idle while others have long queues. The line chief needs to reassign tasks to balance workloads, meet the desired cycle time, and reduce idle time. What method should be used?

1B [4 Points] – A naval drone production unit must process each unit through two sequential operations: assembly and inspection. Each operation takes a different amount of time for each drone type. The operations officer wants to find the sequence that minimizes total completion time and idle time across both stations. What strategy should be used?

1C [4 Points] - A defense contractor compares two drone assembly methods: one uses manual labor with lower setup costs but higher variable costs, and the other uses automated assembly with higher setup costs but lower variable costs. The manager wants to find the exact output quantity where both options cost the same before deciding which to use. What should he use?



1D [4 Points] - At an Army ground vehicle maintenance company, crews report frequent delays during scheduled services due to disorganized toolkits, unclear part storage locations, and repeated backtracking between work areas. Instead of waiting for a full facility renovation, the platoon sergeant begins holding short morning huddles where Soldiers suggest small changes, such as labeling bins, rearranging benches, and creating visual checklists, to reduce wasted time and motion. Over the next month, turnaround time improves by 12%. Which continuous improvement strategy best fits this situation?

1E [4 Points] - A process improvement specialist wants to assess total process time including both work and waiting for a batch of products. What performance measure best captures this total elapsed time in the system?

1F [4 Points] - An Army logistics battalion's data-tracking system for vehicle parts is outdated and unreliable. Rather than developing new software in-house, leadership is considering contracting a civilian tech company that specializes in military logistics systems to design, operate, and maintain it. The logistics battalion is participating in:

1G [6 Points] – Create your own situation which you think you would encounter as a 2LT and provide an answer (from the glossary above – do not use one already used in 1A-F). You will be graded for clarity and correctness. Make sure it is clear why it would be your term and no other terms.

**Section 2: Multiple Choice & Short Answer (No Partial Credit 45 Points)****Record all answers on Page 2, on the answer sheet.**

2A [10 points] You are sequencing five repair jobs through two sequential maintenance stations. The processing times (in hours) are shown below.

Job	Station 1	Station 2
A	8	6
B	4	7
C	5	3
D	7	9
E	2	5

Determine the optimal sequence of jobs that minimizes total makespan. Write the order of jobs from first to last.

2B [5 Points] You are designing a production line for assembling portable radios. Each station has a desired cycle time of 40 seconds. The task list below shows the operation times for each required activity.

Task	Operation	Time (sec)	Requires Special Tool?
A	Mount circuit board	10	No
B	Install speaker	6	No
C	Attach antenna	8	Yes
D	Screw in faceplate	7	No
E	Test sound output	9	Yes
F	Package product	6	No

What is the **minimum number of workstations** required to complete the line without exceeding the cycle time?

- a. 1
- b. 2
- c. 3
- d. 4
- e. 6



2C [5 Points] Which of the following is **always true** when you are dealing with two sequential machines? Write all correct answer(s).

- a. All idle time for Machine 1 is always at the beginning
- b. All idle time for Machine 2 is always at the beginning
- c. All idle time for Machine 1 is always at the end
- d. All idle time for Machine 2 is always at the end
- e. Makespan is determined by the finishing time of the last job on Machine 1
- f. Makespan is determined by the finishing time of the last job on Machine 2

2D [5 points] A logistics officer is selecting the location for a new supply hub to support three forward bases in an open region with no existing facilities. The goal is to minimize total transportation distance, considering how much each base ships per week. Use traditional rounding standards. (round up if .5 or above, round down otherwise). Select the correct strategy and location.

Base	X-Coordinate	Y-Coordinate	Weekly Shipments (tons)
Alpha	4	8	30
Bravo	10	4	50
Charlie	6	12	20

- a. Load Distance; (7,7) New Point
- b. Center of Gravity: (7,7) New Point
- c. Load Distance, (4,8) Alpha
- d. Center of Gravity, (4,8) Alpha

2E [5 points] You're leading a **senior capstone team** working on an engineering design project for a real client. Each cadet on your team approaches the project differently according to the descriptions below

- Cadet A often asks for additional responsibilities and suggests design improvements beyond the assignment.
- Cadet B needs frequent reminders to submit weekly reports and only works when directly supervised.
- Cadet C volunteers to organize the client presentation and stays late to ensure everything is perfect.
- Cadet D avoids taking initiative, saying, "Just tell me exactly what to do."
- Cadet E proactively schedules team meetings and checks in with everyone to help coordinate progress.

Which cadet(s) demonstrate Theory Y behaviors?



2F [5 Points] The table below shows part of a Muther's Relationship Chart for a maintenance facility. The letters indicate the closeness rating between departments. If the engineer can only place two departments side by side due to space limitations, which pair should be prioritized for adjacency based on the chart?

Department	Welding	Paint	Inspection	Storage
Welding	—	A	E	O
Paint	A	—	I	U
Inspection	E	I	—	O
Storage	O	U	O	—

- a. Paint and Welding
- b. Paint and Inspection
- c. Inspection and Storage
- d. Storage and Paint
- e. Any pairing would be adequate

2G [10 Points] A maintenance parts supplier is considering producing its own replacement valves rather than purchasing them. The company would need to invest \$120,000 in new equipment. Each valve would cost \$35 to produce and could be sold for \$95. What is the break-even quantity of valves (in units) the supplier must produce and sell to recover its investment?



Section 3: Operational Readiness (30 Points)

You are managing four repair tasks for a brigade support battalion ahead of a large training exercise. Each task must be completed before deployment to maintain operational readiness. Today's date is May 1st. There are 31 days in May.

Maintenance Task	Repair Time (days)	Required Completion Date
A. Transmission Replacement	7 days	May 10th
B. Optics Calibration	14 days	May 18th
C. Electrical Overhaul	5 days	May 15th
D. Suspension Alignment	20 days	June 5th

3A [3 points] What is the order according to Longest Processing Time?

3B [5 points] What is the order according to SLACK?



3C [3 points] What is the order according to DUE DATE?

3D [12 points] Fill out the following Table (show work under parts A, B and C).

Sequencing Rule	# of Late Assignments	Total Number of Late Days
LPT		
SLACK		
DUE DATE		

3E [3.5 points] The battalion maintenance section assigns one readiness deduction for every three total days of lateness across all repair tasks. For example, if one task is one day late and another task is one day late, their combined two late days still count toward the three-day threshold for a single deduction. Which sequencing rule should you use to minimize the total number of late days and therefore minimize readiness deductions? What is the total number of readiness deductions associated with this rule. Explain your reasoning.

3F [3.5 Points] The battalion's operations officer tracks mission delays based on whether each repair task is completed on time (binary: yes or no). Each late task, regardless of how many days late, counts as one mission delay. Which scheduling strategy minimizes the total number of mission delays? How many mission delays occur with this strategy?



Section 4: Joint Logistics Task Force (25 Points)

You are advising a Joint Logistics Task Force (JLTF) preparing for a multinational exercise. Four regional logistics hubs (Teams A–D) must each take responsibility for one operational task. Your goal is to minimize total project completion time while ensuring that each team is assigned to exactly one task. Below are the time in days that each team takes to do each operational task.

Team	Supply Chain Setup	Equipment Calibration	Site Construction	Vehicle Readiness
A	12	10	14	11
B	8	13	9	12
C	9	11	8	10
D	10	9	13	12

[22.5 points] Use the **Load Assignment Method (Hungarian Method)** to determine which team should be assigned to each task to minimize total project duration.

[2.5 points] If the daily operating cost of each team is \$6,800 per day, what is the total operational cost of these assignments based on your final solution?



Section 5: Drones (20 Points)

A maintenance company sustains 60 tactical drones deployed across multiple sites. To prevent downtime, it uses a pull-based Kanban system for replenishing two types of pre-packed battery modules: Type A for standard operations and Type B for long-range missions. On a typical day, the unit consumes about 1,200 battery modules, with roughly two-thirds being Type A and the rest Type B. Each sealed container can hold 50 modules, though only about 85 percent of that space is usable once padding, labels, and inspection inserts are added. Once a Kanban card is pulled, it takes about eight hours for a full container to be refilled, quality-checked, and returned to the line. To hedge against variability and delays, the maintenance officer applies a 20 percent safety margin.

5A [5 pts] Determine the number of Kanban cards needed for **each** module type to sustain continuous flow without stockouts. Round up, **provide a reason why you would round up in practice.**

5B [10 pts] A new drone model has been introduced that increases total daily battery consumption by 25 percent, while the rest of the system remains unchanged. Recalculate the Kanban requirements for each module type and discuss the operational effects of this higher throughput. Round up.

5C [5 pts] What two countermeasures could help maintain efficiency without simply adding storage space or containers? (Select all that apply.) No partial credit.

- ☐ Shorten resupply lead time through pre-staging or faster transport cycles
- ☐ Reduce container size to increase turnover and responsiveness
- ☐ Increase safety margin from 20 % to 30 %
- ☐ Add an extra storage bay near the maintenance line
- ☐ Increase total on-hand inventory to cover two days of demand

Outstanding Student Work

WPR2

● Graded

WPR 2

Student

Sarah Chernik

Total Points

146.5 / 150 pts

Question 1

Situational Assessment

30 / 30 pts

1.1 — **1A** 4 / 4 pts

✓ - 0 pts Correct

- 4 pts Incorrect Response; Correct Answer is Line Balancing

- 3 pts Incorrect Response; Leveling (Heijunka) concept relates, but it's not the right tool for task reassignment. Correct answer was Line Balancing

1.2 — **1B** 4 / 4 pts

✓ - 0 pts Correct

- 3 pts General sequencing idea, but not the specific 2-machine rule (Johnson's); Correct answer is Johnson's Rule

- 4 pts Incorrect; Correct response is Johnson's Rule

- 4 pts Blank; Correct response is Johnson's Rule

1.3 — **1C** 4 / 4 pts

✓ - 0 pts Correct

- 4 pts Incorrect; Correct response is Point of Indifference

- 3 pts Incorrect; Right domain (cost crossover), but wrong response. Correct response is Point of Indifference

1.4 — **1D** 4 / 4 pts

✓ - 0 pts Correct

- 4 pts Incorrect; correct response is Kaizen

- 2 pts 1/2 Correct; Correct response is Kaizen

1.5 — **1E** 4 / 4 pts

✓ - 0 pts Correct

- 4 pts Incorrect; Correct response is Flow Time

- 3.5 pts Incorrect; Related to timing, but **lead time** is external-order based, not total time in the system. Correct response is Flow Time

- 3 pts Incorrect; **Cycle time** is at the workstation level; it doesn't include waiting like flow time does. Correct response is Flow Time

- 3.5 pts Incorrect; **makespan** applies to scheduling completion time, not total time a unit spends in the system. Correct response is Flow Time

1.6 — **1F** 4 / 4 pts

✓ - 0 pts Correct

- 4 pts Incorrect; Correct response is Outsourcing

✓ - 0 pts Correct; Well articulated response

- 2 pts Generally Correct; Response poorly considered, poorly articulated, or ill-fit to the question
- 3 pts Response does not demonstrate or reasonably articulate understanding of chosen concept
- 4 pts Term but no situation
- 6 pts Blank

Question 2

Multiple Choice and Short Answer

43.5 / 45 pts

2.1 — 2A

10 / 10 pts

✓ - 0 pts Correct

- 8 pts Ordered left to right using shortest job, but ignored station ordering (left if on station 1, right if on station 2). Correct response was E B D A C

- 8 pts Ordered right to left using shortest job, but ignored station ordering (left if on station 1, right if on station 2). Correct response was E B D A C

- 9 pts Incorrect; Correct response was E B D A C

- 9 pts Incorrect; ordered based on lowest time on station 2 only. Correct response was E B D A C

- 5 pts Incorrect; Mild reversal. Correct answer is E B D A C

- 4 pts Added an additional letter, but generally correct. Correct answer is E B D A C

2.2 — 2B

5 / 5 pts

✓ - 0 pts Correct

- 5 pts Incorrect. Likely rounded down rather than up. Minimum was 2 (B)

- 5 pts Incorrect. Minimum was 2 (B)

2.3 — 2C

3.5 / 5 pts

- 0 pts Correct

- 2.5 pts One correct; Either C or F

✓ - 1.5 pts Additional incorrect response; C and F plus a wrong response

- 4 pts Incorrect; correct response is C and F

- 3 pts One correct, Either C or F, plus multiple incorrect responses

2.4 — 2D

5 / 5 pts

✓ - 0 pts Correct

- 4 pts Used the CoG formulation, but identified it as LD. Correct response is Center of Gravity, (7, 7), New Point (B)

$$X = \text{Sum}[WiXi]/\text{Sum}[Wi]$$

$$Y = \text{Sum}[WiYi]/\text{Sum}[Wi]$$

- 4.5 pts Identified correct technique (CoG), but did not correctly apply mathematical formulation. Correct response is Center of Gravity, (7, 7), New Point (B)

$$X = \text{Sum}[WiXi]/\text{Sum}[Wi]$$

$$Y = \text{Sum}[WiYi]/\text{Sum}[Wi]$$

- 5 pts Incorrect. Correct response is Center of Gravity, (7, 7), New Point (B)

$$X = \text{Sum}[WiXi]/\text{Sum}[Wi]$$

$$Y = \text{Sum}[WiYi]/\text{Sum}[Wi]$$

2.5 — **2E** 5 / 5 pts

✓ - 0 pts Correct

- 2.5 pts Incorrect. Response indicates Theory X Cadets. Correct response for Theory Y is CDT A, C, and E

- 5 pts Incorrect. Response mixed both Theory X and Theory Y

2.6 — **2F** 5 / 5 pts

✓ - 0 pts Correct

- 5 pts Incorrect. Murther's Grid is read alphabetically as:

A: Absolutely Necessary

E: Essential

I: Important

O: Okay

U: Unnecessary

X: Undesirable

Correct response indicates 'A-Absolutely Necessary' intersection at Paint and Welding (A)

2.7 — **2G** 10 / 10 pts

✓ - 0 pts Correct

- 1 pt Missing units

- 3 pts Multiplied in the denominator rather than subtracted, resulting in incorrect response. Correct response is 2,000 valves.

- 3 pts Dropped a zero from the fixed cost, resulting in response being off by a factor of 10. Correct response is 2,000 valves.

- 3 pts Subtraction error, resulting in mildly incorrect response. Correct response is 2,000 valves

- 5 pts Incorrect response with acceptable formulation. Unclear error origin. Correct response is 2,000 valves

- 10 pts Incomplete response with incorrect formulation

- 10 pts Incorrect response with incorrect formulation

- 10 pts Incorrect response with no supporting work shown.

Question 3

Operational Readiness

28 / 30 pts

3.1 — 3A

3 / 3 pts

✓ - 0 pts Correct

- 3 pts This is the order according to SPT - Shortest Processing Time. Correct Response is D-B-A-C

3.2 — 3B

5 / 5 pts

✓ - 0 pts Correct

- 4 pts Incorrect; ordered as most slack first. Should have been ordered as least slack first, or A, B, C, D

- 5 pts Incorrect. Correct response would be least slack first A(3)-B(4)-C(10)-D(16)

3.3 — 3C

3 / 3 pts

✓ - 0 pts Correct

- 3 pts Incorrect; Nearest Due Date results in A, C, B, D

3.4 — 3D LPT # Late

2 / 2 pts

✓ - 0 pts Correct

- 2 pts Blank; Should be 3

- 2 pts Incorrect; should be 3

- 0.5 pts Swapped # Late and Late Days, but demonstrated understanding

3.5 — 3D Slack # Late

2 / 2 pts

✓ - 0 pts Correct

- 1 pt Undercounted but generally demonstrated understanding. Correct response is 3

- 2 pts Blank; correct response is 3

- 2 pts Incorrect; correct response is 3

- 2 pts Swapped # of Assignments and Late Days, but was incorrect in either case

- 1.25 pts Follows, but undercounted even given revised order

- 1.5 pts Follows, but overcounted even given revised order

- 1.5 pts Used A's Slack as duration, shifting all other responses left

3.6 — 3D Due Date # Late 2 / 2 pts

✓ - 0 pts Correct

- 1.25 pts Overcount, but demonstrated understanding; correct response is 2
- 1.25 pts Undercount, but demonstrated understanding; correct response is 2
- 2 pts blank
- 2 pts Incorrect; correct response is 2
- 2 pts Swapped; but incorrect value
- 0 pts Incorrect; follows

3.7 — 3D LTP Tardiness 2 / 2 pts

✓ - 0 pts Correct, +/-3 days (counting starting date at day 0 = 78 days; counting starting at day 1 = 81 days)

- 2 pts Blank
- 2 pts Incorrect; correct response is approximately 81 days
- 1.25 pts Incorrect, but technique demonstrated understanding of concept
- 0.75 pts Swapped Number of Late Days and Late Assignments, but correct number
- 0 pts Incorrect, but follows

3.8 — 3D Slack Tardiness 2 / 2 pts

✓ - 0 pts Correct within +/-3 of accepted response

- 2 pts Blank
- 2 pts Incorrect; correct response should be approximately 27 days
- 1.25 pts Incorrect, but response demonstrated understanding of concept
- 0 pts Incorrect, but follows
- 2 pts Swapped responses; Incorrect value due to inputting May 5 for June 5 as due date for D

3.9 — 3D Due Date Tardiness 2 / 2 pts

✓ - 0 pts Correct within +/-3 of 20 days

- 1.25 pts Incorrect, but demonstrates understanding of the concept
- 2 pts Blank
- 2 pts Incorrect; correct response is 20 days
- 0 pts incorrect, but follows
- 2 pts Swapped, but completely incorrect

3.10 3E

1.5 / 3.5 pts

– 0 pts Correct

✓ – 2 pts Correct rule, incorrect reasoning or readiness deduction

– 3.5 pts Incorrect

– 3.5 pts Blank

– 0 pts Incorrect, or partly incorrect, but follows

– 2 pts Incorrect, or partly incorrect, but follows, without sufficient explanation

3.11 3F

3.5 / 3.5 pts

– 0 pts Correct

✓ – 0 pts Outside the box solution, but Great idea! A C D B does result in the fewest late tasks

– 0 pts Incorrect, but follows

– 1.5 pts Related concept applied, but incorrect choice given question

– 1 pt Correct scheduling strategy, incorrect number of mission delay.

– 1 pt Did not express number of mission delays

– 2.5 pts Incorrect use of Critical Ratio; would have resulted in 3 mission delays

– 3 pts Incorrect concept applied. Correct response is Due Date with 2 Mission Delays

– 3.5 pts Blank

Question 4

Joint Logistics Task Force

25 / 25 pts

4.1 — 4A

22.5 / 22.5 pts

✓ — 0 pts Correct

- 2.5 pts Did not explicitly answer the question
- 5 pts Did not arrive at answer or made no indication of selecting an answer
- 5 pts Incorrectly performed row reduction
- 7.5 pts Incorrectly performed column reduction
- 7.5 pts Performed unnecessary matrix modification
- 7.5 pts Performed steps to Assignment, but then did not demonstrate understanding of the output
- 10 pts Performed steps to Assignment, but then selected teams with 1 or more on task rather than 0, resulting in incorrect answer
- 10 pts Performed steps to Assignment, but then selected teams with the maximum value rather than 0, resulting in incorrect answer
- 10 pts Performed column reduction first rather than row reduction, resulting in incorrect response
- 10 pts After completing row reduction, did not perform column reduction

4.2 — 4B

2.5 / 2.5 pts

✓ — 0 pts Correct

- 0 pts Incorrect, but follows. Should have been $(11+8+8+9)\text{days} * (\$6800/\text{day}) = \$244,800$
- 0.5 pts Math error
- 0.5 pts Question asked for total, not individual; Needed to sum
- 1.5 pts Multiplied by reduced matrix rather than original matrix values
- 2 pts Used incorrect value for number of days with indeterminate origin or set all days to 1
- 2 pts Multiplied by reduced matrix rather than original matrix values, but also assigned incorrect number of teams to jobs (should be 4 teams for 4 jobs)
- 2.25 pts Restated question without any demonstration of understanding problem
- 2 pts Incorrect response with no work shown
- 2.5 pts Blank

Question 5

Drones

20 / 20 pts

5.1 — 5A

5 / 5 pts

✓ - 0 pts Correct

- 2 pts Demand was in Days, Lead Time was in Hours; inconsistent units caused a 24x increase in number of kanbans
- 1 pt Safety stock is a percent of the demand during lead time ($d \times L$) better thought of as $(d \times L) \times (1 + S) / c$
- 1 pt Incorrect Container size (needed to adjust to 85% of 50 ~ 42 or 43)
- 0.5 pts Cannot have fractional Kanbans. Instructions say to round up
- 2.5 pts Question asks for Type A and Type B, only answerd one consolidated version
- 2.5 pts Did not factor in lead time in your calculations or used incorrect lead time
- 4.5 pts Used incorrect equation
- 0.5 pts Minor math error leading to slightly incorrect response. Correct response was 8 Kanbans of Type A, 4 Kanbans of Type B
- 1.5 pts Incorrect value for demand
- 2 pts demand was in days, lead time was converted to minutes rather than days, resulting in a 1000x increase in number of kanbans

5.2 — 5B

10 / 10 pts

✓ - 0 pts Correct

- 4 pts Lead time is in hours, demand is in days, need to be converted to the same unit
- 2 pts incorrect safety stock calculation
- 1.5 pts Incorrect container size calculation
- 0 pts Incorrect, but follows
- 9 pts Incorrect equation used or insufficient work shown
- 5 pts Did not split into type A and type B
- 6 pts Did not incorporate Lead Time
- 3.5 pts Incorrectly applied 25% increase.

5.3 — 5C

5 / 5 pts

✓ - 0 pts Correct

- 2.5 pts 1/2 Correct
- 3.3 pts 1/3 Correct
- 1.7 pts 2/3 Correct
- 5 pts Incorrect

EM420 Production and Operations Management
Written Partial Review
Version C

- 1) This is a 75 minute graded event worth 150 points.
- 2) This is a closed book and closed notes exam. You may use one single sided 3x5" index cards and the provided FE book (DO NOT MARK IN IT!) and an authorized scientific calculator. No collaboration is allowed.
- 3) You may disassemble this document if you'd like. You may also use extra paper provided at the front of the classroom. Please ensure you staple the exam back together in the correct order with extra sheets labeled and at the back.
- 4) **SHOW ALL WORK!** Unsupported or missing work earns little or no partial credit.
- 5) **Rounding.** Show dollars and cents for any monetary answer (\$12.75). Show two significant digits beyond the decimal for percentage answers (12.75%) and at least two significant digits beyond the decimal for all other answers (0.13) unless otherwise noted
- 7) Upon completion, sign the acknowledgment statement below:

"I did not use any sources, nor did I receive any assistance while completing this WPR. This WPR will be released from academic security on 21NOV2025. Prior to that time, I may not discuss any aspects of this exam with anyone except an EM420 instructor."

Printed Name: sarah chernikSignature: *Sarah Chernik*Date Signed: 11/17/25

8) GRADING SUMMARY:

Section	Topic	Total Points Available	Total Points Earned
1	Situation Assessment	30	
2	Multiple Choice & Short Answer	45	
3	Operational Readiness	30	
4	Joint Logistics Task Force	25	
5	Drones	20	
	Total	150	



Answer Sheet for Section 1 & Section 2

1A line balancing

1B Johnsons rule

1C point of indifference

1D kaizen

1E flow time

1F outsourcing

1G gantt chart

1G Situation:

As a PM you will have to plan a training event such as a range. This range will take a good amount of time to plan and you will have tasks you need to complete concurrently. To help solve this, you can use a gantt chart to keep organizing your tasks and have a clear schedule, allowing you to stay on track.

2A EBDAC

2B B

2C B, C, F

2D B

2E A, C, E

2F A

2G 2000 units

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Johnson's

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point of indifference

break even analysis



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1G [6 Points] - Create your own situation which you think you would encounter as a 2LT and provide an answer (from the glossary above - do not use one already used in 1A-F). You will be graded for clarity and correctness. Make sure it is clear why it would be your term and no other terms.

As a 2LT you could have to plan a training event such as a range. This range will take a good amount of time to plan and you will have tasks you need to complete concurrently. To help solve this issue you can use a gantt chart to help organize your tasks and make a clear schedule, allowing you to stay on track.

**Section 2: Multiple Choice & Short Answer (No Partial Credit 45 Points)****Record all answers on Page 2, on the answer sheet.**

2A [10 points] You are sequencing five repair jobs through two sequential maintenance stations. The processing times (in hours) are shown below.

Job	Station 1	Station 2
A	8	6
B	4	7
C	5	3
D	7	9
E	2	5

E B D A C

Determine the optimal sequence of jobs that minimizes total makespan. Write the order of jobs from first to last.

E B D A C

2B [5 Points] You are designing a production line for assembling portable radios. Each station has a desired cycle time of 40 seconds. The task list below shows the operation times for each required activity.

Task	Operation	Time (sec)	Requires Special Tool?
A	Mount circuit board	10	No
B	Install speaker	6	No
C	Attach antenna	8	Yes
D	Screw in faceplate	7	No
E	Test sound output	9	Yes
F	Package product	6	No

What is the **minimum number of workstations** required to complete the line without exceeding the cycle time?

$$N = \frac{\sum t_i}{C} = \frac{46}{40} = 1.15 \sim 2$$

- a. 1
- ☒ b. 2
- c. 3
- d. 4
- e. 6



2C [5 Points] Which of the following is **always true** when you are dealing with two sequential machines? Write all correct answer(s).

- a. All idle time for Machine 1 is always at the beginning
- ☒ b. All idle time for Machine 2 is always at the beginning
- ☒ c. All idle time for Machine 1 is always at the end
- d. All idle time for Machine 2 is always at the end
- e. Makespan is determined by the finishing time of the last job on Machine 1
- ☒ f. Makespan is determined by the finishing time of the last job on Machine 2

2D [5 points] A logistics officer is selecting the location for a new supply hub to support three forward bases in an open region with no existing facilities. The goal is to minimize total transportation distance, considering how much each base ships per week. Use traditional rounding standards. (round up if .5 or above, round down otherwise). Select the correct strategy and location.

Base	X-Coordinate	Y-Coordinate	Weekly Shipments (tons)
Alpha	4	8	30
Bravo	10	4	50
Charlie	6	12	20

- a. Load Distance; (7,7) New Point
- ☒ b. Center of Gravity; (7,7) New Point
- c. Load Distance, (4,8) Alpha
- d. Center of Gravity, (4,8) Alpha

$$\frac{(4 \times 30) + (10 \times 50) + (6 \times 20)}{100} = 7.4$$

$$\frac{(8 \times 30) + (4 \times 50) + (12 \times 20)}{100} = 6.8$$

2E [5 points] You're leading a **senior capstone team** working on an engineering design project for a real client. Each cadet on your team approaches the project differently according to the descriptions below

- ☒ a. Cadet A often asks for additional responsibilities and suggests design improvements beyond the assignment.
- ☒ b. Cadet B needs frequent reminders to submit weekly reports and only works when directly supervised.
- ☒ c. Cadet C volunteers to organize the client presentation and stays late to ensure everything is perfect.
- ☒ d. Cadet D avoids taking initiative, saying, "Just tell me exactly what to do."
- ☒ e. Cadet E proactively schedules team meetings and checks in with everyone to help coordinate progress.

Which cadet(s) demonstrate Theory Y behaviors? A, C, E



2F [5 Points] The table below shows part of a Muther's Relationship Chart for a maintenance facility. The letters indicate the closeness rating between departments. If the engineer can only place two departments side by side due to space limitations, which pair should be prioritized for adjacency based on the chart?

Department	Welding	Paint	Inspection	Storage
Welding	—	A	E	O
Paint	A	—	I	U
Inspection	E	I	—	O
Storage	O	U	O	—

- a. Paint and Welding
- b. Paint and Inspection
- c. Inspection and Storage
- d. Storage and Paint
- e. Any pairing would be adequate

2G [10 Points] A maintenance parts supplier is considering producing its own replacement valves rather than purchasing them. The company would need to invest \$120,000 in new equipment. Each valve would cost \$35 to produce and could be sold for \$95. What is the break-even quantity of valves (in units) the supplier must produce and sell to recover its investment?

Invest: 120000
valve cost: \$35 each
valve sold: \$95

$$V = \frac{CF}{P - CV}$$

$$V = \frac{120000}{95 - 35} = 2000 \text{ units}$$

**Section 3: Operational Readiness (30 Points)**

You are managing four repair tasks for a brigade support battalion ahead of a large training exercise. Each task must be completed before deployment to maintain operational readiness. Today's date is May 1st. There are 31 days in May.

Maintenance Task	Repair Time (days)	Required Completion Date
A. Transmission Replacement	7 days	May 10th
B. Optics Calibration	14 days	May 18th
C. Electrical Overhaul	5 days	May 15th
D. Suspension Alignment	20 days	June 5th

3A [3 points] What is the order according to Longest Processing Time?

LPT

Job	Start	processing	completion	due	tardiness
D	may 1	20	may 21	June 5	0
B	may 21	14	June 4	May 18	17
A	June 4	7	June 11	May 10	32
C	June 11	5	June 16	May 15	32
					81

D, B, A, C

21
17
32

3B [5 points] What is the order according to SLACK?

$$\text{SLACK} = (\text{due date} - \text{today's date}) - \text{processing time}$$

$$A \quad 10 - 1 - 7 = 2$$

$$B \quad 18 - 1 - 14 = 3$$

$$C \quad 15 - 1 - 5 = 9$$

$$D \quad 36 - 1 - 20 = 15$$

A B C D

	Start	processing	completion	due	slack
A	may 1	7	may 8	may 10	0
B	may 8	14	may 22	may 18	4
C	may 22	5	may 27	may 15	12
D	may 27	20	June 16	June 5	11

27

**3C [3 points]** What is the order according to DUE DATE?

A C B D

	start	processing	completion	due	readiness
A	1	7	8	10	0
C	8	5	13	15	0
B	13	14	27	18	9
D	27	20	16	5	11

20

3D [12 points] Fill out the following Table (show work under parts A, B and C).

Sequencing Rule	# of Late Assignments	Total Number of Late Days
LPT	3	81
SLACK	3	27
DUE DATE	2	20

3E [3.5 points] The battalion maintenance section assigns one readiness deduction for every three total days of lateness across all repair tasks. For example, if one task is one day late and another task is one day late, their combined two late days still count toward the three-day threshold for a single deduction. Which sequencing rule should you use to minimize the total number of late days and therefore minimize readiness deductions? What is the total number of readiness deductions associated with this rule. Explain your reasoning.

SPT
 C 1 5 6 15 0
 A 6 7 13 10 3
 B 13 14 27 18 9
 D 27 20 16 5 11
 This still gives you 23
 more than Duedate

I would use due date to minimize the number of late days. This would give me 6 readiness deductions, which is the shortest of the other sequences of solving. This is the only way to keep a low amount of late days with the big jump from may to June due dates.

3F [3.5 Points] The battalion's operations officer tracks mission delays based on whether each repair task is completed on time (binary: yes or no). Each late task, regardless of how many days late, counts as one mission delay. Which scheduling strategy minimizes the total number of mission delays? How many mission delays occur with this strategy?

CUST PR
 A 1 7 8 10 0
 C 8 5 13 15 0
 D 15 20 4 5 0
 B 4 14 19 18 31
 June not

I would use CUST PR to minimize the total number of mission delays. This gives me a sequence of A, C, D, B and has only 1 late assignment. Since it is binary, the number of days late were a concern, so this would be the best option.

**Section 4: Joint Logistics Task Force (25 Points)**

You are advising a Joint Logistics Task Force (JLTF) preparing for a multinational exercise. Four regional logistics hubs (Teams A–D) must each take responsibility for one operational task. Your goal is to minimize total project completion time while ensuring that each team is assigned to exactly one task. Below are the time in days that each team takes to do each operational task.

Team	Supply Chain Setup	Equipment Calibration	Site Construction	Vehicle Readiness
A	12	10	14	11
B	8	13	9	12
C	9	11	8	10
D	10	9	13	12

[22.5 points] Use the Load Assignment Method (Hungarian Method) to determine which team should be assigned to each task to minimize total project duration.

-10	12	10	14	11	0	0	0	1	↓	2	0	4	0
-8	8	13	9	12	0	5	1	4		0	5	1	3
-8	9	11	8	10	1	3	0	2		1	3	0	1
-9	10	9	13	12	1	0	4	3		1	0	4	2

team A (vehicle readiness): 11
 team B (supply chain setup): 8
 team C (site construction): 8
 team D (equipment calibration): 9

total duration: 36 days

[2.5 points] If the daily operating cost of each team is \$6,800 per day, what is the total operational cost of these assignments based on your final solution?

$$36(6800) = \$244,800$$

**Section 5: Drones (20 Points)**

A maintenance company sustains 60 tactical drones deployed across multiple sites. To prevent downtime, it uses a pull-based Kanban system for replenishing two types of pre-packed battery modules: Type A for standard operations and Type B for long-range missions. On a typical day, the unit consumes about 1,200 battery modules, with roughly two-thirds being Type A and the rest Type B. Each sealed container can hold 50 modules, though only about 85 percent of that space is usable once padding, labels, and inspection inserts are added. Once a Kanban card is pulled, it takes about eight hours for a full container to be refilled, quality-checked, and returned to the line. To hedge against variability and delays, the maintenance officer applies a 20 percent safety margin.

5A [5 pts] Determine the number of Kanban cards needed for each module type to sustain continuous flow without stockouts. Round up, provide a reason why you would round up in practice.

daily: 1200 battery modules
 type A: $1200(2/3) = 800$
 type B: $1200(1/3) = 400$
 container: $50(0.85) = 42.5$
 lead time: $8/24 = 0.333$
 safety: 20%

$$\text{type A} \\ \frac{800 \times 0.333 \times 1.20}{42.5} = 7.52 \sim 8 \text{ kanban cards}$$

$$\text{type B} \\ \frac{400 \times 0.333 \times 1.20}{42.5} = 3.76 \sim 4 \text{ kanban cards}$$

You would round up in practice to allow

for extra idle time. You can't round down because then the # of Kanban cards won't be sufficient but rounding up allows for idle time and still keeps things

5B [10 pts] A new drone model has been introduced that increases total daily battery consumption by 25 percent, while the rest of the system remains unchanged. Recalculate the Kanban requirements for each module type and discuss the operational effects of this higher throughput. Round up.

$$0.25(8) = 2 \text{ hrs}$$

total 10 hrs

$$10/24 = 0.4167$$

$$\text{type A:} \\ \frac{800 \times 0.4167 \times 1.20}{42.5} = 9.4125 \sim 10 \text{ kanban cards}$$

$$\text{type B} \\ \frac{400 \times 0.4167 \times 1.20}{42.5} = 4.706 \sim 5 \text{ kanban cards}$$

5C [5 pts] What two countermeasures could help maintain efficiency without simply adding storage space or containers? (Select all that apply.) No partial credit.

- ☒ Shorten resupply lead time through pre-staging or faster transport cycles
- ☒ Reduce container size to increase turnover and responsiveness
- ☐ Increase safety margin from 20 % to 30 %
- ☐ Add an extra storage bay near the maintenance line
- ☐ Increase total on-hand inventory to cover two days of demand

Fair-to-Good Student Work

WPR2

● Graded

WPR 2

Student

Harrison Faulkner

Total Points

120.7 / 150 pts

Question 1

Situational Assessment

26 / 30 pts

1.1 — 1A 4 / 4 pts

✓ - 0 pts Correct

- 4 pts Incorrect Response; Correct Answer is Line Balancing

- 3 pts Incorrect Response; Leveling (Heijunka) concept relates, but it's not the right tool for task reassignment. Correct answer was Line Balancing

1.2 — 1B 4 / 4 pts

✓ - 0 pts Correct

- 3 pts General sequencing idea, but not the specific 2-machine rule (Johnson's); Correct answer is Johnson's Rule

- 4 pts Incorrect; Correct response is Johnson's Rule

- 4 pts Blank; Correct response is Johnson's Rule

1.3 — 1C 4 / 4 pts

✓ - 0 pts Correct

- 4 pts Incorrect; Correct response is Point of Indifference

- 3 pts Incorrect; Right domain (cost crossover), but wrong response. Correct response is Point of Indifference

1.4 — 1D 0 / 4 pts

- 0 pts Correct

✓ - 4 pts Incorrect; correct response is Kaizen

- 2 pts 1/2 Correct; Correct response is Kaizen

1.5 — 1E 4 / 4 pts

✓ - 0 pts Correct

- 4 pts Incorrect; Correct response is Flow Time

- 3.5 pts Incorrect; Related to timing, but **lead time** is external-order based, not total time in the system. Correct response is Flow Time- 3 pts Incorrect; **Cycle time** is at the workstation level; it doesn't include waiting like flow time does. Correct response is Flow Time- 3.5 pts Incorrect; **makespan** applies to scheduling completion time, not total time a unit spends in the system. Correct response is Flow Time

1.6 — 1F 4 / 4 pts

✓ - 0 pts Correct

- 4 pts Incorrect; Correct response is Outsourcing

✓ - 0 pts Correct; Well articulated response

- 2 pts Generally Correct; Response poorly considered, poorly articulated, or ill-fit to the question
- 3 pts Response does not demonstrate or reasonably articulate understanding of chosen concept
- 4 pts Term but no situation
- 6 pts Blank

Question 2

Multiple Choice and Short Answer

42.5 / 45 pts

2.1 — 2A

10 / 10 pts

✓ - 0 pts Correct

- 8 pts Ordered left to right using shortest job, but ignored station ordering (left if on station 1, right if on station 2). Correct response was E B D A C

- 8 pts Ordered right to left using shortest job, but ignored station ordering (left if on station 1, right if on station 2). Correct response was E B D A C

- 9 pts Incorrect; Correct response was E B D A C

- 9 pts Incorrect; ordered based on lowest time on station 2 only. Correct response was E B D A C

- 5 pts Incorrect; Mild reversal. Correct answer is E B D A C

- 4 pts Added an additional letter, but generally correct. Correct answer is E B D A C

2.2 — 2B

5 / 5 pts

✓ - 0 pts Correct

- 5 pts Incorrect. Likely rounded down rather than up. Minimum was 2 (B)

- 5 pts Incorrect. Minimum was 2 (B)

2.3 — 2C

5 / 5 pts

✓ - 0 pts Correct

- 2.5 pts One correct; Either C or F

- 1.5 pts Additional incorrect response; C and F plus a wrong response

- 4 pts Incorrect; correct response is C and F

- 3 pts One correct, Either C or F, plus multiple incorrect responses

2.4 — 2D

5 / 5 pts

✓ - 0 pts Correct

- 4 pts Used the CoG formulation, but identified it as LD. Correct response is Center of Gravity, (7, 7), New Point (B)

$$X = \text{Sum}[WiXi]/\text{Sum}[Wi]$$

$$Y = \text{Sum}[WiYi]/\text{Sum}[Wi]$$

- 4.5 pts Identified correct technique (CoG), but did not correctly apply mathematical formulation. Correct response is Center of Gravity, (7, 7), New Point (B)

$$X = \text{Sum}[WiXi]/\text{Sum}[Wi]$$

$$Y = \text{Sum}[WiYi]/\text{Sum}[Wi]$$

- 5 pts Incorrect. Correct response is Center of Gravity, (7, 7), New Point (B)

$$X = \text{Sum}[WiXi]/\text{Sum}[Wi]$$

$$Y = \text{Sum}[WiYi]/\text{Sum}[Wi]$$

2.5 — **2E** 2.5 / 5 pts

✓ — **0 pts** Correct

✓ — **- 2.5 pts** Incorrect. Response indicates Theory X Cadets. Correct response for Theory Y is CDT A, C, and E

— **- 5 pts** Incorrect. Response mixed both Theory X and Theory Y

2.6 — **2F** 5 / 5 pts

✓ — **0 pts** Correct

— **- 5 pts** Incorrect. Murther's Grid is read alphabetically as:

A: Absolutely Necessary

E: Essential

I: Important

O: Okay

U: Unnecessary

X: Undesirable

Correct response indicates 'A-Absolutely Necessary' intersection at Paint and Welding (A)

2.7 — **2G** 10 / 10 pts

✓ — **0 pts** Correct

— **- 1 pt** Missing units

— **- 3 pts** Multiplied in the denominator rather than subtracted, resulting in incorrect response. Correct response is 2,000 valves.

— **- 3 pts** Dropped a zero from the fixed cost, resulting in response being off by a factor of 10. Correct response is 2,000 valves.

— **- 3 pts** Subtraction error, resulting in mildly incorrect response. Correct response is 2,000 valves

— **- 5 pts** Incorrect response with acceptable formulation. Unclear error origin. Correct response is 2,000 valves

— **- 10 pts** Incomplete response with incorrect formulation

— **- 10 pts** Incorrect response with incorrect formulation

— **- 10 pts** Incorrect response with no supporting work shown.

Question 3

Operational Readiness

22 / 30 pts

3.1 — 3A

3 / 3 pts

✓ - 0 pts Correct

- 3 pts This is the order according to SPT - Shortest Processing Time. Correct Response is D-B-A-C

3.2 — 3B

1 / 5 pts

- 0 pts Correct

✓ - 4 pts Incorrect; ordered as most slack first. Should have been ordered as least slack first, or A, B, C, D

- 5 pts Incorrect. Correct response would be least slack first A(3)-B(4)-C(10)-D(16)

3.3 — 3C

3 / 3 pts

✓ - 0 pts Correct

- 3 pts Incorrect; Nearest Due Date results in A, C, B, D

3.4 — 3D LPT # Late

2 / 2 pts

✓ - 0 pts Correct

- 2 pts Blank; Should be 3

- 2 pts Incorrect; should be 3

- 0.5 pts Swapped # Late and Late Days, but demonstrated understanding

3.5 — 3D Slack # Late

2 / 2 pts

✓ - 0 pts Correct

- 1 pt Undercounted but generally demonstrated understanding. Correct response is 3

- 2 pts Blank; correct response is 3

- 2 pts Incorrect; correct response is 3

- 2 pts Swapped # of Assignments and Late Days, but was incorrect in either case

- 1.25 pts Follows, but undercounted even given revised order

- 1.5 pts Follows, but overcounted even given revised order

- 1.5 pts Used A's Slack as duration, shifting all other responses left

3.6 — 3D Due Date # Late 2 / 2 pts

✓ - 0 pts Correct

- 1.25 pts Overcount, but demonstrated understanding; correct response is 2
- 1.25 pts Undercount, but demonstrated understanding; correct response is 2
- 2 pts blank
- 2 pts Incorrect; correct response is 2
- 2 pts Swapped; but incorrect value
- 0 pts Incorrect; follows

3.7 — 3D LTP Tardiness 0 / 2 pts

- 0 pts Correct, +/-3 days (counting starting date at day 0 = 78 days; counting starting at day 1 = 81 days)
- 2 pts Blank

✓ - 2 pts Incorrect; correct response is approximately 81 days

- 1.25 pts Incorrect, but technique demonstrated understanding of concept
- 0.75 pts Swapped Number of Late Days and Late Assignments, but correct number
- 0 pts Incorrect, but follows

3.8 — 3D Slack Tardiness 0 / 2 pts

- 0 pts Correct within +/-3 of accepted response
- 2 pts Blank

✓ - 2 pts Incorrect; correct response should be approximately 27 days

- 1.25 pts Incorrect, but response demonstrated understanding of concept
- 0 pts Incorrect, but follows
- 2 pts Swapped responses; Incorrect value due to inputting May 5 for June 5 as due date for D

3.9 — 3D Due Date Tardiness 2 / 2 pts

✓ - 0 pts Correct within +/-3 of 20 days

- 1.25 pts Incorrect, but demonstrates understanding of the concept
- 2 pts Blank
- 2 pts Incorrect; correct response is 20 days
- 0 pts incorrect, but follows
- 2 pts Swapped, but completely incorrect

3.10 3E

3.5 / 3.5 pts

✓ - 0 pts Correct

- 2 pts Correct rule, incorrect reasoning or readiness deduction
- 3.5 pts Incorrect
- 3.5 pts Blank
- 0 pts Incorrect, or partly incorrect, but follows
- 2 pts Incorrect, or partly incorrect, but follows, without sufficient explanation

3.11 3F

3.5 / 3.5 pts

✓ - 0 pts Correct

- 0 pts Outside the box solution, but Great idea! A C D B does result in the fewest late tasks
- 0 pts Incorrect, but follows
- 1.5 pts Related concept applied, but incorrect choice given question
- 1 pt Correct scheduling strategy, incorrect number of mission delay.
- 1 pt Did not express number of mission delays
- 2.5 pts Incorrect use of Critical Ratio; would have resulted in 3 mission delays
- 3 pts Incorrect concept applied. Correct response is Due Date with 2 Mission Delays
- 3.5 pts Blank

Question 4

Joint Logistics Task Force

13.5 / 25 pts

4.1 — 4A

12.5 / 22.5 pts

- 0 pts Correct
- 2.5 pts Did not explicitly answer the question
- 5 pts Did not arrive at answer or made no indication of selecting an answer
- 5 pts Incorrectly performed row reduction
- 7.5 pts Incorrectly performed column reduction
- 7.5 pts Performed unnecessary matrix modification
- 7.5 pts Performed steps to Assignment, but then did not demonstrate understanding of the output

✓ – 10 pts Performed steps to Assignment, but then selected teams with 1 or more on task rather than 0, resulting in incorrect answer

- 10 pts Performed steps to Assignment, but then selected teams with the maximum value rather than 0, resulting in incorrect answer
- 10 pts Performed column reduction first rather than row reduction, resulting in incorrect response
- 10 pts After completing row reduction, did not perform column reduction

4.2 — 4B

1 / 2.5 pts

- 0 pts Correct
- 0 pts Incorrect, but follows. Should have been $(11+8+8+9)\text{days} * (\$6800/\text{day}) = \$244,800$
- 0.5 pts Math error
- 0.5 pts Question asked for total, not individual; Needed to sum

✓ – 1.5 pts Multiplied by reduced matrix rather than original matrix values

- 2 pts Used incorrect value for number of days with indeterminate origin or set all days to 1
- 2 pts Multiplied by reduced matrix rather than original matrix values, but also assigned incorrect number of teams to jobs (should be 4 teams for 4 jobs)
- 2.25 pts Restated question without any demonstration of understanding problem
- 2 pts Incorrect response with no work shown
- 2.5 pts Blank

Question 5

Drones

16.7 / 20 pts

5.1 — 5A

5 / 5 pts

✓ - 0 pts Correct

- 2 pts Demand was in Days, Lead Time was in Hours; inconsistent units caused a 24x increase in number of kanbans
- 1 pt Safety stock is a percent of the demand during lead time ($d \times L$) better thought of as $(d \times L) \times (1 + S) / c$
- 1 pt Incorrect Container size (needed to adjust to 85% of 50 ~ 42 or 43)
- 0.5 pts Cannot have fractional Kanbans. Instructions say to round up
- 2.5 pts Question asks for Type A and Type B, only answerd one consolidated version
- 2.5 pts Did not factor in lead time in your calculations or used incorrect lead time
- 4.5 pts Used incorrect equation
- 0.5 pts Minor math error leading to slightly incorrect response. Correct response was 8 Kanbans of Type A, 4 Kanbans of Type B
- 1.5 pts Incorrect value for demand
- 2 pts demand was in days, lead time was converted to minutes rather than days, resulting in a 1000x increase in number of kanbans

5.2 — 5B

10 / 10 pts

✓ - 0 pts Correct

- 4 pts Lead time is in hours, demand is in days, need to be converted to the same unit
- 2 pts incorrect safety stock calculation
- 1.5 pts Incorrect container size calculation
- 0 pts Incorrect, but follows
- 9 pts Incorrect equation used or insufficient work shown
- 5 pts Did not split into type A and type B
- 6 pts Did not incorporate Lead Time
- 3.5 pts Incorrectly applied 25% increase.

5.3 — 5C

1.7 / 5 pts

- 0 pts Correct

- 2.5 pts 1/2 Correct

✓ - 3.3 pts 1/3 Correct

- 1.7 pts 2/3 Correct

- 5 pts Incorrect

EM420 Written Partial Review II

Name: Harrison Faulkner

Academic Term 26-1

Hour: H1

EM420 Production and Operations Management
Written Partial Review
Version C

- 1) This is a 75 minute graded event worth 150 points.
- 2) This is a closed book and closed notes exam. You may use one single sided 3x5" index cards and the provided FE book (DO NOT MARK IN IT!) and an authorized scientific calculator. No collaboration is allowed.
- 3) You may disassemble this document if you'd like. You may also use extra paper provided at the front of the classroom. Please ensure you staple the exam back together in the correct order with extra sheets labeled and at the back.
- 4) **SHOW ALL WORK!** Unsupported or missing work earns little or no partial credit.
- 5) **Rounding.** Show dollars and cents for any monetary answer (\$12.75). Show two significant digits beyond the decimal for percentage answers (12.75%) and at least two significant digits beyond the decimal for all other answers (0.13) unless otherwise noted
- 7) Upon completion, sign the acknowledgment statement below:

"I did not use any sources, nor did I receive any assistance while completing this WPR. This WPR will be released from academic security on 21NOV2025. Prior to that time, I may not discuss any aspects of this exam with anyone except an EM420 instructor."

Printed Name: Harrison Faulkner

Signature: 

Date Signed: 11/17/26

8) GRADING SUMMARY:

Section	Topic	Total Points Available	Total Points Earned
1	Situation Assessment	30	
2	Multiple Choice & Short Answer	45	
3	Operational Readiness	30	
4	Joint Logistics Task Force	25	
5	Drones	20	
	Total	150	



Answer Sheet for Section 1 & Section 2

1A Line Balancing

1B Johnson's Rule

1C Point of Indifference

1D Empowerment

1E Flow time

1F Outsourcing

1G Job Rotation

1G Situation: My soldiers are getting bored with their jobs so I switch up their positions to grow experience and enjoyment → Job Rotation

2A EBDAC

2B B

2C C, F

2D B

2E B, D

2F B A

2G 2000 units

**Section 1: Situation Assessment [30 points]**

In the following situations, please choose the BEST problem-solving strategy, chart, or calculation that would aid in making a decision for the problem at hand. A glossary is provided below for your reference. There is one best answer for each situation. No partial credit will be provided. Explanation is not necessary.

Alternative Workplace	FMEA	Load Distance
Block Diagramming	Gantt Chart	Location Factors
Break Even Analysis	Heijunka	Movement
Center of Gravity	Horizontal Enlargement	Muther's Diagram
Commissions	House of Quality	Outsourcing
Cross Training	Hygiene Factors	Point(s) of Indifference
Cycle Time	Individual Incentive	Product Process Matrix
Defects	Inventory	Profit Sharing
Economies of Scale	Job Rotation	Service Blueprint
Ellsberg's Paradox	Job Training	Service-Process Matrix
Empowerment	Johnson's Rule	Simple Sequencing Rules
EVPI	Kaizen	Theory X Employee
Fixed-Position Layout	Kanban System	Theory Y Employee
Flexible Work Schedules	Line Balancing	Vertical Enlargement
Flow Time	Load Assignment Method	Value Stream Mapping

Record all answers on Page 2, on the answer sheet.

1A [4 Points] – A defense electronics plant is assembling communication modules on a high-volume line. Some stations are idle while others have long queues. The line chief needs to reassign tasks to balance workloads, meet the desired cycle time, and reduce idle time. What method should be used?

1B [4 Points] – A naval drone production unit must process each unit through two sequential operations: assembly and inspection. Each operation takes a different amount of time for each drone type. The operations officer wants to find the sequence that minimizes total completion time and idle time across both stations. What strategy should be used?

1C [4 Points] – A defense contractor compares two drone assembly methods: one uses manual labor with lower setup costs but higher variable costs, and the other uses automated assembly with higher setup costs but lower variable costs. The manager wants to find the exact output quantity where both options cost the same before deciding which to use. What should he use?



1D [4 Points] - At an Army ground vehicle maintenance company, crews report frequent delays during scheduled services due to disorganized toolkits, unclear part storage locations, and repeated backtracking between work areas. Instead of waiting for a full facility renovation, the platoon sergeant begins holding short morning huddles where Soldiers suggest small changes, such as labeling bins, rearranging benches, and creating visual checklists, to reduce wasted time and motion. Over the next month, turnaround time improves by 12%. Which continuous improvement strategy best fits this situation?

1E [4 Points] - A process improvement specialist wants to assess total process time including both work and waiting for a batch of products. What performance measure best captures this total elapsed time in the system?

1F [4 Points] - An Army logistics battalion's data-tracking system for vehicle parts is outdated and unreliable. Rather than developing new software in-house, leadership is considering contracting a civilian tech company that specializes in military logistics systems to design, operate, and maintain it. The logistics battalion is participating in:

1G [6 Points] - Create your own situation which you think you would encounter as a 2LT and provide an answer (from the glossary above - do not use one already used in 1A-F). You will be graded for clarity and correctness. Make sure it is clear why it would be your term and no other terms.

**Section 2: Multiple Choice & Short Answer (No Partial Credit 45 Points)****Record all answers on Page 2, on the answer sheet.**

2A [10 points] You are sequencing five repair jobs through two sequential maintenance stations. The processing times (in hours) are shown below.

Job	Station 1	Station 2
A	8	6
B	4	7
C	5	3
D	7	9
E	2	5

E B D A C

Determine the optimal sequence of jobs that minimizes total makespan. Write the order of jobs from first to last.

2B [5 Points] You are designing a production line for assembling portable radios. Each station has a desired cycle time of 40 seconds. The task list below shows the operation times for each required activity.

Task	Operation	Time (sec)	Requires Special Tool?
A	Mount circuit board	10	No
B	Install speaker	6	No
C	Attach antenna	8	Yes
D	Screw in faceplate	7	No
E	Test sound output	9	Yes
F	Package product	6	No

$$\sum t_i = 46$$
$$C_d = 40$$

$$\frac{46}{40} = 1.15 = 2$$

What is the **minimum number of workstations** required to complete the line without exceeding the cycle time?

- a. 1
- b. 2**
- c. 3
- d. 4
- e. 6



2C [5 Points] Which of the following is always true when you are dealing with two sequential machines? Write all correct answer(s).

- a. All idle time for Machine 1 is always at the beginning
- b. All idle time for Machine 2 is always at the beginning
- ☒ c. All idle time for Machine 1 is always at the end
- d. All idle time for Machine 2 is always at the end
- e. Makespan is determined by the finishing time of the last job on Machine 1
- ☒ f. Makespan is determined by the finishing time of the last job on Machine 2

2D [5 points] A logistics officer is selecting the location for a new supply hub to support three forward bases in an open region with no existing facilities. The goal is to minimize total transportation distance, considering how much each base ships per week. Use traditional rounding standards. (round up if .5 or above, round down otherwise). Select the correct strategy and location.

$$4 \cdot 30 + 10 \cdot 50 + 6 \cdot 20 = 74 \quad 240 + 200 + 240 = 680 \quad \frac{680}{100} = 6.8$$

Base	X-Coordinate	Y-Coordinate	Weekly Shipments (tons)
Alpha	4	8	30
Bravo	10	4	50
Charlie	6	12	20

- a. Load Distance; (7,7) New Point
- ☒ b. Center of Gravity; (7,7) New Point
- c. Load Distance, (4,8) Alpha
- d. Center of Gravity, (4,8) Alpha

2E [5 points] You're leading a **senior capstone team** working on an engineering design project for a real client. Each cadet on your team approaches the project differently according to the descriptions below

- Cadet A often asks for additional responsibilities and suggests design improvements beyond the assignment.
- Cadet B needs frequent reminders to submit weekly reports and only works when directly supervised.
- Cadet C volunteers to organize the client presentation and stays late to ensure everything is perfect.
- Cadet D avoids taking initiative, saying, "Just tell me exactly what to do."
- Cadet E proactively schedules team meetings and checks in with everyone to help coordinate progress.

Which cadet(s) demonstrate Theory Y behaviors?



2F [5 Points] The table below shows part of a Muther's Relationship Chart for a maintenance facility. The letters indicate the closeness rating between departments. If the engineer can only place two departments side by side due to space limitations, which pair should be prioritized for adjacency based on the chart?

Department	Welding	Paint	Inspection	Storage
Welding	—	A	E	O
Paint	A	—	I	U
Inspection	E	I	—	O
Storage	O	U	O	—

- a. Paint and Welding
- b. Paint and Inspection
- c. Inspection and Storage
- d. Storage and Paint
- e. Any pairing would be adequate

2G [10 Points] A maintenance parts supplier is considering producing its own replacement valves rather than purchasing them. The company would need to invest \$120,000 in new equipment. Each valve would cost \$35 to produce and could be sold for \$95. What is the break-even quantity of valves (in units) the supplier must produce and sell to recover its investment?

$$V = \frac{120,000}{95 - 35} = 2000 \text{ units}$$

**Section 3: Operational Readiness (30 Points)**

You are managing four repair tasks for a brigade support battalion ahead of a large training exercise. Each task must be completed before deployment to maintain operational readiness. Today's date is May 1st. There are 31 days in May.

Maintenance Task	Repair Time (days)	Required Completion Date	SLACK
A. Transmission Replacement	7 days	May 10th	2
B. Optics Calibration	14 days	May 18th	3
C. Electrical Overhaul	5 days	May 15th	9
D. Suspension Alignment	20 days	June 5th	19

3A [3 points] What is the order according to Longest Processing Time?

D, B, A, C

$\text{May } 1 + 20 = \text{MAY } 21 + 14 = \text{June } 3 + 7 = \text{June } 10 + 5 = \text{June } 15$
 $A = 11 \text{ days late}$
 $B = 3 \text{ days late}$
 $C = 6 \text{ days late}$
 $A = 24 \text{ days late}$
 $C = 19 \text{ days late}$
 $C = 26 \text{ days late}$
 $C = 31 \text{ days late}$
 $5 + 26 + 24 + 3 = 58 \text{ days late}$

3B [5 points] What is the order according to SLACK?

D, C, B, A

$\text{May } 1 + 20 = \text{May } 21 + 5 = \text{May } 26 + 14 = \text{June } 9 + 7 = \text{June } 16$
 $A = 11 \text{ late}$
 $B = 3 \text{ late}$
 $C = 6 \text{ late}$
 $A = 16 \text{ late}$
 $B = 8 \text{ late}$
 $A = 30 \text{ late}$
 $A = 37 \text{ days late}$
 $37 + 8 + 6 = 51 \text{ days late}$



3C [3 points] What is the order according to DUE DATE?

A, C, B, D

May 1 + 7 = May 8 + 5 = May 13 + 14 = May 27 + 20 = June 16
None late None late B = 9 days late D = 11 days late

3D [12 points] Fill out the following Table (show work under parts A, B and C).

Sequencing Rule	# of Late Assignments	Total Number of Late Days
LPT	3	58
SLACK	3	51
DUE DATE	2	20

3E [3.5 points] The battalion maintenance section assigns one readiness deduction for every three total days of lateness across all repair tasks. For example, if one task is one day late and another task is one day late, their combined two late days still count toward the three-day threshold for a single deduction. Which sequencing rule should you use to minimize the total number of late days and therefore minimize readiness deductions? What is the total number of readiness deductions associated with this rule. Explain your reasoning.

Due date, it has the lowest delays/lates which makes sense because you're knocking out the tasks due in their order. We would have 6 readiness reductions.

3F [3.5 Points] The battalion's operations officer tracks mission delays based on whether each repair task is completed on time (binary: yes or no). Each late task, regardless of how many days late, counts as one mission delay. Which scheduling strategy minimizes the total number of mission delays? How many mission delays occur with this strategy?

Due date has the least late tasks with 2 as opposed to 3 which parallels the mission delays.

**Section 4: Joint Logistics Task Force (25 Points)**

You are advising a Joint Logistics Task Force (JLTF) preparing for a multinational exercise. Four regional logistics hubs (Teams A–D) must each take responsibility for one operational task. Your goal is to minimize total project completion time while ensuring that each team is assigned to exactly one task. Below are the time in days that each team takes to do each operational task.

Team	Supply Chain Setup	Equipment Calibration	Site Construction	Vehicle Readiness
A	12 2 2	10 0 0	14 4 4	11 1 0
B	8 0 0	13 5 5	9 1 1	12 4 3
C	9 1 1	11 3 3	8 0	10 2 1
D	10 1 1	9 0 0	13 4 4	12 3 2

[22.5 points] Use the Load Assignment Method (Hungarian Method) to determine which team should be assigned to each task to minimize total project duration.

Row Reduce
Column Reduce
Circle Zeros

A Supply Chain 2
B Site construction 1
C Equipment Calibration 3
D Vehicle Readiness 2

[2.5 points] If the daily operating cost of each team is \$6,800 per day, what is the total operational cost of these assignments based on your final solution?

$$(2+1+3+2) \cdot 6800 = \underline{\underline{\$54,400}}$$

**Section 5: Drones (20 Points)**

A maintenance company sustains 60 tactical drones deployed across multiple sites. To prevent downtime, it uses a pull-based Kanban system for replenishing two types of pre-packed battery modules: Type A for standard operations and Type B for long-range missions. On a typical day, the unit consumes about 1,200 battery modules, with roughly two-thirds being Type A and the rest Type B. Each sealed container can hold 50 modules, though only about 85 percent of that space is usable once padding, labels, and inspection inserts are added. Once a Kanban card is pulled, it takes about eight hours for a full container to be refilled, quality-checked, and returned to the line. To hedge against variability and delays, the maintenance officer applies a 20 percent safety margin.

5A [5 pts] Determine the number of Kanban cards needed for **each** module type to sustain continuous flow without stockouts. Round up, **provide a reason why you would round up in practice.**

$A = 800$
 $B = 400$
 $50 \cdot 85 = 42.5 \approx 42$
 $\frac{1 \text{ day } 800 \text{ Bat}}{24 \text{ hrs } 1 \text{ day}} \approx 33.3 \approx 34$
 $400 / 24 = 16.6 \approx 17$

$$\frac{\bar{d} \cdot L + ts}{C} \rightarrow \frac{348 \cdot 8 \text{ hrs} + 558}{42} = 7.78 = 8^k$$

$$\frac{17 \cdot 8 + 28}{42} = 3.9 = 4 \text{ Kanbans}$$

We would round up to have a safety net against not having enough batteries.

5B [10 pts] A new drone model has been introduced that increases total daily battery consumption by 25 percent, while the rest of the system remains unchanged. Recalculate the Kanban requirements for each module type and discuss the operational effects of this higher throughput. Round up.

$800 + 200 = 1000$
 $\frac{1000}{24} = 42$
 $400 + 100 = 500$
 $\frac{500}{24} = 21$

$\frac{42 \cdot 8 + 68}{42} = 9.6 = 10^k$
 $\frac{21 \cdot 8 + 34}{42} = 4.8 = 5^k$

The higher throughput will result in necessary additions to production if they want to sustain their operations with the new drone.

5C [5 pts] What two countermeasures could help maintain efficiency without simply adding storage space or containers? (Select all that apply.) No partial credit.

- ☒ Shorten resupply lead time through pre-staging or faster transport cycles
- ☐ ~~Reduce container size to increase turnover and responsiveness~~
- ☐ ~~Increase safety margin from 20 % to 30 %~~
- ☒ Add an extra storage bay near the maintenance line
- ☒ Increase total on-hand inventory to cover two days of demand

Poor Student Work

WPR2

● Graded

WPR 2

Student

Dylan Treat

Total Points

68.5 / 150 pts

Question 1

Situational Assessment

21 / 30 pts

1.1 — **1A** 4 / 4 pts

✓ - 0 pts Correct

- 4 pts Incorrect Response; Correct Answer is Line Balancing

- 3 pts Incorrect Response; Leveling (Heijunka) concept relates, but it's not the right tool for task reassignment. Correct answer was Line Balancing

1.2 — **1B** 4 / 4 pts

✓ - 0 pts Correct

- 3 pts General sequencing idea, but not the specific 2-machine rule (Johnson's); Correct answer is Johnson's Rule

- 4 pts Incorrect; Correct response is Johnson's Rule

- 4 pts Blank; Correct response is Johnson's Rule

1.3 — **1C** 4 / 4 pts

✓ - 0 pts Correct

- 4 pts Incorrect; Correct response is Point of Indifference

- 3 pts Incorrect; Right domain (cost crossover), but wrong response. Correct response is Point of Indifference

1.4 — **1D** 0 / 4 pts

- 0 pts Correct

✓ - 4 pts Incorrect; correct response is Kaizen

- 2 pts 1/2 Correct; Correct response is Kaizen

1.5 — **1E** 1 / 4 pts

- 0 pts Correct

- 4 pts Incorrect; Correct response is Flow Time

- 3.5 pts Incorrect; Related to timing, but **lead time** is external-order based, not total time in the system. Correct response is Flow Time

✓ - 3 pts Incorrect; **Cycle time** is at the workstation level; it doesn't include waiting like flow time does. Correct response is Flow Time

- 3.5 pts Incorrect; **makespan** applies to scheduling completion time, not total time a unit spends in the system. Correct response is Flow Time

1.6 — **1F** 4 / 4 pts

✓ - 0 pts Correct

- 4 pts Incorrect; Correct response is Outsourcing

– 0 pts Correct; Well articulated response

✓ – 2 pts Generally Correct; Response poorly considered, poorly articulated, or ill-fit to the question

– 3 pts Response does not demonstrate or reasonably articulate understanding of chosen concept

– 4 pts Term but no situation

– 6 pts Blank

Question 2

Multiple Choice and Short Answer

25 / 45 pts

2.1 — 2A

1 / 10 pts

– 0 pts Correct

– 8 pts Ordered left to right using shortest job, but ignored station ordering (left if on station 1, right if on station 2). Correct response was E B D A C

– 8 pts Ordered right to left using shortest job, but ignored station ordering (left if on station 1, right if on station 2). Correct response was E B D A C

✓ – 9 pts Incorrect; Correct response was E B D A C

– 9 pts Incorrect; ordered based on lowest time on station 2 only. Correct response was E B D A C

– 5 pts Incorrect; Mild reversal. Correct answer is E B D A C

– 4 pts Added an additional letter, but generally correct. Correct answer is E B D A C

2.2 — 2B

0 / 5 pts

– 0 pts Correct

✓ – 5 pts Incorrect. Likely rounded down rather than up. Minimum was 2 (B)

– 5 pts Incorrect. Minimum was 2 (B)

2.3 — 2C

3.5 / 5 pts

– 0 pts Correct

– 2.5 pts One correct; Either C or F

✓ – 1.5 pts Additional incorrect response; C and F plus a wrong response

– 4 pts Incorrect; correct response is C and F

– 3 pts One correct, Either C or F, plus multiple incorrect responses

2.4 — 2D

0.5 / 5 pts

– 0 pts Correct

– 4 pts Used the CoG formulation, but identified it as LD. Correct response is Center of Gravity, (7, 7), New Point (B)

$$X = \text{Sum}[WiXi]/\text{Sum}[Wi]$$

$$Y = \text{Sum}[WiYi]/\text{Sum}[Wi]$$

✓ – 4.5 pts Identified correct technique (CoG), but did not correctly apply mathematical formulation. Correct response is Center of Gravity, (7, 7), New Point (B)

$$X = \text{Sum}[WiXi]/\text{Sum}[Wi]$$

$$Y = \text{Sum}[WiYi]/\text{Sum}[Wi]$$

– 5 pts Incorrect. Correct response is Center of Gravity, (7, 7), New Point (B)

$$X = \text{Sum}[WiXi]/\text{Sum}[Wi]$$

$$Y = \text{Sum}[WiYi]/\text{Sum}[Wi]$$

2.5 — **2E** 5 / 5 pts

✓ - 0 pts Correct

- 2.5 pts Incorrect. Response indicates Theory X Cadets. Correct response for Theory Y is CDT A, C, and E

- 5 pts Incorrect. Response mixed both Theory X and Theory Y

2.6 — **2F** 5 / 5 pts

✓ - 0 pts Correct

- 5 pts Incorrect. Murther's Grid is read alphabetically as:

A: Absolutely Necessary

E: Essential

I: Important

O: Okay

U: Unnecessary

X: Undesirable

Correct response indicates 'A-Absolutely Necessary' intersection at Paint and Welding (A)

2.7 — **2G** 10 / 10 pts

✓ - 0 pts Correct

- 1 pt Missing units

- 3 pts Multiplied in the denominator rather than subtracted, resulting in incorrect response. Correct response is 2,000 valves.

- 3 pts Dropped a zero from the fixed cost, resulting in response being off by a factor of 10. Correct response is 2,000 valves.

- 3 pts Subtraction error, resulting in mildly incorrect response. Correct response is 2,000 valves

- 5 pts Incorrect response with acceptable formulation. Unclear error origin. Correct response is 2,000 valves

- 10 pts Incomplete response with incorrect formulation

- 10 pts Incorrect response with incorrect formulation

- 10 pts Incorrect response with no supporting work shown.

Question 3

Operational Readiness

10.5 / 30 pts

3.1 — 3A

0 / 3 pts

– 0 pts Correct

✓ – 3 pts This is the order according to SPT - Shortest Processing Time. Correct Response is D-B-A-C

3.2 — 3B

5 / 5 pts

✓ – 0 pts Correct

– 4 pts Incorrect; ordered as most slack first. Should have been ordered as least slack first, or A, B, C, D

– 5 pts Incorrect. Correct response would be least slack first A(3)-B(4)-C(10)-D(16)

3.3 — 3C

3 / 3 pts

✓ – 0 pts Correct

– 3 pts Incorrect; Nearest Due Date results in A, C, B, D

3.4 — 3D LPT # Late

2 / 2 pts

✓ – 0 pts Correct

– 2 pts Blank; Should be 3

– 2 pts Incorrect; should be 3

– 0.5 pts Swapped # Late and Late Days, but demonstrated understanding

3.5 — 3D Slack # Late

0 / 2 pts

– 0 pts Correct

– 1 pt Undercounted but generally demonstrated understanding. Correct response is 3

– 2 pts Blank; correct response is 3

✓ – 2 pts Incorrect; correct response is 3

– 2 pts Swapped # of Assignments and Late Days, but was incorrect in either case

– 1.25 pts Follows, but undercounted even given revised order

– 1.5 pts Follows, but overcounted even given revised order

– 1.5 pts Used A's Slack as duration, shifting all other responses left

3.6 — **3D Due Date # Late** 0 / 2 pts

– 0 pts Correct

– 1.25 pts Overcount, but demonstrated understanding; correct response is 2

– 1.25 pts Undercount, but demonstrated understanding; correct response is 2

– 2 pts blank

✓ – 2 pts Incorrect; correct response is 2

– 2 pts Swapped; but incorrect value

– 0 pts Incorrect; follows

3.7 — **3D LTP Tardiness** 0 / 2 pts

– 0 pts Correct, +/-3 days (counting starting date at day 0 = 78 days; counting starting at day 1 = 81 days)

– 2 pts Blank

✓ – 2 pts Incorrect; correct response is approximately 81 days

– 1.25 pts Incorrect, but technique demonstrated understanding of concept

– 0.75 pts Swapped Number of Late Days and Late Assignments, but correct number

– 0 pts Incorrect, but follows

3.8 — **3D Slack Tardiness** 0 / 2 pts

– 0 pts Correct within +/-3 of accepted response

– 2 pts Blank

✓ – 2 pts Incorrect; correct response should be approximately 27 days

– 1.25 pts Incorrect, but response demonstrated understanding of concept

– 0 pts Incorrect, but follows

– 2 pts Swapped responses; Incorrect value due to inputting May 5 for June 5 as due date for D

3.9 — **3D Due Date Tardiness** 0 / 2 pts

– 0 pts Correct within +/-3 of 20 days

– 1.25 pts Incorrect, but demonstrates understanding of the concept

– 2 pts Blank

✓ – 2 pts Incorrect; correct response is 20 days

– 0 pts incorrect, but follows

– 2 pts Swapped, but completely incorrect

3.10 3E

0 / 3.5 pts

– 0 pts Correct

– 2 pts Correct rule, incorrect reasoning or readiness deduction

– 3.5 pts Incorrect

✓ – 3.5 pts Blank

– 0 pts Incorrect, or partly incorrect, but follows

– 2 pts Incorrect, or partly incorrect, but follows, without sufficient explanation

3.11 3F

0.5 / 3.5 pts

– 0 pts Correct

– 0 pts Outside the box solution, but Great idea! A C D B does result in the fewest late tasks

– 0 pts Incorrect, but follows

– 1.5 pts Related concept applied, but incorrect choice given question

– 1 pt Correct scheduling strategy, incorrect number of mission delay.

– 1 pt Did not express number of mission delays

– 2.5 pts Incorrect use of Critical Ratio; would have resulted in 3 mission delays

✓ – 3 pts Incorrect concept applied. Correct response is Due Date with 2 Mission Delays

– 3.5 pts Blank

Question 4

Joint Logistics Task Force

10.5 / 25 pts

4.1 — 4A

10 / 22.5 pts

– 0 pts Correct

– 2.5 pts Did not explicitly answer the question

✓ – 5 pts Did not arrive at answer or made no indication of selecting an answer

– 5 pts Incorrectly performed row reduction

✓ – 7.5 pts Incorrectly performed column reduction

– 7.5 pts Performed unnecessary matrix modification

– 7.5 pts Performed steps to Assignment, but then did not demonstrate understanding of the output

– 10 pts Performed steps to Assignment, but then selected teams with 1 or more on task rather than 0, resulting in incorrect answer

– 10 pts Performed steps to Assignment, but then selected teams with the maximum value rather than 0, resulting in incorrect answer

– 10 pts Performed column reduction first rather than row reduction, resulting in incorrect response

– 10 pts After completing row reduction, did not perform column reduction

4.2 — 4B

0.5 / 2.5 pts

– 0 pts Correct

– 0 pts Incorrect, but follows. Should have been $(11+8+8+9)\text{days} * (\$6800/\text{day}) = \$244,800$

– 0.5 pts Math error

– 0.5 pts Question asked for total, not individual; Needed to sum

– 1.5 pts Multiplied by reduced matrix rather than original matrix values

– 2 pts Used incorrect value for number of days with indeterminate origin or set all days to 1

✓ – 2 pts Multiplied by reduced matrix rather than original matrix values, but also assigned incorrect number of teams to jobs (should be 4 teams for 4 jobs)

– 2.25 pts Restated question without any demonstration of understanding problem

– 2 pts Incorrect response with no work shown

– 2.5 pts Blank

Question 5

Drones

1.5 / 20 pts

5.1 — 5A

0.5 / 5 pts

– 0 pts Correct

– 2 pts Demand was in Days, Lead Time was in Hours; inconsistent units caused a 24x increase in number of kanbans

– 1 pt Safety stock is a percent of the demand during lead time ($\bar{d} \times L$) better thought of as $(\bar{d} \times L) \times (1 + S) / c$

– 1 pt Incorrect Container size (needed to adjust to 85% of 50 ~ 42 or 43)

– 0.5 pts Cannot have fractional Kanbans. Instructions say to round up

– 2.5 pts Question asks for Type A and Type B, only answerd one consolidated version

– 2.5 pts Did not factor in lead time in your calculations or used incorrect lead time

✓ – 4.5 pts Used incorrect equation

– 0.5 pts Minor math error leading to slightly incorrect response. Correct response was 8 Kanbans of Type A, 4 Kanbans of Type B

– 1.5 pts Incorrect value for demand

– 2 pts demand was in days, lead time was converted to minutes rather than days, resulting in a 1000x increase in number of kanbans

5.2 — 5B

1 / 10 pts

– 0 pts Correct

– 4 pts Lead time is in hours, demand is in days, need to be converted to the same unit

– 2 pts incorrect safety stock calculation

– 1.5 pts Incorrect container size calculation

– 0 pts Incorrect, but follows

✓ – 9 pts Incorrect equation used or insufficient work shown

– 5 pts Did not split into type A and type B

– 6 pts Did not incorporate Lead Time

– 3.5 pts Incorrectly applied 25% increase.

5.3 — 5C

0 / 5 pts

– 0 pts Correct

– 2.5 pts 1/2 Correct

– 3.3 pts 1/3 Correct

– 1.7 pts 2/3 Correct

✓ – 5 pts Incorrect

EM420 Written Partial Review II

Name: Dylan Tied

Academic Term 26-1

Hour: 01

EM420 Production and Operations Management
Written Partial Review
Version C

- 1) This is a 75 minute graded event worth 150 points.
- 2) This is a closed book and closed notes exam. You may use one single sided 3x5" index cards and the provided FE book (DO NOT MARK IN IT!) and an authorized scientific calculator. No collaboration is allowed.
- 3) You may disassemble this document if you'd like. You may also use extra paper provided at the front of the classroom. Please ensure you staple the exam back together in the correct order with extra sheets labeled and at the back.
- 4) **SHOW ALL WORK!** Unsupported or missing work earns little or no partial credit.
- 5) **Rounding.** Show dollars and cents for any monetary answer (\$12.75). Show two significant digits beyond the decimal for percentage answers (12.75%) and at least two significant digits beyond the decimal for all other answers (0.13) unless otherwise noted
- 7) Upon completion, sign the acknowledgment statement below:

"I did not use any sources, nor did I receive any assistance while completing this WPR. This WPR will be released from academic security on 21NOV2025. Prior to that time, I may not discuss any aspects of this exam with anyone except an EM420 instructor."

Printed Name: Dylan Tied

Signature: [Signature]

Date Signed: 17 Nov. 25

8) GRADING SUMMARY:

Section	Topic	Total Points Available	Total Points Earned
1	Situation Assessment	30	
2	Multiple Choice & Short Answer	45	
3	Operational Readiness	30	
4	Joint Logistics Task Force	25	
5	Drones	20	
	Total	150	



Answer Sheet for Section 1 & Section 2

- 1A lin. Valencia
1B Johnson's lot
1C Point of indifference
1D empowerment
1E Cycle time
1F outsourcing
1G Center of gravity

1G Situation: Let's say I happen to be a FDC officer & I want to optimize average firing distance between 3 different targets to get best effect for fire

- 2A E, B, C, D, A
2B a
2C B, C, f
2D D
2E Cader A & C & E
2F A
2G 2000 units

**Section 1: Situation Assessment [30 points]**

In the following situations, please choose the BEST problem-solving strategy, chart, or calculation that would aid in making a decision for the problem at hand. A glossary is provided below for your reference. There is one best answer for each situation. No partial credit will be provided. Explanation is not necessary.

Alternative Workplace	FMEA	Load Distance
Block Diagramming	Gantt Chart	Location Factors
Break Even Analysis	Heijunka	Movement
Center of Gravity	Horizontal Enlargement	Muther's Diagram
Commissions	House of Quality	Outsourcing
Cross Training	Hygiene Factors	Point(s) of Indifference
Cycle Time	Individual Incentive	Product Process Matrix
Defects	Inventory	Profit Sharing
Economies of Scale	Job Rotation	Service Blueprint
Ellsberg's Paradox	Job Training	Service-Process Matrix
Empowerment	Johnson's Rule	Simple Sequencing Rules
EVPI	Kaizen	Theory X Employee
Fixed-Position Layout	Kanban System	Theory Y Employee
Flexible Work Schedules	Line Balancing	Vertical Enlargement
Flow Time	Load Assignment Method	Value Stream Mapping

Record all answers on Page 2, on the answer sheet.

1A [4 Points] – A defense electronics plant is assembling communication modules on a high-volume line. Some stations are idle while others have long queues. The line chief needs to reassign tasks to balance workloads, meet the desired cycle time, and reduce idle time. What method should be used?

line balancing

1B [4 Points] – A naval drone production unit must process each unit through two sequential operations: assembly and inspection. Each operation takes a different amount of time for each drone type. The operations officer wants to find the sequence that minimizes total completion time and idle time across both stations. What strategy should be used?

Johnson's Rule

1C [4 Points] – A defense contractor compares two drone assembly methods: one uses manual labor with lower setup costs but higher variable costs, and the other uses automated assembly with higher setup costs but lower variable costs. The manager wants to find the exact output quantity where both options cost the same before deciding which to use. What should he use?



1D [4 Points] - At an Army ground vehicle maintenance company, crews report frequent delays during scheduled services due to disorganized toolkits, unclear part storage locations, and repeated backtracking between work areas. Instead of waiting for a full facility renovation, the platoon sergeant begins holding short morning huddles where Soldiers suggest small changes, such as labeling bins, rearranging benches, and creating visual checklists, to reduce wasted time and motion. Over the next month, turnaround time improves by 12%. Which continuous improvement strategy best fits this situation?

1E [4 Points] - A process improvement specialist wants to assess total process time including both work and waiting for a batch of products. What performance measure best captures this total elapsed time in the system?

1F [4 Points] - An Army logistics battalion's data-tracking system for vehicle parts is outdated and unreliable. Rather than developing new software in-house, leadership is considering contracting a civilian tech company that specializes in military logistics systems to design, operate, and maintain it. The logistics battalion is participating in:

1G [6 Points] - Create your own situation which you think you would encounter as a 2LT and provide an answer (from the glossary above – do not use one already used in 1A-F). You will be graded for clarity and correctness. Make sure it is clear why it would be your term and no other terms.

**Section 2: Multiple Choice & Short Answer (No Partial Credit 45 Points)****Record all answers on Page 2, on the answer sheet.**

2A [10 points] You are sequencing five repair jobs through two sequential maintenance stations. The processing times (in hours) are shown below.

Job	Station 1	Station 2	
A	8	6	→ 14
B	4	7	→ 11
C	5	3	→ 8
D	7	9	→ 16
E	2	5	→ 7

Determine the optimal sequence of jobs that minimizes total makespan. Write the order of jobs from first to last.

2B [5 Points] You are designing a production line for assembling portable radios. Each station has a desired cycle time of 40 seconds. The task list below shows the operation times for each required activity.

Task	Operation	Time (sec)	Requires Special Tool?
A	Mount circuit board	10	No
B	Install speaker	6	No
C	Attach antenna	8	Yes
D	Screw in faceplate	7	No
E	Test sound output	9	Yes
F	Package product	6	No

What is the **minimum number of workstations** required to complete the line without exceeding the cycle time?

- a. 1
- b. 2
- c. 3
- d. 4
- e. 6



2C [5 Points] Which of the following is **always true** when you are dealing with two sequential machines? Write all correct answer(s).

- ☐ a. All idle time for Machine 1 is always at the beginning
- ☒ b. All idle time for Machine 2 is always at the beginning
- ☐ c. All idle time for Machine 1 is always at the end
- ☐ d. All idle time for Machine 2 is always at the end
- ☐ e. Makespan is determined by the finishing time of the last job on Machine 1
- ☐ f. Makespan is determined by the finishing time of the last job on Machine 2

2D [5 points] A logistics officer is selecting the location for a new supply hub to support three forward bases in an open region with no existing facilities. The goal is to minimize total transportation distance, considering how much each base ships per week. Use traditional rounding standards. (round up if .5 or above, round down otherwise). Select the correct strategy and location.

Base	X-Coordinate	Y-Coordinate	Weekly Shipments (tons)
Alpha	4	8	30
Bravo	10	4	50
Charlie	6	12	20

- ☒ a. Load Distance; (7,7) New Point
- ☐ b. Center of Gravity; (7,7) New Point
- ☐ c. Load Distance, (4,8) Alpha
- ☐ d. Center of Gravity, (4,8) Alpha

2E [5 points] You're leading a **senior capstone team** working on an engineering design project for a real client. Each cadet on your team approaches the project differently according to the descriptions below

- Cadet A often asks for additional responsibilities and suggests design improvements beyond the assignment.
- Cadet B needs frequent reminders to submit weekly reports and only works when directly supervised.
- Cadet C volunteers to organize the client presentation and stays late to ensure everything is perfect.
- Cadet D avoids taking initiative, saying, "Just tell me exactly what to do."
- Cadet E proactively schedules team meetings and checks in with everyone to help coordinate progress.

Which cadet(s) demonstrate Theory Y behaviors?



2F [5 Points] The table below shows part of a Muther's Relationship Chart for a maintenance facility. The letters indicate the closeness rating between departments. If the engineer can only place two departments side by side due to space limitations, which pair should be prioritized for adjacency based on the chart?

Department	Welding	Paint	Inspection	Storage
Welding	—	A	E	O
Paint	A	—	I	U
Inspection	E	I	—	O
Storage	O	U	O	—

- a. Paint and Welding
- b. Paint and Inspection
- c. Inspection and Storage
- d. Storage and Paint
- e. Any pairing would be adequate

2G [10 Points] A maintenance parts supplier is considering producing its own replacement valves rather than purchasing them. The company would need to invest \$120,000 in new equipment. Each valve would cost \$35 to produce and could be sold for \$95. What is the break-even quantity of valves (in units) the supplier must produce and sell to recover its investment?

$$V = \frac{120,000}{95 - 35} \rightarrow V = 2000 \text{ units that must be sold to recover investment of } 120,000$$

**Section 3: Operational Readiness (30 Points)**

You are managing four repair tasks for a brigade support battalion ahead of a large training exercise. Each task must be completed before deployment to maintain operational readiness. Today's date is May 1st. There are 31 days in May.

Maintenance Task	Repair Time (days)	Required Completion Date
A. Transmission Replacement	7 days	May 10th
B. Optics Calibration	14 days	May 18th
C. Electrical Overhaul	5 days	May 15th
D. Suspension Alignment	20 days	June 5th

3A [3 points] What is the order according to Longest Processing Time?

C, A, B, D,

A = 7
B = 14
C = 5
D = 20
45 days total
45

3B [5 points] What is the order according to SLACK?

A. May 1st + 7 = May 8th 2 slack days

B. May 1st + 14 = May 15th 3 slack days

C. May 1st + 5 = May 6th 9 slack days

D. May 1st + 20 = May 21st 16 slack days

order of slack = A, B, C, D



3C [3 points] What is the order according to DUE DATE?

A, C, B, D

A = 9 days

C = 15 days

B = 18 days

D = 36 days

3D [12 points] Fill out the following Table (show work under parts A, B and C).

Sequencing Rule	# of Late Assignments	Total Number of Late Days
LPT	3	2
SLACK	0	0
DUE DATE	0	0

3E [3.5 points] The battalion maintenance section assigns one readiness deduction for every three total days of lateness across all repair tasks. For example, if one task is one day late and another task is one day late, their combined two late days still count toward the three-day threshold for a single deduction. Which sequencing rule should you use to minimize the total number of late days and therefore minimize readiness deductions? What is the total number of readiness deductions associated with this rule. Explain your reasoning.

3F [3.5 Points] The battalion's operations officer tracks mission delays based on whether each repair task is completed on time (binary: yes or no). Each late task, regardless of how many days late, counts as one mission delay. Which scheduling strategy minimizes the total number of mission delays? How many mission delays occur with this strategy?

Johnson's rule

**Section 4: Joint Logistics Task Force (25 Points)**

You are advising a Joint Logistics Task Force (JLTF) preparing for a multinational exercise. Four regional logistics hubs (Teams A–D) must each take responsibility for one operational task. Your goal is to minimize total project completion time while ensuring that each team is assigned to exactly one task. Below are the time in days that each team takes to do each operational task.

Team	Supply Chain Setup	Equipment Calibration	Site Construction	Vehicle Readiness
A	12	10	14	11
B	8	13	9	12
C	9	11	8	10
D	10	9	13	12

[22.5 points] Use the Load Assignment Method (Hungarian Method) to determine which team should be assigned to each task to minimize total project duration.

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[2.5 points] If the daily operating cost of each team is \$6,800 per day, what is the total operational cost of these assignments based on your final solution?

$$2 + 3 + 2 + 2 + 1 + 1 + 3 + 1 (6,800)$$

$$15 (6,800) = \$102,000$$

**Section 5: Drones (20 Points)**

A maintenance company sustains 60 tactical drones deployed across multiple sites. To prevent downtime, it uses a pull-based Kanban system for replenishing two types of pre-packed battery modules: Type A for standard operations and Type B for long-range missions. On a typical day, the unit consumes about 1,200 battery modules, with roughly two-thirds being Type A and the rest Type B. Each sealed container can hold 50 modules, though only about 85 percent of that space is usable once padding, labels, and inspection inserts are added. Once a Kanban card is pulled, it takes about eight hours for a full container to be refilled, quality-checked, and returned to the line. To hedge against variability and delays, the maintenance officer applies a 20 percent safety margin.

5A [5 pts] Determine the number of Kanban cards needed for each module type to sustain continuous flow without stockouts. Round up, **provide a reason why you would round up in practice.**

$$K = \frac{\text{Capacity} + \text{Safety} \quad (\text{Usable})}{\text{Sealed Capacity}} = \frac{\text{Type B (UCC)}}{42.5} = 1.89 \quad \text{or } 2 \text{ Kanbans}$$

$$\text{Type A} \quad K = \frac{0.2(800)}{42.5} = 3.76 \text{ Kanbans} \quad \text{or } 4 \text{ Kanbans Type A}$$

Round up because ideally you want each item delivered in exact amounts exactly when needed for full optimization

5B [10 pts] A new drone model has been introduced that increases total daily battery consumption by 25 percent, while the rest of the system remains unchanged. Recalculate the Kanban requirements for each module type and discuss the operational effects of this higher throughput. Round up.

New Demand = 1500 modules

$$\text{Type A} \quad K = \frac{0.2(1000)}{42.5} = 11.76 \text{ Kanbans} \quad \text{or } 12$$

$$\text{Type B} \quad K = \frac{0.2(500)}{42.5} = 5.89 \text{ or } 6 \text{ Kanbans}$$

5C [5 pts] What two countermeasures could help maintain efficiency without simply adding storage space or containers? (Select all that apply.) No partial credit.

- ☐ Shorten resupply lead time through pre-staging or faster transport cycles
- ☐ Reduce container size to increase turnover and responsiveness
- ☒ Increase safety margin from 20 % to 30 %
- ☒ Add an extra storage bay near the maintenance line
- ☐ Increase total on-hand inventory to cover two days of demand



UNITED STATES MILITARY ACADEMY
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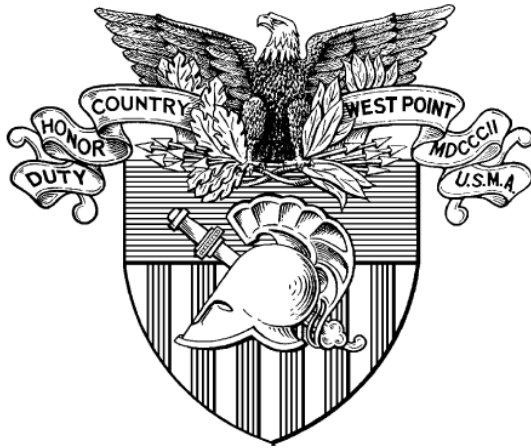
TAB I TERM END EXAMINATION

Term End Examinations are kept in a secure location and can be accessed
by request to the Program Director.



UNITED STATES MILITARY ACADEMY
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TAB J FINAL COURSE GRADES



Department of Systems Engineering
Final Grade Approval
AT 26-1



EM420 Production Operations
Management

<i>Department Head:</i>	<i>COL Coxen</i>
<i>Program Director:</i>	<i>COL Thompson</i>
<i>Course Director:</i>	<i>Dr. Sanders</i>

For Faculty Viewing Only

Secure When Not In Use
Caution: Not For Cadet Viewing!



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

22 December 2025

MEMORANDUM THRU COL Brandon Thompson, Engineering Management Program

Director FOR COL Julia Coxen, Professor and Head, Department of Systems Engineering

SUBJECT: EM420 Production Operations Management Final Grade Approval for Academic Term 26-1

1. **RECOMMENDATION.** Recommend approval of EM420 final grade results for Academic Year 26-1. Enclosed in this report is the course order of merit list with recommended cut lines.

2. **COURSE DATA.**

- a. **Course Enrollment:** 90 Students
- b. **Course Average:** 78.16%
- c. Table 1 shows this semester's course average alongside the previous 13 semester averages. Table 2 shows this semester's course average by cadet major.

Academic Term	Average
AT26-1	78.16%
AT25-1	83.11%
AT24-1	85.59%
AT23-1	87.42%
AT22-1	84.79%
AT20-1	86.21%
AT18-2	86.08%
AT18-1	83.86%
AT17-2	83.74%
AT17-1	85.59%
AT16-2	85.91%
AT16-1	83.80%

Table 1: Historical Course Averages



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Table 2: Course Average by Major

Major	# of Cadets	Average
SDS0	24	72.2%
ENM1	48	78.0%
ASD0	1	76.9%
MGN0	3	81.1%
CEN2	3	84.1%
ORE2	8	90.9%
ECN2	1	83.3%
SEN1	2	88.0%

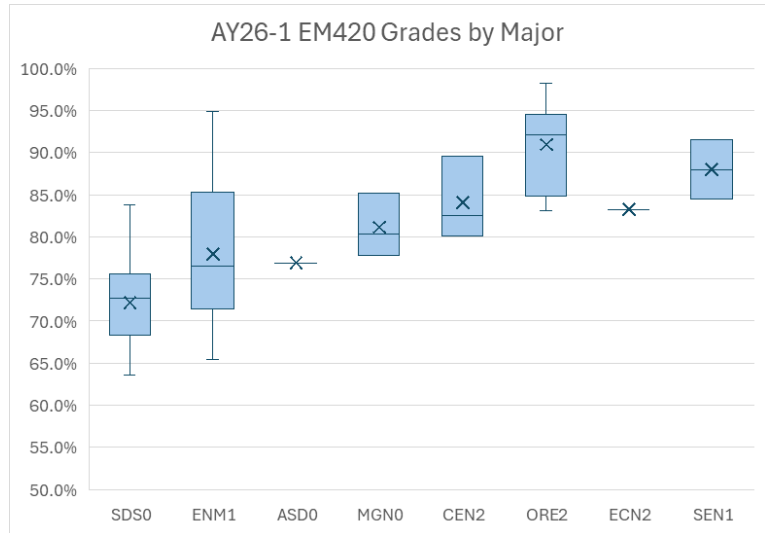


Figure 1: EM420 Grade Distribution by Major

3. OTHER STATISTICS. The grade distribution table and histogram for the term is depicted below. We recommend changes to the A-, C+, and C.

Table 2. Recommended Grade Distribution

	Standard Cut Line Count	Cut Line (Standard)	Modified Cut Line Count
A+	1	97.5	1
A	4	92.5	4
A-	7	89.4 (90.0)	9
B+	2	87.5	0
B	14	82.5	14
B-	8	80	8
C+	6	77.0 (77.5)	7
C	22	72.2 (72.5)	24
C-	7	70	5
D	16	65	16
F	2	0	2



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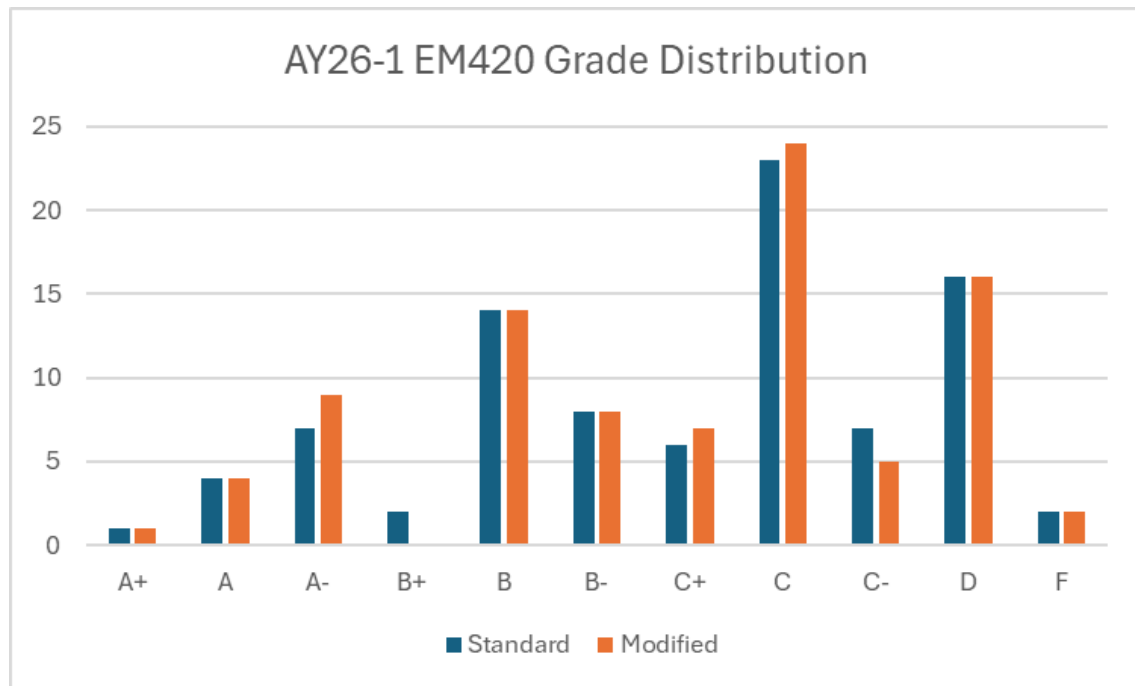


Figure 2. Grade Distribution Before and After Adjustment

Table 3. Course Graded Event Summary

Graded Event	Points	Mean Percent	Note
Problem Sets	90	87.2%	
Individual Assignments	100	73.3%	
Group Assignments	75	72.4%	
WPRs (x2)	300	78.9%	
Case Study	90	66.5%	
TEE	300	80.2%	
Instructor Points	45	86.5%	

4. **FAILURES.** There were two failures in this course.
5. The point of contact for this memorandum is COL Brandon Thompson at x3180.

Enclosures
A: Course OML with Cut Lines

BRANDON S. THOMPSON
COL, MI/FA47
EM Program Director



UNITED STATES MILITARY ACADEMY
WEST POINT.

SECTION III

ASSIGNMENTS



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB K PROBLEM SETS



UNITED STATES MILITARY ACADEMY
WEST POINT.

PROBLEM SET #1



EM 420 Production Operations Management, Problem Set #1
Operations and Supply Chain Management



EM420 Problem Set #1

Name: _____

Academic Term 26-1

Hour: _____

DOCUMENTATION. This problem set is an individual graded assignment. Any assistance received must be documented in detail. Document all sources in accordance with the Office of the Dean Pamphlet “Documentation of Academic Work,” (July 2017), Appendix E, and course guidance. Documentation must be turned in at the time of submission. The problem set is considered late until all portions of the assignment and the documentation are submitted.

DUE: NLT 23:59 21 September 2025

1. This individual graded assignment consists of 6 sections worth a total of 45 pts. **The last day for AI is September 20 September at 16:00.**
2. Document any assistance received by the Dean’s Policy for the Documentation of Written Work. Professionally present your work, including ***a cover sheet with a signed acknowledgement statement.***
3. Round any number to three decimal places, and round any dollar amount to the nearest cent.
4. ChatGPT or any Generative AI is NOT permitted on this assignment.
5. You must turn in a PDF for this assignment. If you use Excel, you should submit that as a secondary document.
6. Submitted Microsoft Excel ® files ***must be documented appropriately if any assistance is received*** and should be named *LastName_First_Name_EM420_PS1_Hour(GI).xlsx* and submitted via Teams by the due date.

GRADING SUMMARY:

Problem	Possible Points	Points Earned
1 – Productivity Analysis	5	
2 – Decision Analysis I	5	
3 – Decision Analysis II	5	
4 – Decision Analysis III	10	
5 – Quality Tools	5	
6 – Statistical Process Control	15	
Total:	45	



1. **Productivity Analysis (5 Pts)**

You've started a company that produces **custom surfboards**. Each year, your production changes based on design trends and material costs.

Year	# Boards Produced	Labor Costs	Material Costs	Capital Costs	Transport Costs	Overhead Costs
2021	14,000	\$142,350	\$161,220	\$173,500	\$112,300	\$125,700
2022	32,800	\$119,900	\$135,480	\$188,100	\$156,200	\$140,600
2023	29,600	\$128,400	\$175,300	\$169,800	\$123,400	\$116,900

Given the above financial information, what is your:

Single-Factor Productivity for Transportation in 2021, 2022 & 2023? How is this rate changing over time? Provide a hypothesis for why there is a change. **(2 pt)**

Which year has the highest Total Factor Productivity? What is it? Give a business reason why this year had a higher TFP.**(3 pts)**



2. Decision Analysis I (5 pts)

You are planning a weekend adventure with friends. Because reservations must be made in advance, once you commit, you cannot change. Weather at Lake Tahoe is unpredictable. The table below shows the **value** you get from each activity depending on conditions (100 = best, 0 = worst).

Make a recommended workout using the criteria prescribed below. Show your work.

Activity	Sunny	Cloudy	Light Rain	Heavy Rain	Windy
Kayaking	95	80	25	5	10
Hiking	90	85	65	30	40
Mountain Biking	98	92	70	20	15
Fishing	75	88	90	60	55
Rock Climbing	100	96	40	15	5
Camping	85	82	50	25	35

a. Maximax (1 pts)

b. Minimax Regret (1 pts)

c. Hurwicz ($\alpha = 0.7$) (1 pts)



EM 420 Production Operations Management, Problem Set #1
Operations and Supply Chain Management



- d. Which decision do you recommend and why? Include a discussion on the impact of the management's optimism or pessimism on your recommendation. **(2 pts)**



3. **Decision Analysis II (5 pts)**

As fate would have it, Dr. Sanders has an advanced degree in weather patterns and through her research, she's informed you that the probabilities of the weather in Problem 2 are:

- Sunny: 20%
- Cloudy: 25%
- Light Rain: 20%
- Heavy Rain: 15%
- Windy: 20%

Given these probabilities:

- a. What is the best activity to choose using expected value? What is that expected value? **(2 pts)**

- b. A Local Guide company states that his predictions are **perfectly accurate**. He can tell you, a week in advance, what the weather will be. He demands \$100 for his predictions. Would you pay that much? Why? What's the most you would pay? Assume you and your friends would pay \$10 for a single "value". (e.x. you would pay \$300 for a value of 30 like weights in the wet, cold) **(3 pts)**



4. **Decision Analysis III (10 pts)**

You and a classmate have been tossing around the idea of starting a drone delivery company together. One evening over coffee, you sketch out some options. To even open your doors, you'd need to sink a six-figure investment — around \$120,000 — just to get things off the ground. You estimate that only about one in four startups like yours really succeed, but if it flops, you might still scrape together roughly \$25,000 in income before shutting down.

On the other hand, if the company *does* take off, you'll face a big decision: keep operations local or attempt a national expansion. Staying local would be safe, and you project about \$450,000 in revenue if things continue steadily. But expansion would mean raising and spending an additional \$180,000 right away.

Expansion has its risks. From what you've researched, there's about a one-in-four chance it really takes off nationwide, which could pull in close to \$950,000 in revenue. Still, there's also about a one-in-five chance things backfire, dragging income down to only \$300,000. More likely than not, though (just over half the time), expansion would do neither better nor worse than staying local, leaving you back at the same \$450,000 outcome you would have had without expanding.

Task: Based on this scenario, draw a decision tree, roll back the tree to calculate expected values, and recommend a course of action. Be sure to clearly state which option you would choose and why.



5. **Quality Indexes (5 Pts)**

Democratic Designs is considering three different options of printing T-Shirts: Screen Printing, Vinyl Graphics, and Direct-to-Garment. Each option has 3 steps to print a single shirt, each with different reliabilities.

- a. You're testing 3 methods of drone assembly: Automated Line, Semi-Auto, and Manual. Compute overall reliability of each method, which is the most reliable (2 pts).

Step	Automated	Semi-Auto	Manual
1	97%	96%	95%
2	98%	94%	96%
3	99%	95%	92%

- b. You've decided that you will choose the option with the cheapest Product Cost no matter what. Also, the information from Part A above has been updated. Given the information below, calculate the Product Cost for each option. Which option will you choose? If producing **5,000 drones**, compute product cost for each and choose the cheapest. (3 pts)

Option	% Initially Good	% Reworkable	Direct Cost/unit	Rework Cost/unit
Automated	95%	85%	\$20	\$8
Semi-Auto	93%	80%	\$18	\$7
Manual	90%	70%	\$16	\$6



EM 420 Production Operations Management, Problem Set #1
Operations and Supply Chain Management





6. Statistical Process Control (15 Pts)

To ensure reliable drone performance, the battery pack must meet strict quality standards. Because even small deviations in battery weight can indicate defects in assembly or inconsistencies in cell balancing, you've decided to implement **Statistical Process Control (SPC)**. By taking repeated samples of five battery packs across multiple production lots, you can monitor variation in the manufacturing process and detect whether the system remains stable over time. The following dataset provides battery pack weights, measured in grams, for 20 samples of size five. Your task is to use these measurements to construct **$\bar{X}$ and R charts**, evaluate whether the process is in control, and determine whether the process is capable of consistently meeting customer specifications.

Sample	Weights of Battery Pack in g		
1	505	513	486
2	503	489	495
3	490	497	494
4	500	499	512
5	509	503	508
6	511	499	510
7	510	514	492
8	495	506	497
9	492	496	510
10	501	496	485
11	489	490	503
12	491	512	494
13	499	497	493
14	506	497	494
15	504	497	512
16	489	499	494
17	514	498	510
18	503	510	512
19	505	489	504
20	510	494	499



EM 420 Production Operations Management, Problem Set #1
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a. Construct **X-bar and R charts**. Include σ and 2σ lines on X-bar only. (10 pts)

b. Is the process in control? Why or why not? (2 pts)



EM 420 Production Operations Management, Problem Set #1
Operations and Supply Chain Management



c. Specs require **500g $\pm$ 20g**, with $\sigma = 6.5\text{g}$. Is process capable (C_p/C_{pk})? (3 pts)

Instructor Solution

1. Productivity Analysis (5 pts)

a) Single-Factor Productivity for Transportation

Formula:

$$\text{SFP}_{\text{Transportation}} = \frac{\text{Output}}{\text{Transportation Cost}} = \frac{14,000}{112,300} = 0.125 \text{ boards per \$}$$

- 2021: $14,000 / 112,300 = 0.125$ boards per \$
- 2022: $32,800 / 156,200 = 0.210$ boards per \$
- 2023: $29,600 / 123,400 = 0.240$ boards per \$

Trend: Transportation productivity improves steadily ($0.125 \rightarrow 0.210 \rightarrow 0.240$).

Hypothesis: Likely due to economies of scale and improved logistics efficiency as production expanded.

b) Total Factor Productivity (TFP)

Formula:

$$\text{TFP} = \frac{\text{Output}}{\text{Total Input Cost}} = \frac{14,000}{715,070} = 0.0196$$

Total Input = Labor + Material + Capital + Transport + Overhead

- 2021: Costs = $142,350 + 161,220 + 173,500 + 112,300 + 125,700 = \$715,070$
TFP = $14,000 / 715,070 = 0.0196$
- 2022: Costs = $119,900 + 135,480 + 188,100 + 156,200 + 140,600 = \$740,280$
TFP = $32,800 / 740,280 = 0.0443$
- 2023: Costs = $128,400 + 175,300 + 169,800 + 123,400 + 116,900 = \$713,800$
TFP = $29,600 / 713,800 = 0.0415$

Highest TFP: 2022 (0.0443).

Reason: Output increased substantially relative to input costs—likely reflecting strong demand and efficient resource use.

2. Decision Analysis I (5 pts)

a) Maximax (Optimistic)

Pick highest possible payoff per activity:

- Kayaking 95, Hiking 90, Mountain Biking 98, Fishing 90, Rock Climbing 100, Camping 85.

Choice: Rock Climbing (100).

b) Minimax Regret

Construct regret table (difference between payoff and best in each state). Summarize:

- Kayaking worst regret = 25
- Hiking worst regret = 45
- Mountain Biking worst regret = 30
- Fishing worst regret = 25
- Rock Climbing worst regret = 70
- Camping worst regret = 50

Choice: Kayaking or Fishing (both = 25).

c) Hurwicz ($\alpha = 0.7$)

Formula: $H = \alpha \times \text{max payoff} + (1 - \alpha) \times \text{min payoff}$
 $H = 0.7 \times \text{max payoff} + 0.3 \times \text{min payoff}$

- Kayaking: $0.7(95) + 0.3(5) = 68$
- Hiking: $0.7(90) + 0.3(30) = 72$
- Mountain Biking: $0.7(98) + 0.3(15) = 74.9$
- Fishing: $0.7(90) + 0.3(55) = 79.5$
- Rock Climbing: $0.7(100) + 0.3(5) = 73.5$

- Camping: $0.7(85) + 0.3(25) = 67$

Choice: Fishing (79.5).

d) Recommendation

Fishing balances risk and optimism—it performs well across conditions. Optimists may choose Rock Climbing, pessimists prefer Kayaking/Fishing.

3. Decision Analysis II (5 pts)

Probabilities: Sunny 0.20, Cloudy 0.25, Light Rain 0.20, Heavy Rain 0.15, Windy 0.20.

a) Expected Values

Compute EV for each:

- Kayaking = $0.20(95) + 0.25(80) + 0.20(25) + 0.15(5) + 0.20(10) = 48.5$
- Hiking = $0.20(90) + 0.25(85) + 0.20(65) + 0.15(30) + 0.20(40) = 63.25$
- Mountain Biking = $0.20(98) + 0.25(92) + 0.20(70) + 0.15(20) + 0.20(15) = 63.05$
- Fishing = $0.20(75) + 0.25(88) + 0.20(90) + 0.15(60) + 0.20(55) = 75.15$
- Rock Climbing = $0.20(100) + 0.25(96) + 0.20(40) + 0.15(15) + 0.20(5) = 55.9$
- Camping = $0.20(85) + 0.25(82) + 0.20(50) + 0.15(25) + 0.20(35) = 57.35$

Best: Fishing (75.15).

b) Value of Perfect Information (VPI)

Max payoff per weather: Sunny 100, Cloudy 96, Light Rain 90, Heavy Rain 60, Windy 55.

EV with perfect info = $0.20(100) + 0.25(96) + 0.20(90) + 0.15(60) + 0.20(55) = 82.45$.

VPI = $82.45 - 75.15 = 7.3$ “value” units.

At \$10/value unit → worth up to **\$73**.

Since guide charges \$100, not worth paying.

4. Decision Analysis III (10 pts)

Step 1: Open vs. Don’t Open

- Don’t open = \$0.

- Open: 25% success, 75% fail.

Failure payoff = $25,000 - 120,000 = -95,000$.

Success continues to decision node.

Step 2: If Success

- Local: $450,000 - 120,000 = 330,000$.
- Expand: costs extra 180,000. Payoffs:
 - Big success (25%): $950,000 - 300,000 = 650,000$
 - Loss (20%): $300,000 - 300,000 = 0$
 - No change (55%): $450,000 - 300,000 = 150,000$
 $\rightarrow EV = 0.25(650k) + 0.20(0) + 0.55(150k) = 162,500 + 0 + 82,500 = 245,000$

Local = 330,000 vs. Expand = 245,000 $\rightarrow$ better to stay local.

Step 3: Roll Back EV of Opening

$EV = 0.25(330,000) + 0.75(-95,000) = 82,500 - 71,250 = 11,250$.

Recommendation: Open and stay local (positive EV).

5. Quality Tools (5 pts)

a) Overall Reliability (multiply steps)

- Automated: $0.97 \times 0.98 \times 0.99 = 0.941$ (94.1%)
- Semi-Auto: $0.96 \times 0.94 \times 0.95 = 0.858$ (85.8%)
- Manual: $0.95 \times 0.96 \times 0.92 = 0.840$ (84.0%)

Most Reliable: Automated.

b) Product Cost (5,000 units)

Effective yield:

Effective Units = Good + (Bad $\times$ Rework)
 $\text{Effective Units} = \text{Good} + (\text{Bad} \times \text{Rework})$

- Automated: Good = 95%, Rework = 85% → effective = $0.95 + (0.05 \times 0.85) = 0.9925$.
Units produced = $5000 / 0.9925 = 5038$. Cost = $5038 \times 20 + (0.05 \times 5038 \times 0.85 \times 8) \approx$
\$101,460.
- Semi-Auto: $0.93 + (0.07 \times 0.80) = 0.986$. Units = 5071. Cost = $5071 \times 18 +$
 $(0.07 \times 5071 \times 0.8 \times 7) \approx$ **\$95,960.**
- Manual: $0.90 + (0.10 \times 0.70) = 0.97$. Units = 5155. Cost = $5155 \times 16 +$
 $(0.10 \times 5155 \times 0.7 \times 6) \approx$ **\$86,890.**

Cheapest: Manual (even though least reliable).

6. Statistical Process Control (15 pts)

a) $\bar{X}$ and R Charts

- Compute sample means and ranges. ($\bar{X} \sim 500\text{g}$, $R \sim 20\text{g}$).
- Control limits use $\bar{X}$, $\bar{R}$, and factors (A2, D3, D4).
- Plot $\bar{X}$ with centerline, $\pm 1\sigma$, $\pm 2\sigma$.
- Plot R with control limits.

b) Process in Control?

If all sample means/ranges fall within limits with no runs or trends, then **yes, in control.**

c) Capability (Cp, Cpk)

Specs = $500 \pm 20 = (480-520)$. Range = 40.

$\sigma = 6.5$. $C_p = (USL - LSL) / (6\sigma) = 40 / (39) \approx 1.03 \rightarrow$ barely capable.

$C_{pk} = \min[(520 - 500)/(3\sigma), (500 - 480)/(3\sigma)] = 20 / 19.5 \approx 1.03$.

Process is just capable.



UNITED STATES MILITARY ACADEMY
WEST POINT.

PROBLEM SET #2



EM420 Problem Set #2

Name: _____

Academic Term 26-1

Hour: _____

DOCUMENTATION. This problem set is an individual graded assignment. Any assistance received must be documented in detail. Document all sources in accordance with the Office of the Dean Pamphlet “Documentation of Academic Work,” (July 2017), Appendix E, and course guidance. Documentation must be turned in at the time of submission. The problem set is considered late until all portions of the assignment and the documentation are submitted.

DUE: NLT 1159 on 6 November 2025

1. This is an individual graded assignment consisting of 5 sections worth a total of 45 points. The use of Microsoft Excel is *optional* for all problems.
2. Document any assistance received in accordance with the Dean’s Policy for the Documentation of Written Work. Present your work in a professional manner to ***include a cover sheet with a signed acknowledgement statement and works cited as a separate attachment from the problem set.***
3. Round any number to three decimal places. Round any dollar amount to the nearest cent. You must use units (i.e. \$, widgets/hr, etc.) in your final answer if appropriate.
4. The deliverable for this problem set is a ***written or digitally scanned copy of all problems, and, if used, an electronic copy of Microsoft Excel ® work.*** To submit the written portion, print this document and ***show all work*** in provided sections. Use extra paper if need be. Partial credit cannot be awarded if your thought process is not evident. Microsoft Excel ® work needs to be professionally presented in an organized, easy to follow, and properly identified manner. Please ensure to label each tab with the question it refers to and only complete one question per tab. **All results require explanation and substantive interpretation to demonstrate your knowledge and understanding of your work.** Thus if you cannot clearly articulate your work and knowledge in Microsoft Excel ® the problem must be completed by hand.

Submitted Microsoft Excel ® files ***must be documented appropriately if any assistance is received*** and should be named *LastName_First_Name_EM420_PS2.xlsx* and with your problem set on canvas prior to the due date.

GRADING SUMMARY:

Problem	Possible Points	Points Earned
1 – In-Class Activity Evaluation	7.5	
1 – In-Class Activity Alternative	7.5	
2 – Process and Technology	7.5	
2 – Process and Service Layout	10	
3 – Product Layout	10	
4 – Facility Location Model	10	
Total:	45	



1. **In-Class Activity Evaluation (7.5 Pts)**

If you were present during Lesson 9, complete this version of problem 1, if you were absent, complete the alternative version of problem 1. You may not choose.

1a. Please place an x in the box that corresponds to your presence/absence for Lesson 9: Service Design, during which we conducted the in-class product/service design activity.

- ☐ I was Present for Lesson 9
- ☐ I was Absent for Lesson 9

1b. (1 pt): Product/Service Description. Write a brief description of your Product or Service

1c. (4 pts): Pre Service/Product Analysis. Think back to your Service or Product before you conducted the in-class activity. Choose **one (1)** of the following to justify how you made decisions (pricing, quantity, service rate, etc.) using either Decision Making Under Risk, or Decision Making Under Uncertainty. Place an x in the box next to the set of tools you are using:

- ☐ Maximax, Maximin, Hurwicz, etc. (must use at least 3 different tools)
- ☐ Decision Tree

Use the area below for your work.



1d. (2.5 pts): Post Service/Product Analysis. Think back to your Service or Product after you conducted the in-class activity. Choose **two (2)** of the following sets of tools to assess your product or service performance. Use the data you collected during the exercise. Place an x in the box next to the set of tools you are using:

- ☐ Quality Indices, Product Yield, QPR
- ☐ Statistical Process Control Chart (P, C, Xbar and R)
- ☐ Capability (Cp, Cpk)
- ☐ Pareto Analysis

Use the area below for your work.



1(Alt). **In-Class Activity Alternative (7.5 Pts)**

If you were absent during Lesson 9, complete this version of problem 1, if you were present, complete the previous version of problem 1. **You may not choose.**

Alternative 1a. Please place an x in the box that corresponds to your presence/absence for Lesson 9: Service Design, during which we conducted the in-class product/service design activity.

- ☐ I was Present for Lesson 9
- ☐ I was Absent for Lesson 9

Alternative 1b. (1 pt): You will develop a service in this problem. Provide a short description of your service incorporating aspects of service design.

Alternative 1c. (1.5 pts): Analyze all seven dimensions of quality with respect to your service design.

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.



Alternative 1d. (2 pts): Build and draw a service blueprint for your service and be sure to identify: Line of Influence, Line of Interaction, Line of Visibility, and Line of Support.



Alternative 1e. (3 pts): You just started a small side business making **custom vinyl decals** for laptops and water bottles. You need to decide whether to buy **pre-cut decals** from a supplier or invest in a **vinyl-cutter kit** to make your own. Given the information below, what should you do? Build/Draw the decision tree and evaluate the Expected Value. Make a decision(s), write them down, and explain your reasoning.

- Pre-cut decals cost **\$5.00**, and a vinyl-cutter kit costs **\$7.00**.
- In the first week there is an **80%** chance the pre-cut decals remain usable (adhesive doesn't degrade).
- In the first week there is a **70%** chance the vinyl-cutter kit operates without defects.
- In the second week there is a **30%** chance the pre-cut decals remain usable.
- In the second week, you can decide whether to buy a **blade/alignment upgrade** for your vinyl-cutter which costs **\$2.00**.
- If the vinyl-cutter has the upgrade, there is a **90%** chance it continues operating without issues in the second week.
- If you do not buy the upgrade, the chance remains the same as the first week (**70%**).
- If your decals spoil or your cutter fails in the **first week**, you make **\$5** in profit. If it happens in the **second week**, you make **\$10** in profit.
- If it is still going in the **third week**, you make **\$15**, and if you have the vinyl-cutter (not the pre-cut decals), you can also **resell the cutter for \$5**.

Expected Value: _____



2. Process and Technology (10 Pts)

2a. (3 Points) AeroLink Systems is projecting sales of 30,000 autonomous drones in 2025 at an average selling price of \$80,000 per drone. Forecast fixed operating costs for 2025 are \$1.20 billion. What variable cost per drone must AeroLink achieve to break even in 2025? Show all work.

Variable Cost _____



2b. (4 pts): AeroLink currently manufactures its own flight-control modules. The in-house module line has fixed costs of \$150 million and variable costs of \$15,000 per module. Management is evaluating two outsourcing options:

- Option 1 – SkyTech Robotics (reservation contract): \$16,500 per module + \$90 million annual fixed fee.
- Option 2 – Stratos Components (spot purchases): \$18,000 per module, no fixed fee.

How many modules would AeroLink need to buy from SkyTech (Option 1) to be indifferent between manufacturing in-house and Option 1? Show all work.

Number of modules _____



2c. (3 Points) At what annual volume would AeroLink be indifferent between manufacturing in-house and Option 2 (Stratos at \$18,000/unit, no fixed fee)? Show all work.

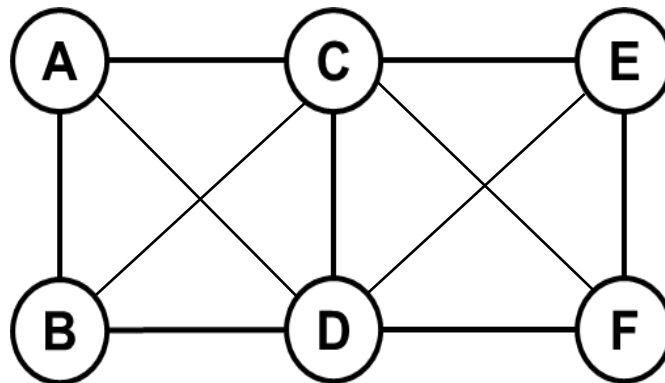
Annual Volume _____



3. Process and Service Layout (10 Pts)

3a. (2.5 Points) CatalystCo is commissioning a pilot plant for specialty catalyst production. Engineers proposed the 2×3 layout below and recorded weekly load movements (pallets/containers) between unit operations. Adjacency means sharing a side *or* a corner (diagonals count as adjacent). Calculate the number of non adjacent loads in the current (initial) design, and draw them on the diagram below.

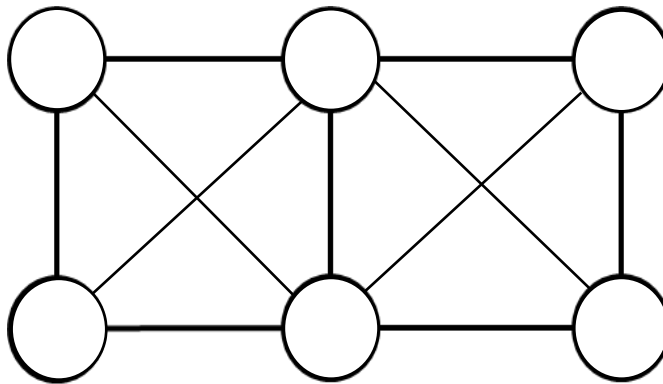
	A	B	C	D	E	F
A - Reactor	-	240	60	80	180	0
B – Distillation	200	-	40	30	160	10
C – QC Lab	40	20	-	10	0	90
D – Utilities/Heat Exchange	40	20	10	-	40	0
E – Solvent Recover	160	140	0	30	-	40
F – Drum Warehouse	0	0	80	0	60	-



Initial Nonadjacent loads _____



3b. (2.5 Points) Design and draw a new 2×3 layout (reassign A–F to the six cells) that minimizes the number of non-adjacent loads. Write the total nonadjacent loads in the block below.

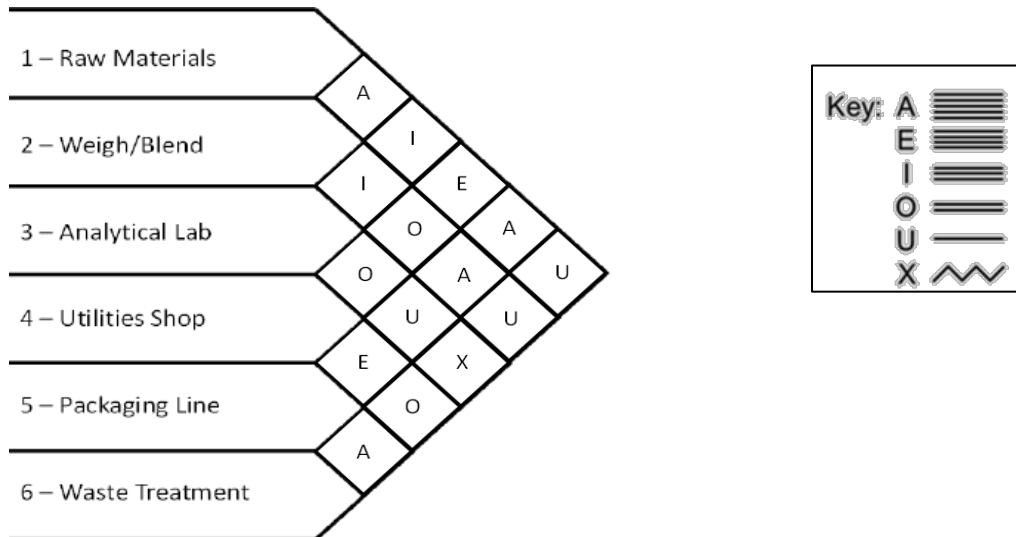


New Layout Nonadjacent loads _____



3c. (5 Points) After completing the pilot-plant optimization, CatalystCo plans to build a materials-handling annex to support catalyst scale-up. This new area will include several support departments that interact regularly but lack detailed load data. Managers therefore developed a Muther's Relationship Chart (below) to guide the initial layout planning. Relationships are based on frequency of contact, safety, and material compatibility.

Design and draw a 2×3 facility layout that best satisfies the relationships shown with all relevant connections shown.



**4. Product Layout (10 Pts)**

Rook Defense assembles a light armored reconnaissance vehicle (LARV) in small to medium batches. The company wants to scale up using a paced assembly line. Rook plans to produce 120 vehicles per week. The plant operates 8 hours/day for 5 days/week. The following table lists tasks, times, and precedence requirements.

<u>Task</u>	<u>Predecessor</u>	<u>Time (minutes)</u>
A – Cut and form hull plates		6
B – Machine suspension/wheel units		8
C – Build wiring harness & control unit		7
D – Install drivetrain into hull	A, B	9
E – Mount battery/fuel & cooling	D	6
F – Bench-test electronics	C	5
G – Calibrate systems software	E, F	8
H – Install external armor panels	D	8
I – Final inspection	G, H	4

4a (2 Points). Draw and label a precedence diagram.



4b. (2 Points). What is the desired cycle time in minutes per vehicle? Show all work.

Desired cycle time _____

4c. (2 Points). What is the minimum number of workstations Rook would need to support their production goal? Show all work.

Minimum number of workstations _____



4d. (2 Points) Group elements into workstations on your precedence diagram following the precedence rules and the desired cycle time. Then calculate the line efficiency using the precedence table, efficiency formulations, and your workstation grouping. Show all work.

Efficiency _____

4e. (2 Points). If labor cost and space were not a concern and Rook made every task its own workstation, would efficiency increase, decrease, or remain the same as in part 4d? Calculate and compare. Show all work.

Proposed Efficiency (Every Task a Workstation) _____

Increase, Decrease, or Same? _____

**5. Facility Location Model (10 pts)**

ValorX Fulfillment is a national e-commerce retailer planning a Mid-Atlantic fulfillment center to support next-day delivery on the East Coast. The new site will line-haul to five parcel sortation hubs and wants to minimize line-haul fuel costs. The coordinates below are a planning grid in miles (straight-line distance). Monthly “deliveries” are the number of outbound line-haul trips to each hub.

Parcel Sort Hubs	Coordinates		Monthly deliveries
	x	y	
A – Northeast Hub	620	760	300
B – Southeast Hub	160	180	180
C – Midwest Hub	240	420	220
D – Mid Atlantic Hub	520	160	260
E – New England Hub	300	740	140

5a. (5 Points) Determine the best site using the center-of-gravity technique with straight-line distances. Show all work.

Center of gravity coordinates _____



5b. (5 Points) ValorX is concerned the pure center of gravity may be far from interstate/rail spurs. The team identified three highway/rail-adjacent alternatives (given as x,y):

- Site 1: (540, 580)
- Site 2: (360, 520)
- Site 3: (240, 360)

Using the load-distance technique with the monthly deliveries as loads and straight-line distances, determine the best site for the new facility.

Best site for new facility _____

Instructor Solution



EM420 Problem Set #2

Name: _____

Academic Term 26-1

Hour: _____

DOCUMENTATION. This problem set is an individual graded assignment. Any assistance received must be documented in detail. Document all sources in accordance with the Office of the Dean Pamphlet “Documentation of Academic Work,” (July 2017), Appendix E, and course guidance. Documentation must be turned in at the time of submission. The problem set is considered late until all portions of the assignment and the documentation are submitted.

DUE: NLT 1159 on 6 November 2025

1. This is an individual graded assignment consisting of 5 sections worth a total of 45 points. The use of Microsoft Excel is *optional* for all problems.
2. Document any assistance received in accordance with the Dean’s Policy for the Documentation of Written Work. Present your work in a professional manner to ***include a cover sheet with a signed acknowledgement statement and works cited as a separate attachment from the problem set.***
3. Round any number to three decimal places. Round any dollar amount to the nearest cent. You must use units (i.e. \$, widgets/hr, etc.) in your final answer if appropriate.
4. The deliverable for this problem set is a ***written or digitally scanned copy of all problems, and, if used, an electronic copy of Microsoft Excel ® work.*** To submit the written portion, print this document and ***show all work*** in provided sections. Use extra paper if need be. Partial credit cannot be awarded if your thought process is not evident. Microsoft Excel ® work needs to be professionally presented in an organized, easy to follow, and properly identified manner. Please ensure to label each tab with the question it refers to and only complete one question per tab. **All results require explanation and substantive interpretation to demonstrate your knowledge and understanding of your work.** Thus if you cannot clearly articulate your work and knowledge in Microsoft Excel ® the problem must be completed by hand.

Submitted Microsoft Excel ® files ***must be documented appropriately if any assistance is received*** and should be named *LastName_First_Name_EM420_PS2.xlsx* and with your problem set on canvas prior to the due date.

GRADING SUMMARY:

Problem	Possible Points	Points Earned
1 – In-Class Activity Evaluation	7.5	
1 – In-Class Activity Alternative	7.5	
2 – Process and Technology	7.5	
2 – Process and Service Layout	10	
3 – Product Layout	10	
4 – Facility Location Model	10	
Total:	45	



1. **In-Class Activity Evaluation (7.5 Pts)**

If you were present during Lesson 9, complete this version of problem 1, if you were absent, complete the alternative version of problem 1. You may not choose.

1a. Please place an x in the box that corresponds to your presence/absence for Lesson 9: Service Design, during which we conducted the in-class product/service design activity.

- ☐ I was Present for Lesson 9
- ☐ I was Absent for Lesson 9

1b. (1 pt): Product/Service Description. Write a brief description of your Product or Service

1c. (4 pts): Pre Service/Product Analysis. Think back to your Service or Product before you conducted the in-class activity. Choose **one (1)** of the following to justify how you made decisions (pricing, quantity, service rate, etc.) using either Decision Making Under Risk, or Decision Making Under Uncertainty. Place an x in the box next to the set of tools you are using:

- ☐ Maximax, Maximin, Hurwicz, etc. (must use at least 3 different tools)
- ☐ Decision Tree

Use the area below for your work.



1d. (2.5 pts): Post Service/Product Analysis. Think back to your Service or Product after you conducted the in-class activity. Choose **two (2)** of the following sets of tools to assess your product or service performance. Use the data you collected during the exercise. Place an x in the box next to the set of tools you are using:

- ☐ Quality Indices, Product Yield, QPR
- ☐ Statistical Process Control Chart (P, C, Xbar and R)
- ☐ Capability (Cp, Cpk)
- ☐ Pareto Analysis

Use the area below for your work.



1(Alt). **In-Class Activity Alternative (7.5 Pts)**

If you were absent during Lesson 9, complete this version of problem 1, if you were present, complete the previous version of problem 1. **You may not choose.**

Alternative 1a. Please place an x in the box that corresponds to your presence/absence for Lesson 9: Service Design, during which we conducted the in-class product/service design activity.

- ☐ I was Present for Lesson 9
- ☐ I was Absent for Lesson 9

Alternative 1b. (1 pt): You will develop a service in this problem. Provide a short description of your service incorporating aspects of service design.

Alternative 1c. (1.5 pts): Analyze all seven dimensions of quality with respect to your service design.

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.



Alternative 1d. (2 pts): Build and draw a service blueprint for your service and be sure to identify: Line of Influence, Line of Interaction, Line of Visibility, and Line of Support.



Alternative 1e. (3 pts): You just started a small side business making **custom vinyl decals** for laptops and water bottles. You need to decide whether to buy **pre-cut decals** from a supplier or invest in a **vinyl-cutter kit** to make your own. Given the information below, what should you do? Build/Draw the decision tree and evaluate the Expected Value. Make a decision(s), write them down, and explain your reasoning.

- Pre-cut decals cost **\$5.00**, and a vinyl-cutter kit costs **\$7.00**.
- In the first week there is an **80%** chance the pre-cut decals remain usable (adhesive doesn't degrade).
- In the first week there is a **70%** chance the vinyl-cutter kit operates without defects.
- In the second week there is a **30%** chance the pre-cut decals remain usable.
- In the second week, you can decide whether to buy a **blade/alignment upgrade** for your vinyl-cutter which costs **\$2.00**.
- If the vinyl-cutter has the upgrade, there is a **90%** chance it continues operating without issues in the second week.
- If you do not buy the upgrade, the chance remains the same as the first week (**70%**).
- If your decals spoil or your cutter fails in the **first week**, you make **\$5** in profit. If it happens in the **second week**, you make **\$10** in profit.
- If it is still going in the **third week**, you make **\$15**, and if you have the vinyl-cutter (not the pre-cut decals), you can also **resell the cutter for \$5**.

Expected Value: _____



2. Process and Technology (10 Pts)

2a. (3 Points) AeroLink Systems is projecting sales of 30,000 autonomous drones in 2025 at an average selling price of \$80,000 per drone. Forecast fixed operating costs for 2025 are \$1.20 billion. What variable cost per drone must AeroLink achieve to break even in 2025? Show all work.

Variable Cost _____



2b. (4 pts): AeroLink currently manufactures its own flight-control modules. The in-house module line has fixed costs of \$150 million and variable costs of \$15,000 per module. Management is evaluating two outsourcing options:

- Option 1 – SkyTech Robotics (reservation contract): \$16,500 per module + \$90 million annual fixed fee.
- Option 2 – Stratos Components (spot purchases): \$18,000 per module, no fixed fee.

How many modules would AeroLink need to buy from SkyTech (Option 1) to be indifferent between manufacturing in-house and Option 1? Show all work.

Number of modules _____



2c. (3 Points) At what annual volume would AeroLink be indifferent between manufacturing in-house and Option 2 (Stratos at \$18,000/unit, no fixed fee)? Show all work.

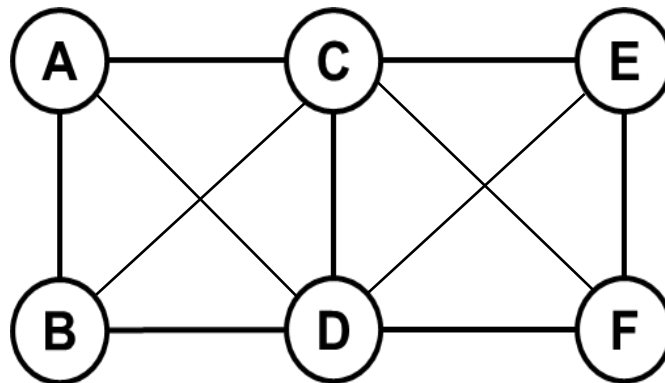
Annual Volume _____



3. Process and Service Layout (10 Pts)

3a. (2.5 Points) CatalystCo is commissioning a pilot plant for specialty catalyst production. Engineers proposed the 2×3 layout below and recorded weekly load movements (pallets/containers) between unit operations. Adjacency means sharing a side *or* a corner (diagonals count as adjacent). Calculate the number of non adjacent loads in the current (initial) design, and draw them on the diagram below.

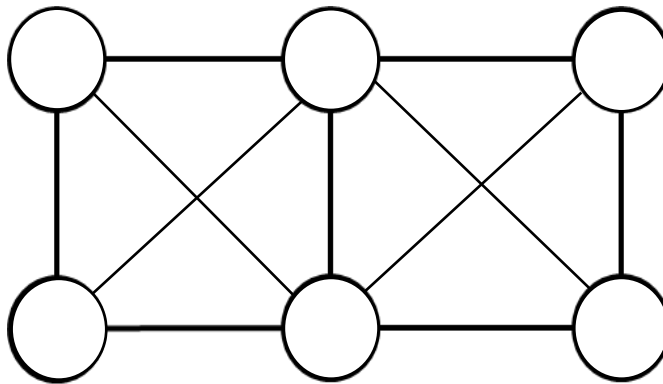
	A	B	C	D	E	F
A - Reactor	-	240	60	80	180	0
B – Distillation	200	-	40	30	160	10
C – QC Lab	40	20	-	10	0	90
D – Utilities/Heat Exchange	40	20	10	-	40	0
E – Solvent Recover	160	140	0	30	-	40
F – Drum Warehouse	0	0	80	0	60	-



Initial Nonadjacent loads _____



3b. (2.5 Points) Design and draw a new 2×3 layout (reassign A–F to the six cells) that minimizes the number of non-adjacent loads. Write the total nonadjacent loads in the block below.

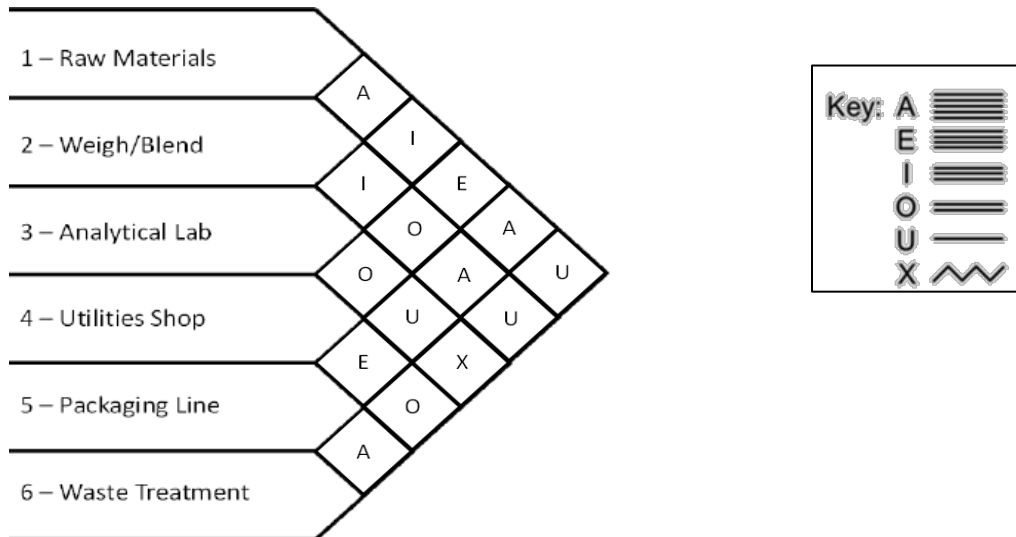


New Layout Nonadjacent loads _____



3c. (5 Points) After completing the pilot-plant optimization, CatalystCo plans to build a materials-handling annex to support catalyst scale-up. This new area will include several support departments that interact regularly but lack detailed load data. Managers therefore developed a Muther's Relationship Chart (below) to guide the initial layout planning. Relationships are based on frequency of contact, safety, and material compatibility.

Design and draw a 2×3 facility layout that best satisfies the relationships shown with all relevant connections shown.



**4. Product Layout (10 Pts)**

Rook Defense assembles a light armored reconnaissance vehicle (LARV) in small to medium batches. The company wants to scale up using a paced assembly line. Rook plans to produce 120 vehicles per week. The plant operates 8 hours/day for 5 days/week. The following table lists tasks, times, and precedence requirements.

<u>Task</u>	<u>Predecessor</u>	<u>Time (minutes)</u>
A – Cut and form hull plates		6
B – Machine suspension/wheel units		8
C – Build wiring harness & control unit		7
D – Install drivetrain into hull	A, B	9
E – Mount battery/fuel & cooling	D	6
F – Bench-test electronics	C	5
G – Calibrate systems software	E, F	8
H – Install external armor panels	D	8
I – Final inspection	G, H	4

4a (2 Points). Draw and label a precedence diagram.



4b. (2 Points). What is the desired cycle time in minutes per vehicle? Show all work.

Desired cycle time _____

4c. (2 Points). What is the minimum number of workstations Rook would need to support their production goal? Show all work.

Minimum number of workstations _____



4d. (2 Points) Group elements into workstations on your precedence diagram following the precedence rules and the desired cycle time. Then calculate the line efficiency using the precedence table, efficiency formulations, and your workstation grouping. Show all work.

Efficiency _____

4e. (2 Points). If labor cost and space were not a concern and Rook made every task its own workstation, would efficiency increase, decrease, or remain the same as in part 4d? Calculate and compare. Show all work.

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Best site for new facility _____